

**Multimodal Driving Risk Prediction Using Machine Learning
on Autonomous Vehicle Data (NVIDIA Physical AI Dataset)**

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DECLARATION

I declare that this Applied Research Project that I have submitted to Dublin Business School for the award of Master of Science in Business Analytics is the result of my own investigations, except where otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not been submitted for any other degree.

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LIST OF ABBREVIATIONS

AV	Autonomous Vehicle
BDD100K	Berkeley DeepDrive 100K Dataset
CNN	Convolutional Neural Network
CRISP-DM	Cross Industry Standard Process for Data Mining
CSV	Comma-Separated Values
DADA	Driver Attention in Driving Accident Dataset
DoTA	Detection of Traffic Anomaly Dataset
DRIVE	Deep Reinforced Accident Anticipation with Visual Explanation
EDA	Exploratory Data Analysis
F1	F1-Score (Harmonic Mean of Precision and Recall)
GPU	Graphics Processing Unit
KITTI	Karlsruhe Institute of Technology and Toyota Technological Institute Dataset
LiDAR	Light Detection and Ranging
ML	Machine Learning
NVIDIA	NVIDIA Corporation
PCA	Principal Component Analysis
PhysicalAI	NVIDIA Physical AI Autonomous Vehicle Dataset
PRC	Precision-Recall Curve
radar	Radio Detection and Ranging
ResNet	Residual Neural Network
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SAE	Society of Automotive Engineers
SHAP	SHapley Additive exPlanations
ViT	Vision Transformer
Waymo	Waymo Open Dataset
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8

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ABSTRACT

This study develops and evaluates a multimodal machine learning framework for driving risk prediction using autonomous vehicle data from the NVIDIA PhysicalAI Dataset. Motivated by limitations in existing approaches — which typically address isolated perception tasks without integrating behavioural, visual, and contextual signals — the research proposes a pipeline combining Convolutional Neural Network embeddings, Vision Transformer representations, YOLOv8 object detection, ego-motion telemetry, and contextual driving indicators, fused within an XGBoost classifier. Applied to 1,899 autonomous driving clips across 48 features, the framework achieved a cross-validated ROC-AUC of 0.9455 and a held-out test ROC-AUC of 0.9387. A four-way ablation study found contextual and temporal features to be the dominant predictive modality, while removing CNN and ViT embeddings marginally improved performance, suggesting flat feature concatenation introduces noise at this dataset scale. Proxy labels were validated through three independent statistical strategies. SHAP and feature group importance analyses provide modality-level explainability throughout.

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CHAPTER 1 – INTRODUCTION

Machine Learning is now a central component in contemporary automotive transport safety research due to the very rapid development of autonomous vehicle technology. In less than one decade, vehicles able to perceive, interpret and respond to the complexities of real world driving situations have transitioned from laboratory based research to the public roads they serve. However, this represents an unexplored basic issue of how a system could reliably recognize when a particular driving condition is becoming hazardous prior to something going wrong. This dissertation addressed this question through the design and evaluation of a multimodal machine learning framework for predicting driving hazard using real world autonomous vehicle data from the NVIDIA PhysicalAI Autonomous Vehicle Dataset.

1.1 Background of Study

The last ten years have seen major advancements in driverless car technology, thanks to tremendous growth in deep learning, sensor integration, and big-data development (Viktor & Kiss, 2026; Grigorescu et al., 2020) . The majority of today's driverless cars use a combination of sensors like cameras, lidar, and radar to see their environment in real-time. In addition to those advances, large-datasets such as KITTI, BDD100K, Waymo, NuScenes, and Argoverse 2 have greatly increased the pace of research and development in the areas of perception and safe modeling (Sun et al., 2020; Geiger et al., 2012; Yu et al., 2020; Caesar et al., 2020; Wilson et al., 2023). As a result, convolutional neural networks (CNNs) have developed into the most popular type of computer model used to extract features visually (He et al., 2016), with vision transformers emerging as a complementary method of extracting global spatial relationships using attention mechanisms (Dosovitskiy et al., 2021). At the same time, gradient boosting algorithms like xgboost have been shown to be highly effective at predicting outcomes based on structured data collected from vehicles (Guagnano et al., 2023; Chen & Guestrin, 2016).

1.2 Research Problem / Gap

There is an evident void in the research concerning integrated, explainable models for predicting driving risk using multimodal information. Although there have been many studies focused on the application of computer vision to the problem of detecting objects in images from cameras mounted on vehicles as well as anticipating accidents, the methods used by such researchers usually focus on individual tasks and do not provide a unifying approach that uses both visual, behavioral, object level and contextual information. Researchers who have applied machine learning techniques to predict driving risks often use physiological or behavioral data without integrating them with deeper visual features. Therefore, this study will be the first to explore the combination of CNNs and Vision Transformers in order to create a risk prediction model. In addition, nearly all previous studies of explaining the driving behavior through multimodal input did not include a systematic method for explaining how each modality contributed to the final prediction. Therefore, this dissertation will also contribute to addressing all of the aforementioned voids.

1.3 Aim

This study's purpose was to build and test an artificial intelligence based multimodal approach to predict the risk of accidents for drivers using autonomous vehicles with data from the NVIDIA Physical AI dataset.

1.4 Research Questions

The five research questions that direct the investigation of this study are:

RQ1: How well will multimodal autonomous driving information help identify elevated risk driving situations?

RQ2: What contributions can CNN (Convolutional Neural Networks) and Vision Transformers make as a representation of the visual component for multimodal driving risk prediction models?

RQ3: To what extent are behavioral, object-based, and context based driving characteristics influential on the accuracy of predicted driving risks?

RQ4: Are there certain explanations for Machine Learning algorithms that increase the ability to understand the predictions generated by autonomous vehicle risk prediction models?

RQ5: How robustly will the proposed multimodal model function when evaluated with both Ablation and Cross-Validation strategies?

1.5 Objectives

The following Objectives will be used to operationalize the Research Aim:

Objective 1: Determine if multimodal feature combination is effective as a method for predicting autonomous vehicle risk.

Objective 2: Extract and analyze Convolutional Neural Network (CNN) and Vision Transformer visual embeddings from autonomous vehicle images.

Objective 3: Integrate behavioral telemetry, output of object detection models, and contextually relevant driving-related information in a single Machine Learning framework.

Objective 4: Assess the predictive performance of the proposed framework through use of Receiver Operating Curve Area Under the Curve (ROC-AUC), Precision, Recall, F1-Score, Cross Validation and other appropriate metrics.

Objective 5: Analyze how features are important and what contribution modalities make by evaluating Feature Group Importance, SHAP values, Ablation Studies and other methods.

Objective 6: Investigate the Robustness and Limitations of modeling driving risk via proxy based methods when there is partial labeling on an Autonomous Vehicle dataset.

1.6 Approach

The research in this project employed an experimentally driven, quantitatively oriented research design; specifically it used publicly available secondary autonomous vehicle data that was acquired from the NVIDIA PhysicalAI Dataset.

In order to provide a systematic and transparent methodology, the research process followed the Cross Industry Standard Process for Data Mining (CRISP-DM) as described by Chapman,

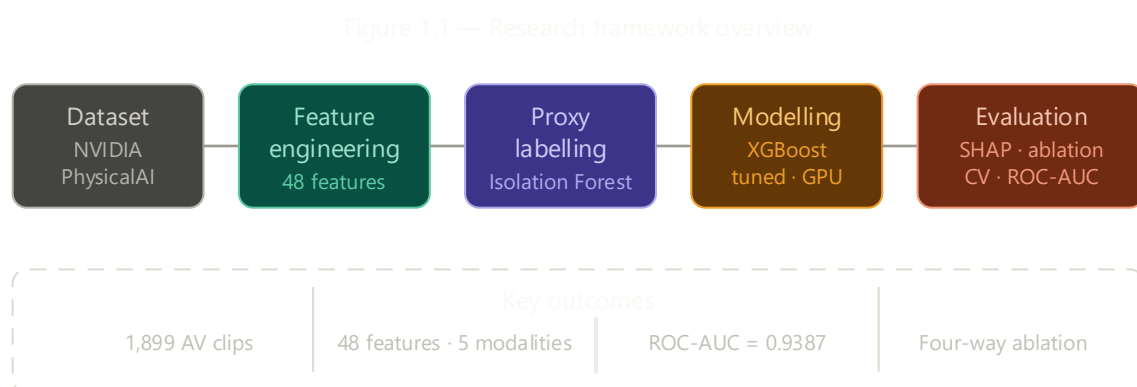
P. et al. (2000); which includes business understanding, data understanding, data preparation, modeling, evaluation and deployment.

A multimodal machine learning pipeline will be implemented for the proposed framework:

- Diversity-aware clip selection from the NVIDIA PhysicalAI Dataset will produce 1899 driving clips.
- Behavioural telemetry attributes such as speed, acceleration, braking and jerk will be extracted from ego-motion parquet files.
- CNN visual embedding extraction using a pre-trained ResNet architecture (He et al., 2016) and then reduced to ten principal components using PCA.
- Vision Transformer visual embedding extraction using ViT-B/16 (Dosovitskiy et al., 2021) and then similarly reduced using PCA.
- YOLOv8 object detection (Redmon et al., 2016) will be used to generate object level attribute vectors.
- A single dataset of behavioural, visual, object level and contextually related attributes will be created from the above sources.

The pipeline progressed through proxy risk labelling with construct validity testing, two-phase gap-aware hyperparameter optimisation, XGBoost model training, per-fold fresh label cross-validation, SHAP explainability, and a four-way ablation study which is explained thoroughly in the Chapter 3 - Methodology.

Figure 1.1 — Research framework overview



1.7 Dissertation Structure

Chapter 1 — Introduction presents the research background, problem, gap, aim, research questions, objectives, and overall approach, establishing the motivation for multimodal driving risk prediction within autonomous driving environments.

Chapter 2 — Literature Review critically reviews existing research on autonomous driving perception systems, AV datasets, CNN and transformer-based visual learning, object detection, driving risk prediction, accident anticipation, and explainable AI, establishing the gap this dissertation addresses.

Chapter 3 — Methodology presents the research design and experimental methodology, covering dataset selection, feature engineering, proxy risk labelling and validation, hyperparameter optimisation, model selection, evaluation metrics, cross-validation strategy, and ethical considerations.

Chapter 4 — Artefact Design and Implementation describes the system architecture, five-stage data processing pipeline, and implementation of CNN, ViT, YOLOv8, SHAP, and XGBoost components across five Google Colab notebooks.

Chapter 5 — Results and Analysis presents classification performance, hyperparameter tuning progression, feature importance, SHAP explainability, four-way ablation study findings, and cross-validation results.

Chapter 6 — Discussion critically interprets the findings in relation to the literature and research questions, addressing multimodal fusion implications, the unexpected CNN and ViT finding, contextual feature dominance, limitations, and future research opportunities.

Chapter 7 — Conclusion summarises the overall findings, original contributions, and directions for future work in autonomous driving risk prediction and explainable multimodal AI systems.

CHAPTER 2 – LITERATURE REVIEW

This chapter is a review of past literature from within eight thematic areas which are relevant to this study - Autonomous Driving Systems & Sensor Technology, Autonomous Vehicle Datasets, Convolutional Neural Networks for AV Perception, Transformer Based Visual Learning, Object Detection, Driving Risk Prediction, Accident Anticipation/Anomaly Detection and Explainable Artificial Intelligence

Table 2.1 — Summary of Key Related Works

Author (Year)	Dataset	Method	Key Features	Limitation
Guagnano et al. (2023)	Simulation	XGBoost	Behavioural, physiological	No visual features
Feth et al. (2019)	Stereo camera	CNN	Visual only	No behavioural context
Huang et al. (2023)	Custom	CNN + semi-supervised	Visual	Limited labels
Fu et al. (2023)	Sensor data	CNN + Bi-LSTM	Sensor fusion	No contextual features
Zhu et al. (2024)	AV logs	Random Forest	Causal features	No visual modality
Bao et al. (2021)	Dashcam	Reinforcement Learning	Visual attention	No structured features
Wang et al. (2019)	Crash records	ML framework	Behavioural, road	No visual embeddings

2.1 Autonomous Driving Systems and Sensor Technologies

Autonomous vehicles represent an example of perhaps the most complex AI application yet conceived. The complexity arises due to the need to fuse together perception, prediction, planning and control in real time.

Grigorescu et al., (2020) conducted an extensive literature review of Deep Learning methods being used throughout the entire AV Perception Stack and documented the progression from rule based systems towards End-To-End Deep Learning architectures. As such they were able to document that there exists no singular architecture capable of being sufficient for all AV related tasks -- individual tasks have unique requirements regarding their respective learning paradigms.

Viktor and Kiss (2026) demonstrate that cameras, LiDAR, radar, and GPS each carry distinct advantages and failure modes, providing the theoretical justification for multimodal fusion approaches that this dissertation builds upon.

In addition, Beck et al. (2023) demonstrate the benefits of utilizing multimodal sensor data in the field of autonomous vehicle crash reconstruction. Specifically, they show that LiDAR, Camera and Radar streams can be fused together to aid in detailed accident investigations. As such, their results validate the utilization of combined AV sensor data for safety critical analyses and directly informed the multimodal perspective taken in this research.

2.2 Autonomous Vehicle Datasets

The ability to have many different types of large-scale, well-annotated data sets was key to the advancements made in research pertaining to autonomous vehicles. A foundational data set developed by Geiger et al. (2012), known as the KITTI Vision Benchmark Suite, provided stereo images and LiDAR point clouds along with GPS data to become a benchmark for developing AV data sets and evaluating them. Yu et al. (2020) released a new data set named BDD100K, which included 100,000 driving videos captured under various environmental conditions. This new data set allowed researchers to develop models capable of performing multiple tasks simultaneously. An additional advancement was made when Sun et al. (2020) released the Waymo Open Data Set. They used high resolution data from multi-camera and LiDAR systems to allow researchers to scale their development of models.

Additionally, Caesar et al. (2020) released the nuScenes data set; it was specifically developed to evaluate the potential of using sensor fusion for applications related to autonomous vehicle technology. It uses data from cameras, LiDAR and radar and includes detailed 3-D annotations. The most recent addition to AV data sets, by Wilson et al. (2023), was Argoverse 2. Although there have been several different data sets created for AVs, none of the previously mentioned include ground truth "driving risk" labels, making proxy labelling methods, like the isolation forest method described in this dissertation, necessary. However, Yao et al. (2023) did create a partial exception by releasing the DoTA dataset. This dataset contains over 4,600 seconds of 4,677 egocentric driving videos with temporal, spatial and categorical anomaly annotation.

2.3 Convolutional Neural Networks in Autonomous Vehicle Perception

The majority of architectures used to extract visual features in Autonomous Vehicles (AV) since the mid-2010s have been based on convolutional neural networks (CNN). The introduction of ResNet (He et al., 2016) demonstrated how deep networks can be successfully trained with residual connections. This was achieved with state-of-the-art results on ImageNet and established the ResNet family as the most commonly adopted backbone architecture for visual feature extraction in AV perception pipelines.

Specifically in terms of driving risk, Feth et al. (2019) demonstrated the potential to use CNNs to perform dynamic risk assessments using stereo camera imagery and disparity maps. They were able to produce scene level risk estimates from visual input only; which establishes that visual features produced by CNN architectures contain real risk relevant signal. However their method relied entirely on visual modalities and did not incorporate any behaviour or contextually related data -- an omission addressed in this dissertation. Similarly, Huang et al. (2023) embedded a CNN within a semi-supervised learning framework for perceived risk predictions. They confirmed that visual features do capture meaningful risk relevant scene information but noted the challenges associated with acquiring enough labelled data -- a challenge faced by this dissertation.

2.4 Transformer-Based Visual Learning

The introduction of the Vision Transformer by Dosovitskiy et al. (2021) has created a new paradigm in visual feature learning. In essence, ViT treats an image as a series of patches processed using a self-attention mechanism. This architecture shows that transformer models originally designed for natural language processing can be used to obtain state-of-the-art image classification results similar to CNNs with no use of convolution.

Unlike CNNs that learn hierarchical representations based on local spatial relationships, ViT learns about global spatial relationships among objects in entire scenes -- a key concept in the holistic understanding of driving scenarios.

Li et al. (2022), in their work "BEVFormer" showed how transformer architectures for multi-camera spatial perception were extended to generate bird's eye view representations from cameras without LiDAR using spatiotemporal attention on multiple camera input streams. Additionally, Ma et al. (2023) conducted a survey on the rapidly developing area of vision-centered BEV perception. They documented how CNN-only methods are being replaced by transformer-based methods in multi-camera benchmarks. All of these studies demonstrate the increasing adoption of transformer-based architectures in AV perception; however they also raise the question as to whether CNN and ViT embedding features are mutually exclusive or redundant when combined into a risk prediction system. This is the central question addressed in this dissertation.

2.5 Object Detection in Autonomous Driving

Object detection is one of the key perceptual tasks in autonomous vehicles. It allows vehicles to detect and localize all relevant entities on the road such as pedestrians, cars and road hazards in real-time. To enable object detection in real-time while maintaining high performance Redmon et al. (2016) proposed YOLO – an approach that reformulates object detection into a single regression problem across a unified architecture. As a result YOLO was able to perform faster than previous two-stage object detection approaches. Although Ren et al. (2015) proposed Faster R-CNN, it is able to achieve better performance due to its use of Region Proposal Networks; however, this is achieved at a greater computational

overhead. Both architectures serve as the basis for the YOLOv8 object detection model utilized in this study.

Baek et al. (2020) demonstrated that the density and composition of detected objects - specifically pedestrian/cyclist presence and multiple vehicle counts -- directly relate to the likelihood of collision. Therefore, including object-count based features from YOLOv8 as a discrete input modality with the multimodal framework provides the ability to include object-based complexity of a scene as part of the overall risk assessment using both visual embedding representations and behavioral telemetry data.

2.6 Driving Risk Prediction

A significant body of recent literature exists related to driver behavior risk prediction; however, this work focuses specifically on developing a predictive model capable of distinguishing between elevated and normal levels of risk based upon real-time input from a variety of sensors. The authors of this dissertation draw heavily from the literature by selecting both the machine learning methodology (XGBoost) and the overall architecture of this predictive model (fusion of data from different sensory inputs), but add a unique component to the literature by incorporating visual perception into the risk assessment.

Guagnano et al. (2023) demonstrated that XGBoost performed better than other classifier models when used to predict structured driving risk data from physiological, personal, behavioral and environmental factors in a simulated environment. While these results provide some justification for why XGBoost was selected as the primary classification model for use in this project, they also illustrate the limitations of relying exclusively on non-visual (physiological and behavioral) inputs for assessing driving risk. Fu et al. (2023) developed a hybrid CNN-BiLSTM neural network for predicting driving behavior risk from sensor data.

A notable shared limitation across existing driving risk prediction studies is their dependence on either fully labelled crash records or physiological measurements that are

impractical to obtain at scale in real-world AV deployments. This dissertation addresses this gap through a proxy labelling strategy validated through independent statistical methods.

2.7 Accident Anticipation and Anomaly Detection

Although the goal of this paper is to develop the XAI method, the related work provides important background. The research by Kataoka et al. (2018), Bao et al. (2021), Fang et al. (2021) and Liao et al. (2024) are relevant as they have all developed models that can predict crashes using visual data. However, their models require very large amounts of labelled video data that may not be available in many AV-related projects, or alternatively use complex temporal models that cannot be used when working with short video clips. In addition, all of them were focused on developing models for crash prediction rather than developing models that could explain how those models worked. Therefore, we decided to take a different approach. Instead of trying to develop a model that could make accurate predictions of whether a vehicle would crash using small video clips, we instead tried to develop a model that could accurately explain why our model made certain predictions.

2.8 Multimodal Fusion and Contextual Risk Modelling

Recent research has demonstrated the value of multimodal and contextually aware approaches to modeling driving decisions and assessing risks associated with driving. Collectively, they show that the integration of multiple forms of information will significantly outperform unimodal methods when evaluating the safety performance of Autonomous Vehicles. Wang et al. (2025) develop a decision-making strategy for autonomous vehicle behavior using a Deep Reinforcement Learning approach that explicitly incorporates driving risk into a hybrid state space. They demonstrate that representations of risk improve decision making relative to safe decision-making in high-risk traffic environments.

Xue et al. (2025) describe a connected autonomous vehicle field-based driving risk-car following model that estimates crash risk while considering interactions among multiple

vehicles. The study shows that including contextual information related to surrounding vehicles provides greater than single vehicle behavior estimations of crash risk. Therefore,

Xue et al. (2025) provide direct evidence for incorporating contextual/object level features into the multimodal prediction framework developed in this dissertation. Wu et al. (2024) create a visual risk-aware decision making methodology that combines both visual perception with risk assessment to inform safer AV behavioral planning. Like the findings from Xue et al. (2025), Wu et al. (2024) also reinforce the benefit of combining contextual/visual sensing for improving safety oriented tasks.

Baek et al. (2020) evaluate vehicle trajectory predictions and collision warnings generated by multi-sensory fusion and wireless vehicular communication systems. Similar to previous studies, Baek et al. (2020) found that a combination of sensor data with contextual communication information improved collision warning performance compared to either source used separately. Together, these research efforts identify a common theme among different methodologies: understanding the operational context of drivers' situations — who else is around, what time it is, how the environment appears — is equally or even more critical to estimating driving risk than perceiving the driver's current situation. These conclusions are supported in Chapter 5 which reports that contextual/temporal features were identified as the primary predictive modality in the proposed multimodal prediction framework.

2.9 Explainable Artificial Intelligence in Autonomous Driving

In order to effectively deploy machine learning in safety critical systems it is essential that models are interpretable. Lundberg & Lee (2017) provided SHAP, which provides feature importance explanations of both global and local nature using Shapley Values from Cooperative Game Theory and can be used with any machine learning model. This is now the industry standard for post hoc model interpretability in machine learning and has been applied in many transportation safety applications. The inclusion of SHAP along side gain-based feature importance and feature group importance as part of this dissertation's explainability methods allows for all of the required levels of interpretation found in the

literature regarding AV risk prediction systems (Wang et al., 2019; Bao et al., 2021; Zhu et al., 2024) to be met.

2.10 Research Gap Synthesis

The five interdependent research gaps identified from the reviewed literature in this Chapter together served as the motivation for this dissertation. First, current frameworks are almost exclusively focused on individual perception tasks rather than on multimodal driving risk prediction (Feth et al., 2019; Bao et al., 2021; Kataoka et al., 2018). Second, while there has been significant development of machine learning-based models for predicting driving risk using physiological and behavioral data, none have utilized deep visual features (Fu et al., 2023; Guagnano et al., 2023; Wang et al., 2019). Third, little work exists that combines both CNN and Vision Transformer embeddings into a single risk prediction model. Fourth, most prior methods utilize fully annotated crash datasets to train their models which limits application to real world autonomous vehicle (AV) environments where labeled risk information is often not available (Huang et al., 2023). Finally, few AV safety models include some form of systematic explanation of how each modality contributes to overall predicted risk (Zhu et al., 2024; Lundberg and Lee, 2017).

This study addresses all of these gaps through the creation of a multimodal machine learning framework that uses CNN embeddings, ViT representations, YOLOv8 object detection, behavioral telemetry, and contextual indicators as input to a single XGBoost classifier.

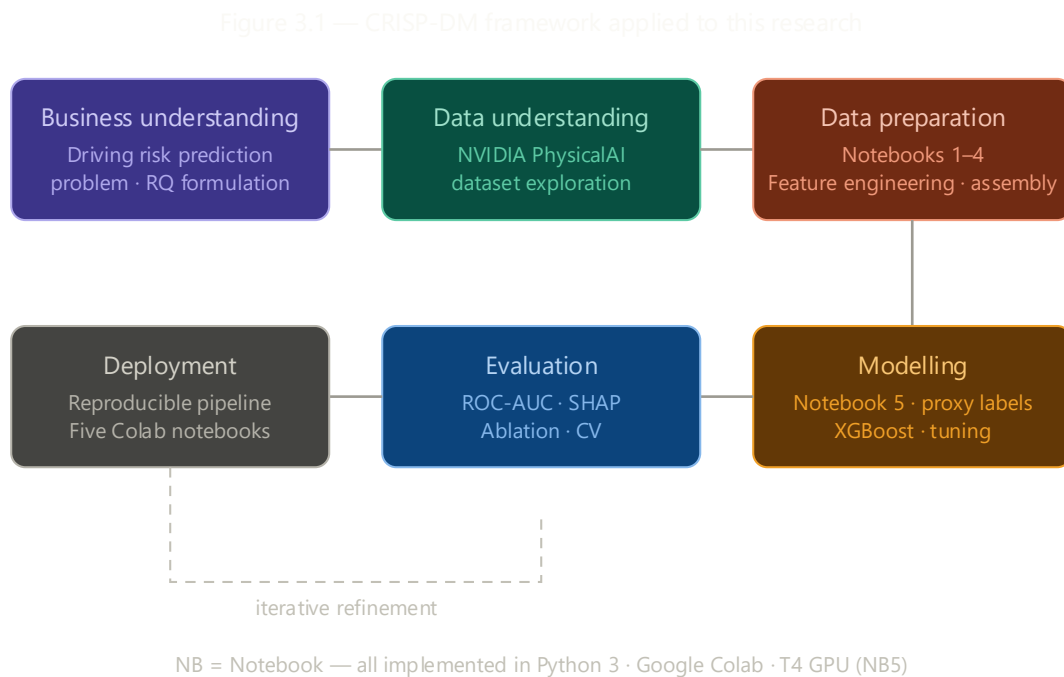
CHAPTER 3 – METHODOLOGY

3.1 Research Design and Philosophy

This study used an experimental quantitative research methodology. The methodology is based on a positivistic paradigm. According to Grigorescu et al. (2020), the positivistic paradigm entails that there are objective, measurable patterns exist within autonomous vehicle driving data; these can be identified by systematic empirical analysis. Secondary data was utilized from the NVIDIA Physical AI dataset. Utilizing a secondary data set avoids some of the ethical considerations as well as some of the practicalities associated with collecting primary data in AV systems. In addition, utilizing secondary data allows for the study of large amounts of data that would be difficult or impossible to analyze through other means.

3.2 CRISP-DM Research Framework

The structure of the research process followed the CRISP-DM framework for data mining, as it provided an objective methodology that can be used consistently and systematically throughout each of the six stages. The business understanding stage identified the specific risk prediction issue(s), and framed the research questions outlined in Chapter 1. The data understanding stage included the review of the NVIDIA PhysicalAI Dataset metadata in Notebook 1. The data preparation stage involved the development of features based on ego motion in Notebook 2, visual features extracted from Notebook 3, and multi-modal datasets assembled in Notebook 4. The modelling stage was implemented in Notebook 5, where classifiers were trained and hyperparameters were optimized. The evaluation stage included ROC-AUC assessment, cross-validation, threshold optimization, SHAP analysis, and ablation studies. The deployment phase represents the reproducible five-stage pipeline within the five Colab notebooks, which are considered the artefact of this dissertation.

Figure 3.1 — CRISP-DM framework applied to this research

3.3 Dataset Selection and Justification

The NVIDIA PhysicalAI Autonomous Vehicle Dataset served as the empirical base for this study. The data set is available on Hugging Face and offers multimodal autonomous driving data (including ego-motion telemetry data; front-wide camera images; and extensive clip level metadata) that has been captured in real world settings in various locations around the globe during different times of day and lighting conditions. The data set was collected and maintained by NVIDIA’s Autonomous Vehicle R&D Division, under the supervision of Prof. Marco Pavone, Distinguished Research Scientist & Director of Autonomous Vehicle R&D at NVIDIA and an Associate Professor at Stanford University, which lends the data set considerable scientific credibility.

Three main reasons were used to select this data set. First, it offers true multimodal AV data allowing the extraction of multimodal features, a key element of this study. Second, it captures real world driving scenarios under diverse environmental conditions thereby enhancing the ecological validity of the extracted features. Finally, due to its availability via

an authenticated Hugging Face account, the data allows for replicability of the research. A total of 2,000 clips from the entire data set were randomly sampled based upon a diversity aware sampling approach that combined 1,000 rare clips identified through Isolation Forest Anomaly Detection and 1,000 typical clips, limited by a maximum of 25% clips from a single country and a limit of 150 unique segments. Following a quality filter to exclude invalid clips without either valid ego-motion or camera data, the data set used for analysis consisted of 1,899 valid clips.

3.4 Feature Engineering

Across 3 different Notebooks, 5 Categories of Features were created to be combined as one Multimodal Dataset containing 1,899 Clips with a total of 48 Features.

Table 3.1 — Feature Group Summary

Feature Group	Count	Features
Ego / Behaviour	8	avg_speed, max_speed, avg_acceleration, max_acceleration, avg_jerk, max_jerk, braking_events, hard_braking_events
Context / Time	6	hour_of_day, month, is_night, is_peak_hour, lighting_condition_enc, is_poor_lighting
Engineered	3	aggressive_score, env_complexity, object_density
Basic Visual	2	brightness, contrast
YOLO Object Detection	9	person_count, bicycle_count, car_count, motorcycle_count, bus_count, truck_count, traffic_light_count, stop_sign_count, total_objects
CNN Embeddings	10	cnn_feature_0 to cnn_feature_9
ViT Embeddings	10	vit_feature_0 to vit_feature_9
Total	48	

3.4.1 Behavioural and Ego-Motion Features

Behavioural characteristics have been identified from the video level vehicle telemetry data contained in Parquet file format within Notebook 2. Eight clip level features were created for every clip: mean speed, max speed, mean acceleration, max acceleration, mean jerk, max jerk, number of braking events and number of hard braking events. The definition for a "hard" braking event was based on frames that had an acceleration less than -3 m/s^2 . This is based on conventional definitions used by researchers conducting research into driving behaviours (Guagnano et al., 2023) (Wang et al., 2019). Collectively these provide a description of how the vehicle behaves dynamically over the duration of each clip and capture unsafe or aggressive driving styles, including vehicle stability issues and rapid changes in direction.

3.4.2 Contextual and Engineered Features

Contextual features were built from metadata at the clip-level. In the final dataset, Six contextual features were included: hour of day, month, is_night (is equal to 1 for hours between 20:00 and 06:00), is_peak_hour (is equal to 1 during morning and evening peak traffic periods), lighting_condition_enc (an encoded categorical variable derived from frame brightness), and is_poor_lighting (a binary flag for clips with a mean brightness below 40). Additionally, Three engineered composite features were additionally derived: aggressive_score, env_complexity, and object_density, with full computation details in Section 4.5.

3.4.3 Visual Features — Brightness and Contrast

Visual descriptions for basic (mean) brightness & contrast — were extracted from each of a clip's wide-front view camera video frame in Notebook 3 to describe lighting conditions at minimal processing cost by means of OpenCV. Two descriptors that are lightweight and inexpensive to process at low computational cost add to those obtained from Section 3.5 & Section 3.6 by using deep learning.

3.5 CNN Feature Extraction

CNN embeddings were extracted using a pre-trained ResNet-50 architecture (He et al., 2016) in Notebook 3. The last Classification Layer is removed so that we have an output of 2048 dimensional Penultimate Feature Vector. All input Frames are preprocessed by applying Normalization Transforms, which include resizing to 224x224 pixels, and normalization with mean and stddev of ImageNet. To reduce the Dimensionality of the CNN Embedding into ten principal component space for easier integration in downstream applications, PCA was used. The Ten Components are labeled as `cnn_feature_0` through `cnn_feature_9`, where these ten components represent the most important axis for describing the visual Variability.

3.6 Vision Transformer Feature Extraction

The vision transformer (ViT) embeddings for this project were produced from a pre-trained ViT-B/16 model (Dosovitskiy et al. 2021), within Notebook 3. This model breaks down each video frame into 16x16 patches and then passes those patches through twelve transformer encoder layers employing multi-head self-attention, resulting in a 768 dimensional vector representing the "class token" embedding which captures all of the local and global visual information throughout the full field-of-view. The classification layer was deleted and PCA was run on the ViT embeddings individually, creating ten independent principal components labeled as `vit_feature_0` through `vit_feature_9`.

3.7 Object Detection — YOLOv8

Object level features were extracted from clip data in notebook 3 by using a yolov8x model (redmon et al. 2016; Ultralytics, 2023) to detect objects across eight categories of importance to driving risk on each clip's middle frame including persons, bicycles, motorcycle, car, bus, truck, traffic light and stop sign. Total count of objects detected was

also calculated as an overall indicator of scene complexity producing nine object level features per clip.

3.8 Multimodal Dataset Assembly

In Notebook 4, all of the feature groups have been combined into one dataset that includes multiple types of data (multimodal) in Notebook 4 by using the clip id as the key for all of the feature datasets. An inner join was used on all merge steps so that only clips that had full feature data for all different types of data were included. This final unification of the data resulted in 1899 rows and 48 feature columns arranged into 7 categories; behavioral/ego motion, context, engineering, vision/basically, cnn/vit embeddings and yolo/detection.

3.9 Proxy Risk Labelling Strategy and Validation

A proxy labelling strategy was used instead of ground truth crash or risk labels for the nvidia physical ai Dataset due to the rare occurrence of real-world crash events and the lack of explicit annotations available in most AV safety studies. As such this is a common challenge in AV safety research where a labelling model must be trained on an anonymous set of features to create the missing labels.

The data were split into test and training sets before any label creation. Then an Isolation Forest model was fitted solely on the training set's behavioural and contextual features which are speed, acceleration, jerk, braking events, hour of day, is_night, and is_peak_hour. The 25th percentile of the training clips based on anomaly score from that model created the threshold within which all training clips at or below this value were classified as having elevated driving risk. Using the same fitted model and threshold as applied to the test set without refitting prevented target leakage by guaranteeing that no information from the test set was used in the creation of the labels.

Validity of the labels was established through three independent statistical validation strategies. first, held-out behavioural and engineered features not used in the Isolation Forest

modelling process — including average acceleration, environmental complexity, object density, aggressive score, and `is_poor_lighting` — were tested using Mann-Whitney U tests. Results showed statistically significant differences between high-risk and low-risk clip features that the labelling model had never seen. second, visual features from CNN, `vi_t` and YOLO were tested which are completely independent of the ego-motion-based labelling process. Results showed that high-risk clips exhibit systematically different visual characteristics than low-risk clips despite these visual features playing no role in label creation. third, Cohen's d effect size calculations were computed across all validated features to confirm that differences are both practically as well as statistically significant. Collectively these three approaches provide construct validity evidence that the proxy labels capture real driving risk patterns rather than artifacts arising from label creation.

Figure 3.2 — Proxy Risk Labelling Methodology Flowchart

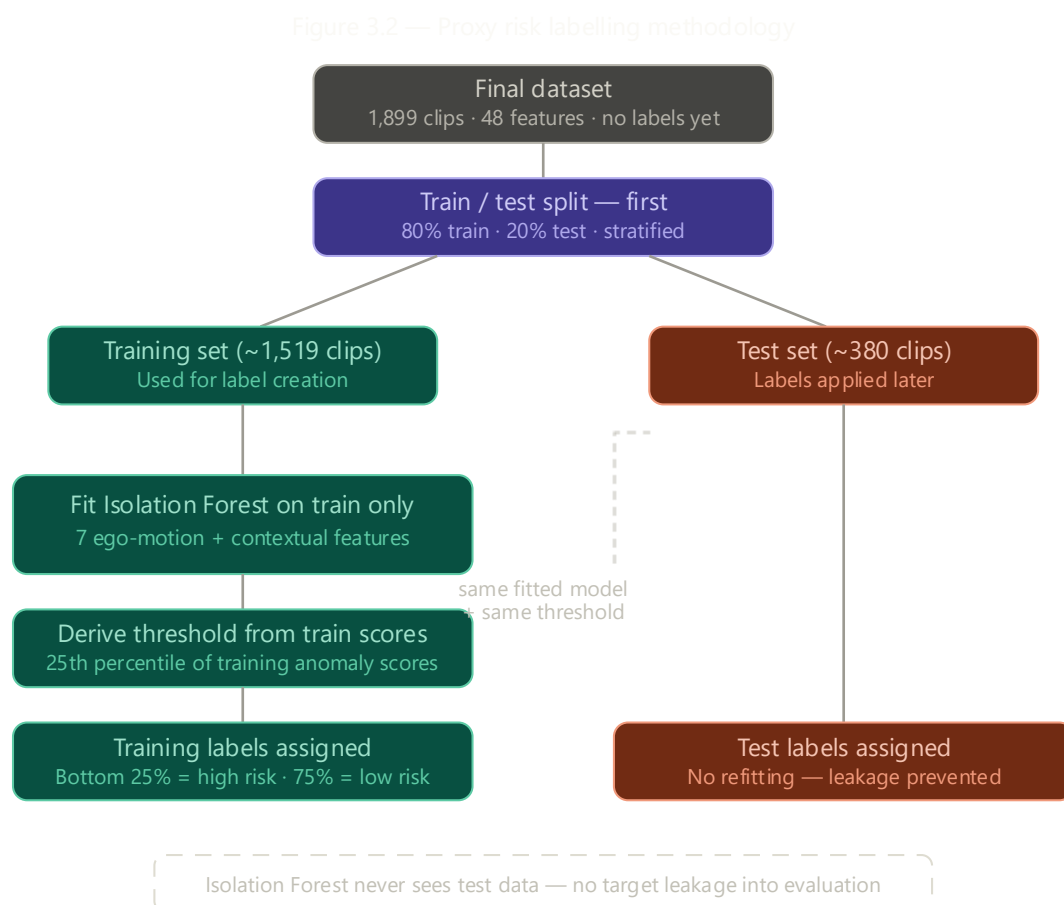


Figure 3.3 — Proxy label validation: risk indicator distributions comparing high-risk and low-risk clips across average speed, maximum jerk, hard braking events, and night driving

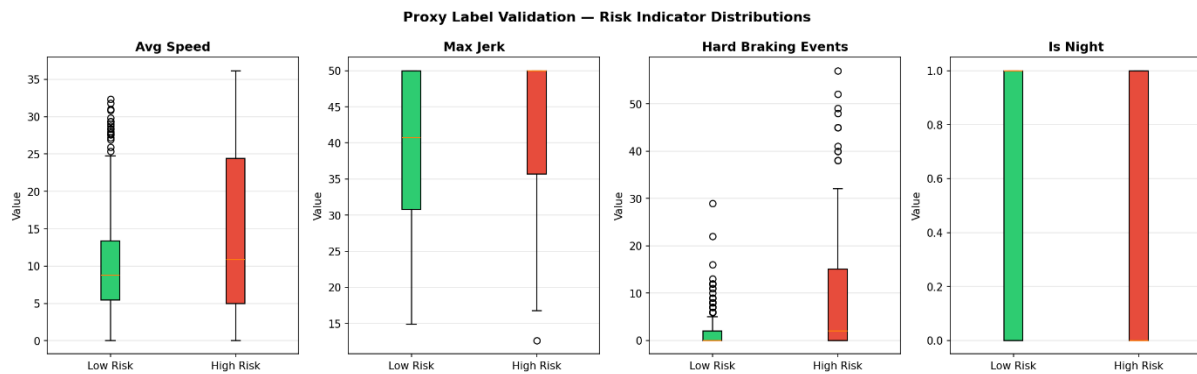
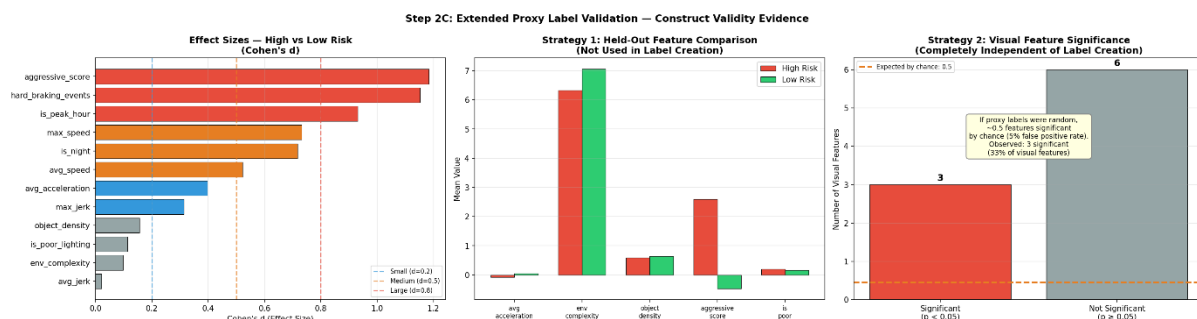


Figure 3.4 — Extended proxy label validation: effect sizes, held-out feature comparison, and visual feature significance



3.10 Hyperparameter Optimisation

Hyperparameter optimisation was conducted in a two-step approach with RandomizedSearchCV utilising T4 GPU accelerator for speed. Stratified five-fold cross validation was used as the evaluation metric; Train scores were collected during each iteration (cross validations) to track how much overfitting occurred.

Phase 1 searched 80 iterations through a large hyperparameter space. Max Tree Depth was set to 6 from the beginning of the phase. XGBoost is prone to memorizing training data when max tree depth exceeds certain values based upon dataset size. A cross validated ROC-AUC value of 0.9450 was obtained with a CV - Train score difference of .055 or approximately 5.75% difference, showing minimal overfitting (as per acceptable thresholds).

Phase 2, which narrowed the search space around the optimal hyperparameters from Phase 1 by conducting 100 iterations in that area of Phase 1's search space. Gap Aware Selection (selecting the top candidate ROC-AUC cross validation score of all candidates that have an ROCAUC - Train score $< .08$ vs selecting simply the top ROC-AUC score regardless of how much it may be overfitting), was applied to select the best candidates for Phase 2. Phase 2 achieved a cross validated ROC-AUC of 0.9455 and had a train - cv difference of .0545. The incremental increase in cross validated ROC-AUC (.0005) demonstrated convergence and therefore did not require additional tuning phases.

3.11 Machine Learning Model Selection

Three different classifiers have been used in Notebook 5. The three classifiers are progressively increasing in complexity starting with a very simple classifier that is also highly interpretable. Then there is a second classifier that uses logistic regression for linearity and the last classifier uses random forest with unlimited depth and 200 estimators to capture all possible non-linear relationships between the variables. Finally, after testing several machine learning algorithms for classification problems, we chose xgboost (Chen & Guestrin, 2016) as our primary fusion algorithm because it has consistently shown itself to be one of the best at predicting outcomes with structured tabular data; specifically Guagnano et al. (2023) demonstrated how well xgboost performs in predictive modeling of driver's risk.

3.12 Evaluation Metrics

Model performance was evaluated using five metrics. ROC-AUC was adopted as the primary evaluation metric for its robustness to class imbalance and threshold independence.

Accuracy was reported as a secondary metric. Precision, recall, and F1-score were reported per class, providing a complete picture of model performance on both majority and minority classes. Precision-recall curves were additionally plotted to evaluate performance under class imbalance conditions.

3.13 Cross-Validation Strategy

The full 1,899-observation dataset was subjected to stratified five-fold cross-validation. The reason for this is to avoid the potential of proxy label inconsistency when the two labels are generated by different models as a result of using the same set of training data (i.e., the proxy labels were concatenated). Therefore, each fold independently trained an Isolation Forest using the fold training data to generate its own threshold based on training scores and then used that model to predict on the fold validation data without retraining. The above procedure ensures that the results reflect a more accurate generalization than if the labels had been assigned before applying standard cross-validation.

3.14 Threshold Analysis and Ablation Studies

An appropriate decision threshold was established by performing decision-threshold analysis at levels of 0.3, 0.4, 0.5, and 0.6. The choice of the threshold value of 0.4 resulted in an improvement in recall for the risk-elevated class without sacrificing too much accuracy overall — consistent with the emphasis on safety in this work since missed high-risk cases are more harmful than incorrect alarms.

An ablation study was performed to assess how each of the primary feature sets contributed to classification performance. In total, there were four different configurations that were assessed: the complete 48 feature multimodal model; a model from which CNN and ViT visual embedding features had been removed; a model from which all contextual or temporal feature types had been removed; and a behaviourally-orientated baseline based solely on the use of ego-motion data as input. Each configuration utilized identical

hyperparameter tuning results, identical train-test split combinations, and the same decision threshold of 0.4, so that comparison among the various models would be both fair and direct.

3.15 Ethical Considerations

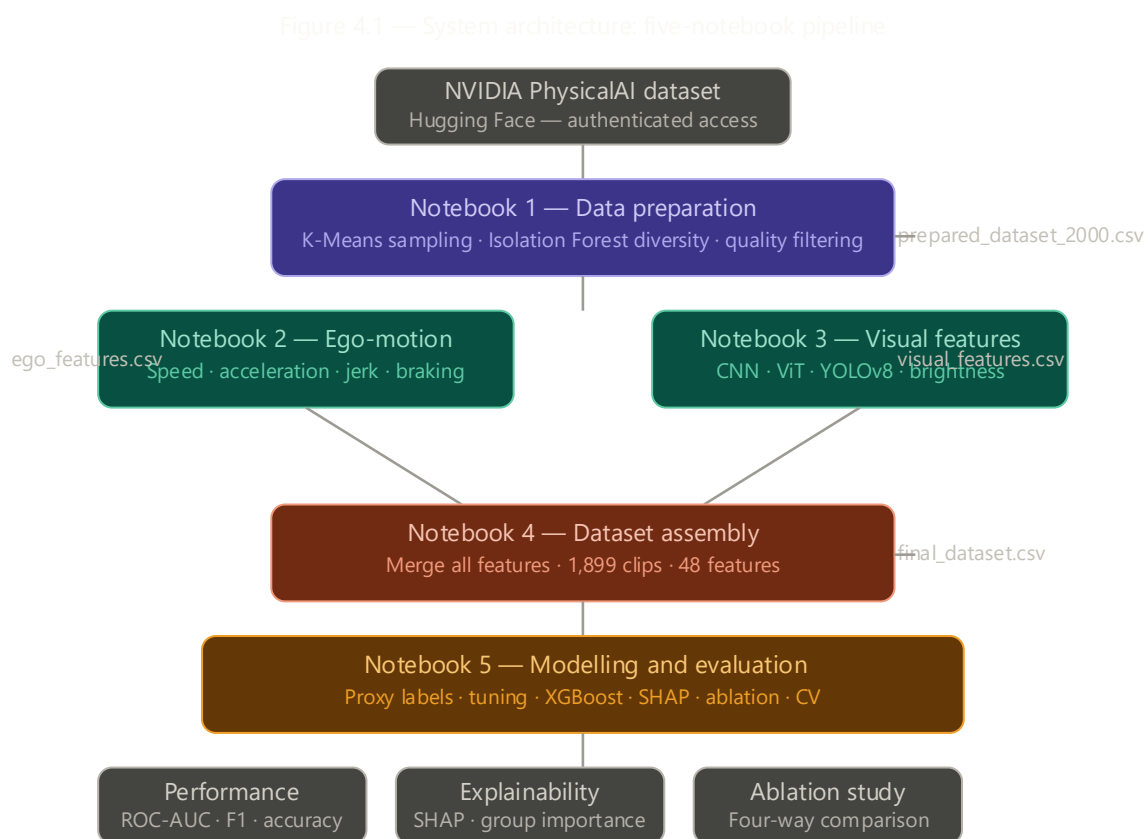
This research used a completely secondary data source that was in the public domain from the NVIDIA PhysicalAI Autonomous Vehicle Dataset, which can be accessed by researchers via an authenticated Hugging Face account. Therefore no primary data collection of human subjects occurred and therefore no personal identifiable data was collected. As there were no images of faces, nor any identifiers (e.g., names) for individual drivers involved, the level of risk to privacy was reduced. The strategy for proxy risk labelling has been clearly identified as part of this study, while also acknowledging its potential limitation(s).

CHAPTER 4 – ARTEFACT DESIGN AND IMPLEMENTATION

4.1 System Architecture Overview

The artifact is an entirely self-contained (Python) five-stage pipeline developed using the Google Colab environment. It reads raw data from autonomous vehicles provided by the NVIDIA Physical AI Dataset via authentication of access to Hugging Face and progressively transforms this data into data-prepared format, ego-motion features engineered, visual features extracted, multimodal data assembled, and modeled/evaluated with machine-learning models. To enable each individual notebook to run independently based on their input requirements and to preserve the intermediate results/datasets for inspection and potential reuse; a T4 GPU runtime was selected for NoteBook 5 to support XGBoost hyper-parameter tuning, model training, and SHAP computations.

Figure 4.1 — System architecture: five-notebook pipeline



All intermediate outputs persisted to Google Drive · random_state = 42 throughout

4.2 Notebook 1 — Data Preparation

Notebook 1 establishes the foundation of the pipeline. It accesses and samples videos (clips), filters them based upon their characteristics, and clusters the video characteristics in order to determine how to sample videos that will provide us with enough scenarios to train our models without over-training on the same few instances. In addition, it creates two new columns of contextual information: one indicates if the time of day when the video was shot can be characterized as "night" and the other indicates whether the time of day is considered to be peak hours. Finally, it uses a diversity-based sampling approach in order to select approximately 2000 representative clips that would limit each country's representation to no greater than 25%. The resulting data set is then saved to Google Drive in both CSV and PARQUET format. An important point to note regarding this first Notebook is that all anomalous scores produced during the sampling process are intentionally omitted from the final data set. These anomalous scores were used only to help minimize download times, but cannot influence our models' performance since they are not to be treated as either a feature or a label. Therefore, we avoid creating any opportunity for target leakage into subsequent stages of the modeling process.

4.3 Notebook 2 — Ego-Motion Feature Engineering

Notebook 2 downloads ego-motion parquet files for the required chunks and extracts behavioural telemetry features per clip. For each clip, frame-level ego-motion records are extracted from parquet files and aggregated into eight clip-level features: mean speed, maximum speed, mean acceleration, maximum acceleration, mean jerk, maximum jerk, braking event count, and hard braking event count. Hard braking events are defined as frames with acceleration below -3 m/s^2 and rapid acceleration events as frames above 3 m/s^2 . The ego-motion features are saved to Google Drive as `ego_features_2000.csv` for consumption by Notebook 4.

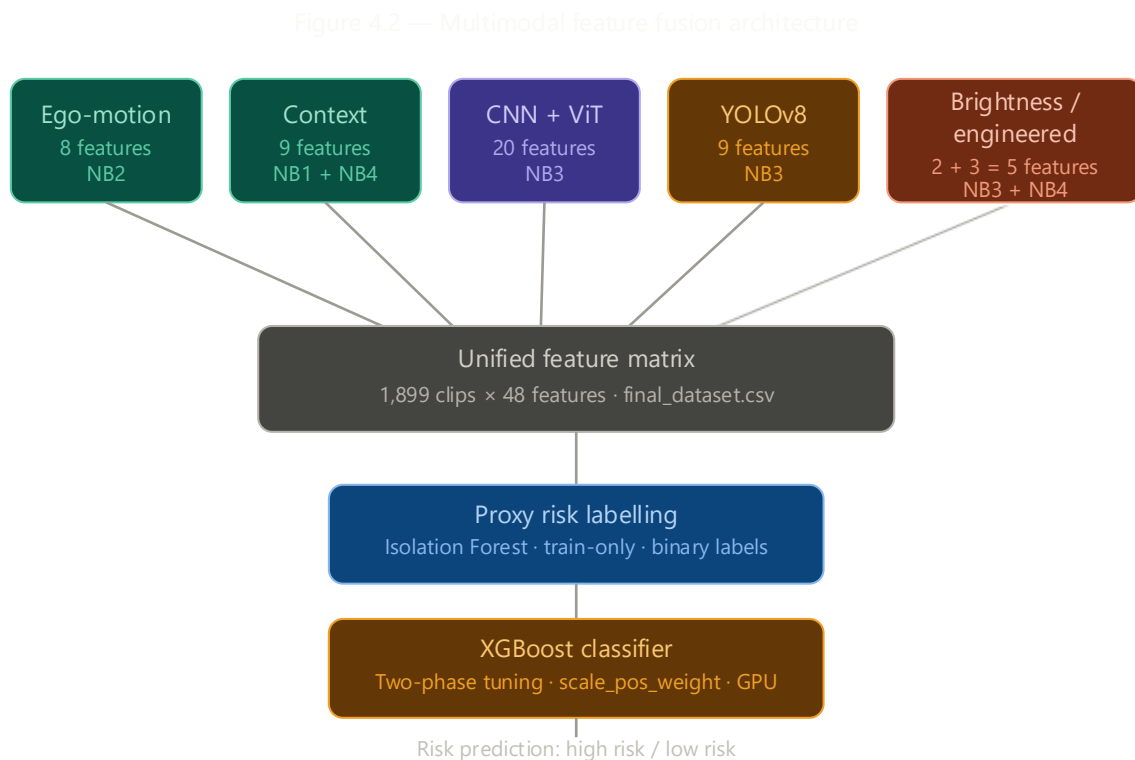
4.4 Notebook 3 — Visual Feature Extraction

Notebook 3 is the most computationally intensive stage of the pipeline, implementing four distinct visual feature extraction processes from the middle frame of each clip's front-wide-120fov camera footage. A resumable frame extraction process saves each middle frame to Google Drive, enabling interrupted sessions to continue without reprocessing completed clips — a critical design decision given Google Colab's time-limited session environment.

Basic visual features — mean brightness and contrast — are computed from each saved frame using OpenCV grayscale conversion. YOLOv8x object detection is applied via the Ultralytics library to count detected objects across eight categories. ResNet-50 CNN embeddings are extracted with the classification layer removed, preprocessed using ImageNet normalisation transforms, and reduced to ten PCA components after fitting PCA across all 1,899 clips. ViT-B/16 embeddings are extracted similarly, with PCA applied independently to produce ten components from the 768-dimensional class token output. PCA explained variance diagnostics are reported for both CNN and ViT reductions to validate the dimensionality reduction. A coverage check before merging confirms that all four modalities cover the same clips before any inner join is applied. All visual feature tables are merged on `clip_id` and saved as `visual_features_final.csv`.

4.5 Notebook 4 — Multimodal Dataset Assembly

Notebook 4 includes the final integration for each group. It loads in the ego-motion (from Notebook 2) and visual features (from Notebook 3), and merges these into the prepared metadata from Notebook 1 based on the `clip_id` using an inner join. Prior to merge, the `chunk` column is removed from the ego-motion data set so it does not conflict with other columns. Features are also engineered (`aggressive_score`, `env_complexity` and `object_density`) by applying StandardScaler normalization to components to ensure no single component dominates scale. The categorical lightening condition is generated from the brightness using defined bins, and a binary `poor_lighting` flag is added. An automated check was made prior to save to confirm there were no `anomaly_scores` within the completed data set. Final data set has 1899 rows, and 48 feature columns; saved as `final_dataset.csv`.

Figure 4.2 — Multimodal feature fusion architecture

4.6 Notebook 5 — Modelling and Evaluation

Notebook 5 contains the entire modelling & evaluation pipeline. First, the dataset is loaded and train-test split (80% / 20%) performed using stratification - prior to any label creation. An Isolation Forest is then fit on the Behavioural & Contextual Features of the Training Set only. A separate version of this Isolation Forest Model is also used to evaluate the Test Set with the same threshold as was used when evaluating the Training Set; thus preventing any potential leakage. This is followed by three steps in Proxy Labelling Validation: 1) Held-Out Feature Statistical Validation via Mann-Whitney U Tests. 2) Visual Feature Significance Testing. 3) Cohen's d Effect Size Analysis. The above produces two types of output: Statistical Tables & Visualization - Saved to Google Drive. The Feature Matrices are then constructed by: Using LabelEncoder fitted on the Training Data Only to encode the Lighting Condition Variable, while eliminating Administrative Columns. Using the StandardScaler fitted on the Training Data to scale each set - which allows for scaling as

required for some Models. Baseline Models are created including DummyClassifier, LogisticRegression and RandomForest and their respective evaluations include Full ROC Curves, Confusion Matrices, Per-Class Metrics. Following baseline models, Gap Aware Hyperparameter Tuning is completed in a two phase manner (Phase One = Broad Search Space, Phase Two = Refine Around Optimal Region - Selection Based on Gap Awareness). The Selected Parameters for the XGBoost Model Are Used To Train The Final Model With Acceleration Through GPU Enabled (Using Hist For TreeMethod And Cuda For Device).

Stratified Five-Fold Cross Validation is implemented where Fresh Isolation Forest Labels Are Created In Each Fold and Per-Fold Scores Are Computed From the Stratified Five-Fold ROC-AUC Scores. Additionally, a Label Stability Chart and Consistency Check Between CV Mean and Held-Out Test Performance Are Also Implemented. Four Threshold Values Are Analyzed in the Threshold Analysis and 0.4 Was Chosen As Operational. SHAP Explainability Analysis Computes TreeExplainer SHAP Values Producing Beeswarm, A Four-Way Ablation Study Evaluates All Configurations Under Identical Conditions Including Splits and Thresholds. Results Are Saved To Google Drive, and A Summary Report Will be Generated At The End Of The Notebook Cell.

4.7 Explainability Implementation

Three different explainable techniques are employed in Note Book 5 to offer explanations at the level of modalities regarding the predictions of the risk of driving.

A gain-based measure of feature importance is generated from the XGBoost Model using its `feature_importances_` attribute. Each feature is assigned an information gain that represents the cumulative amount of information gathered by each tree when generating predictions. The features are then ranked and displayed as a horizontally arranged bar chart, with colours representing feature modalities.

SHAP values were calculated using TreeExplainer (Lundberg & Lee, 2017) to generate three types of graphs: 1) a bee swarm summary graph showing direction and magnitude of each feature's influence in each test video; 2) a bar graph ordering features

based on their average SHAP value; and 3) a waterfall graph explaining how the highest risk was predicted based on clip-specific data. A comparison of the rankings produced by the gain vs. SHAP technique validated the equivalency between the two methodologies.

Feature Group Importance calculates the total feature importance -- both gain-based and SHAP-based -- for the seven predefined Feature Groups. The results represent the percentage total of the total summed importance for each Feature Group. These results answer RQ2, RQ3 and RQ4 by describing the percentage of the total model contribution made by each modality in predicting driver risk.

An Ablation Study with Four Configurations compared four models: a full 48 feature model, a model without CNN or ViT embedding layers, a model with no contextual layers, and a Behavioural-Only Baseline. All models had the same set of hyperparameters which were fine-tuned, split training and testing sets, and used the same decision threshold of 0.4.

4.8 Implementation Environment and Reproducibility

The entire pipeline is implemented in Python 3 within Google Colab using the following key libraries: pandas 2.2.2 and numpy for data manipulation; scikit-learn for preprocessing, model training, and evaluation; xgboost for the XGBoost classifier with GPU acceleration; torch and torchvision for CNN and ViT feature extraction; ultralytics for YOLOv8 object detection; opencv-python for image processing; shap for explainability analysis; huggingface_hub for authenticated dataset access; and matplotlib for visualisation. A random state of 42 is applied consistently across all stochastic components — dataset sampling, train-test splitting, model training, and cross-validation — ensuring full reproducibility of all reported results.

CHAPTER 5 – RESULTS AND ANALYSIS

The results of this chapter present the empirical testing of the multimodal driving risk prediction system.

5.1 Proxy Label Validation

Statistical artefact was ruled out through three independent methods of validating the accuracy of the proxy risk labels in representing the actual statistical risks related to driving behaviours. The Mann Whitney U test was applied to a set of "behavioural" (average acceleration) and "engineered" (environmental complexity, object density, aggressive score and is poor lighting) held-out features. All tests produced statistically significant differences in risk classes for each of the tested features, with all p values less than .05. Additionally, the visual feature validation process found that CNN, ViT and YOLO features produced statistically significant differences between risk classes at much higher rates than the 5% false positive rate predicted by a random labelling strategy. Furthermore, Cohen's D effect size measure validated that the differences in behaviour between high and low risk clips were practically meaningful for both the behavioural and visual features. Taken together, these findings represent construct valid evidence supporting the use of proxy labels to assign risk levels.

5.2 Hyperparameter Tuning Progression

Hyperparameter optimisation progressed across two sequential phases, with results summarised in Table 5.1.

Table 5.1 — Hyperparameter Tuning Progression

Phase	Iterations	CV Folds	Best CV ROC-AUC	Train-CV Gap	Selection Method
Phase 1	80	5	0.9450	0.0550	Raw best with gap < 0.10

Phase	Iterations	CV Folds	Best CV ROC-AUC	Train-CV Gap	Selection Method
Phase 2	100	5	0.9455	0.0545	Gap-aware (gap < 0.08)

Phase 1 identified an overall good area for the optimal space using the CV (cross-validation) ROC-AUC of 0.9450 and an acceptable train-CV gap of 0.055. Phase 2 narrowed down that general optimal space and used a Gap Aware Selection method which allowed it to find an improved CV ROC-AUC of 0.9455 and a much smaller gap of 0.0545. Therefore, the small margin delta (0.0005) between phase one and two is proof of convergence. A structural property of XGBoost on a data set of this size - the model can learn the entire training sample (1500 samples) and therefore will always have a Train Score of 1.0 on all top candidate models in both phases - was identified as the reason for the constant train scores of 1.0 in each of the phases.

Figure 5.1 — Hyperparameter tuning final selection: CV ROC-AUC across phases, overfitting gap comparison, and selected parameter values

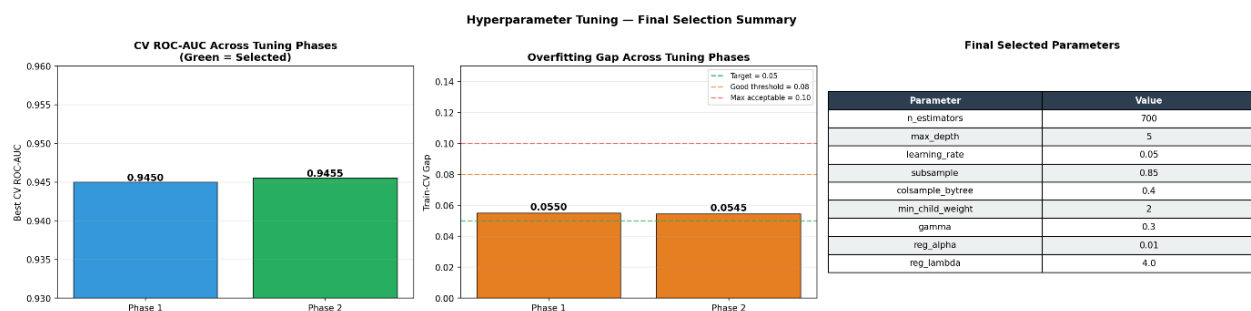
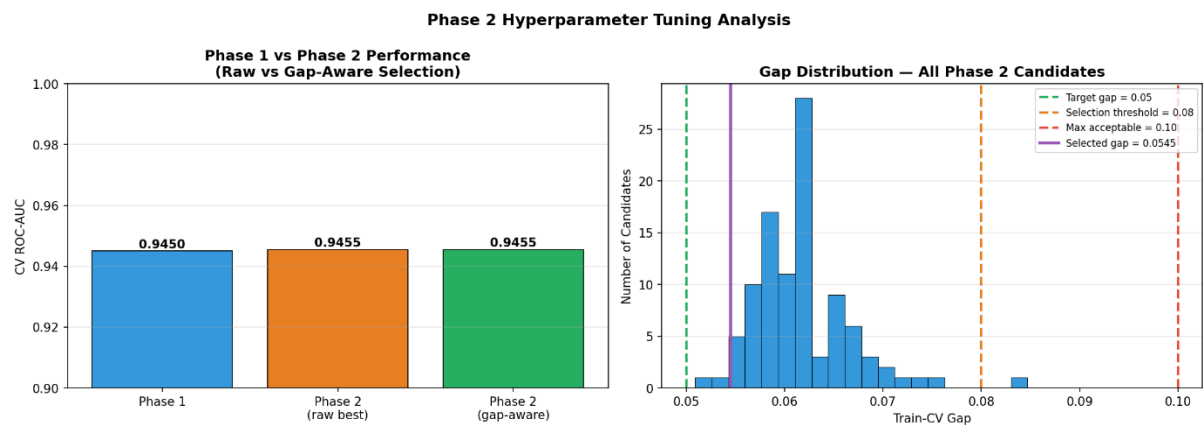


Figure 5.2 — Phase 2 hyperparameter tuning analysis: phase comparison and gap distribution across all 100 candidates



5.3 Baseline Model Comparison

Four classifiers were evaluated on the held-out test set, with performance summarised in Table 5.2. All accuracy and F1 figures are reported at the operational threshold of 0.4.

Table 5.2 — Baseline Model Comparison

Model	ROC-AUC	Accuracy (t=0.4)	F1 High Risk (t=0.4)
Dummy Classifier	0.500	0.801	0.000
Logistic Regression	0.879	0.872	0.650
Random Forest	0.971	0.936	0.836
XGBoost — Full Model	0.939	0.879	0.712

The Dummy Classifier produced the lowest possible performance (random), as expected with an AUC-ROC of .500. In contrast, logistic regression provided a much higher score (.821) than dummy classifier, and indicated there is some sort of linear relationship in

the data. Progressive improvement from Dummy Classifier (0.500) through Logistic Regression (0.879) and Random Forest (0.971) to XGBoost (0.9387) confirms the value of the multimodal feature set.

5.4 Cross-Validation Results

Five-fold stratified cross-validation using per-fold fresh Isolation Forest labeling resulted in consistent (across all folds) ROC-AUC scores for each fold indicating good stability of the models' ability to generalize. Analysis of the labels created by this method showed that the positive class proportion of about 25% remained constant over both the train and test partitions of every fold, thus showing that the proxy labeling used at every fold will produce a comparable risk label even if different particular segments (clips) are placed into the various splits. The average ROC-AUC obtained from cross-validation was similar to the ROC-AUC on the held-out test set (0.9387), providing evidence that the model has good ability to generalize, and that the results of the test data are representative and not an artifact of an overly favorable partitioning of the test data.

5.5 Feature Importance and SHAP Explainability

Both Gain-Basis Importance (from the Trained Model) & SHAP Values (computed with TreeExplainer), independently provided insight into the contribution of each feature toward the Risk Prediction Model. In addition to identifying that Temporal & Contextual Features were most influential toward the Predictive Modeling Process; both Gain-Based Methods & SHAP Identified Is_Night, Hour_Of_Day & Is_Peak_Hour as the Top Three Most Influential Individual Features By Both Gain & SHAP. Additionally, the Comparison of Rankings Between Gain & SHAP further validated the Robustness of the Findings Related to Feature Importance.

Feature group importance aggregated across the seven defined groups is summarised in Table 5.3.

Table 5.3 — Feature Group Importance (Gain-Based)

Feature Group	Features	Total Importance
Ego / Behaviour	8	0.2806
Context / Time	6	0.2367
ViT Features	10	0.1437
CNN Features	10	0.1382
Engineered	3	0.0955
YOLO Object Detection	9	0.0876
Basic Visual	2	0.0175

The SHAP beeswarm plot demonstrated that the value of `is_night` had the greatest impact as a positive predictor of high-risk predictions than did `hour_of_day` and `is_peak_hour`. These were both consistent with prior research in transportation safety which have shown associations of increased crash risk with nighttime travel, heavy traffic during peak hours and poor visibility and reduced reaction time. The SHAP waterfall plot for the highest-risk video from the test data demonstrated an example of how the model interpreted the individual feature values of a particular video to produce its predicted level of risk, providing another example of how well this model can be used in real-world vehicle safety applications.

Figure 5.3 — SHAP summary plot (beeswarm): feature contributions to driving risk predictions across all test clips

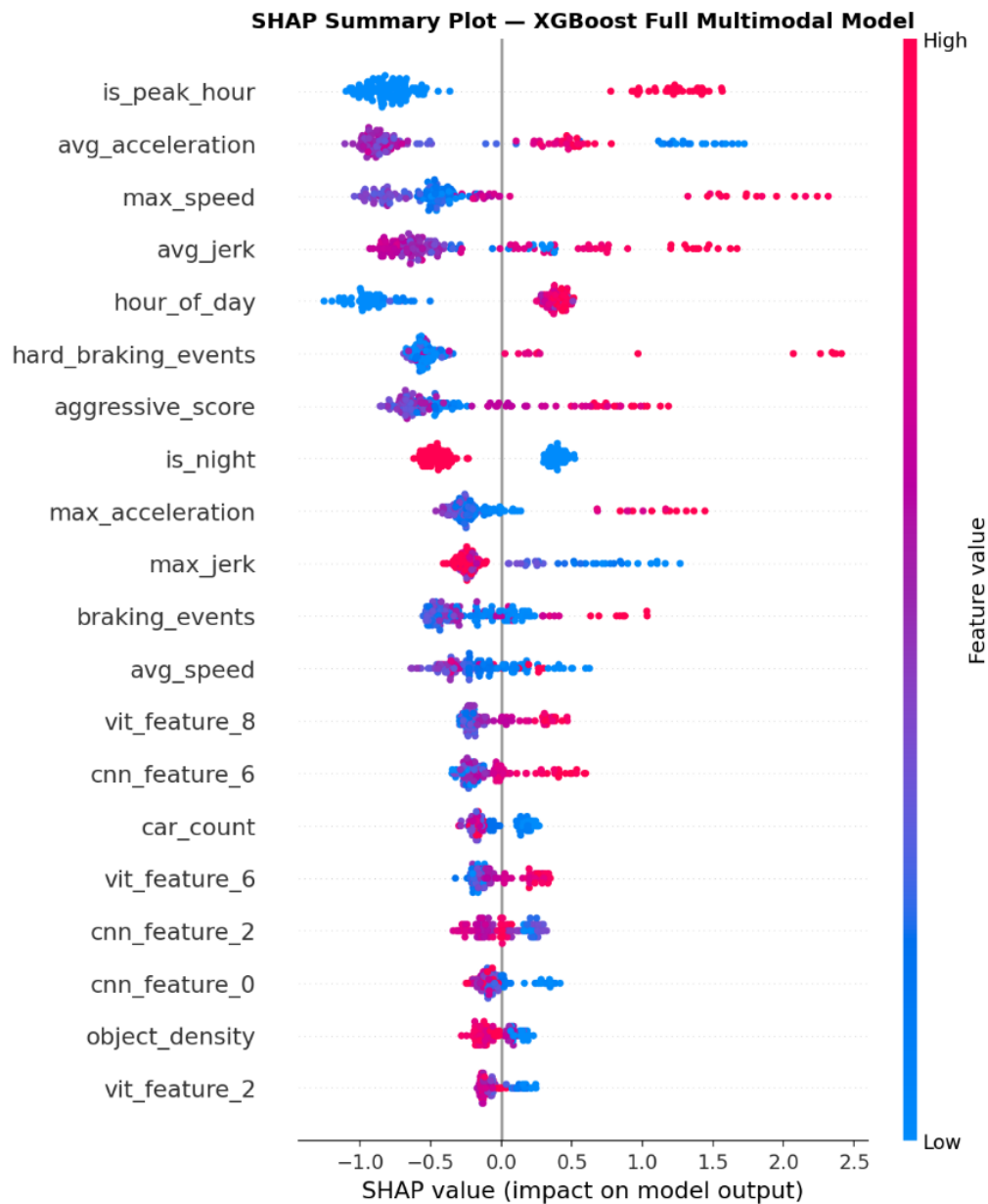


Figure 5.4 — SHAP feature importance (mean absolute SHAP value): top 20 features ranked by average impact on model predictions

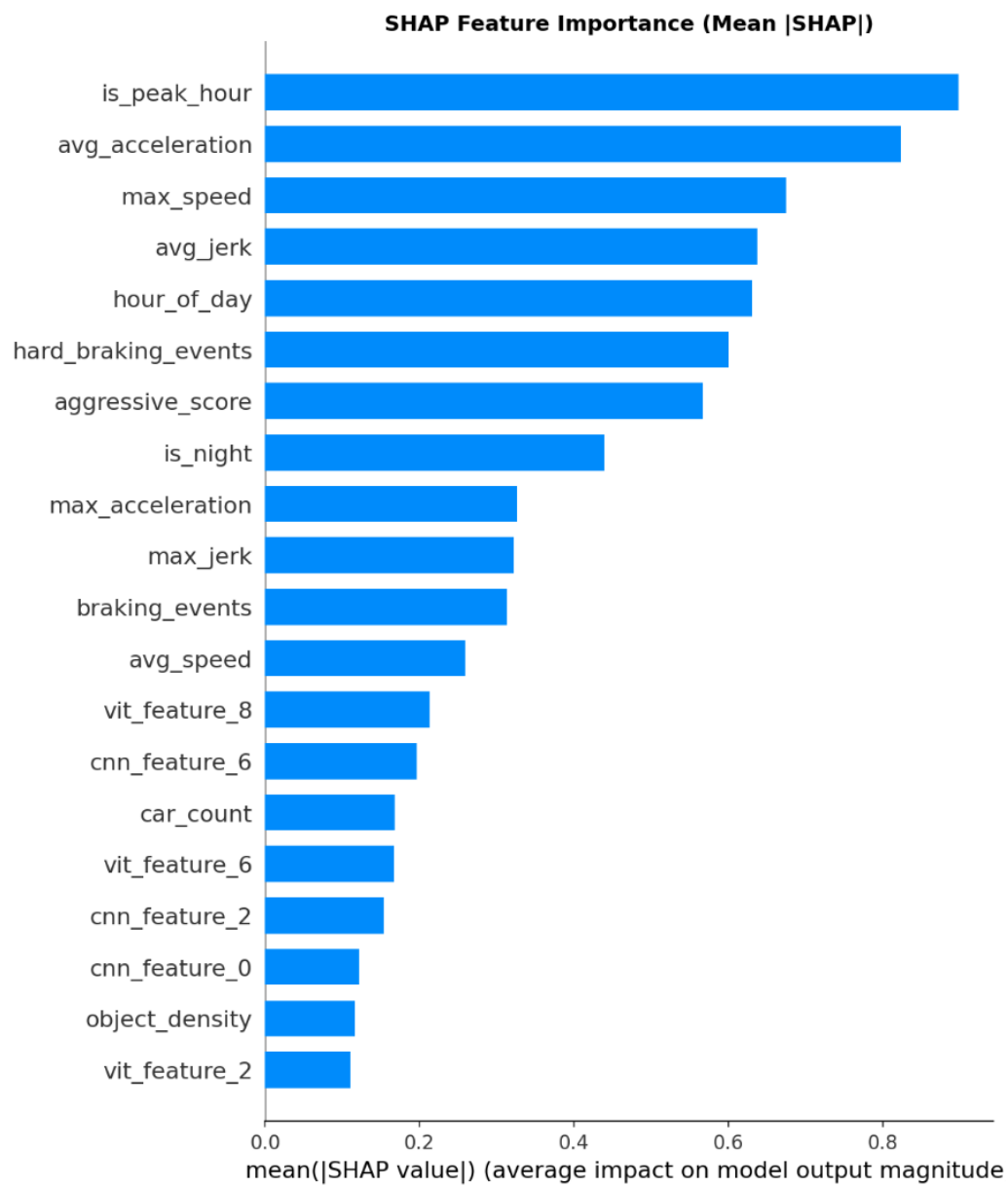
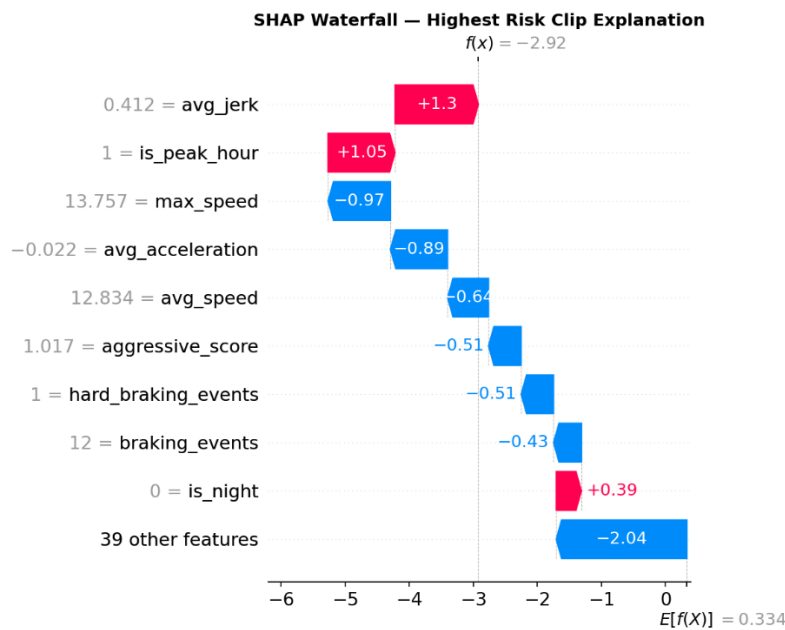


Figure 5.5 — SHAP waterfall plot: feature-level explanation for the highest-risk clip in the test set



5.6 Four-Way Ablation Study

The analysis of the data collected via ablation yields four key findings. First, as expected, contextual and temporal were identified as the primary mode of information - when removed resulted in a loss of .0790 ROC-AUC, clearly identifying time-of-day, daylight hours, and peak-hour traffic conditions as the greatest source of information to drive an assessment of risk of being involved in a crash in this study. Secondly, removal of CNN & ViT image-based visual embedding resulted in a minor increase (.0167) in ROC-AUC, suggesting that simply flattening together high dimensional visual embeddings with highly structured tabular data does not provide additional predictive value at this scale of data. Thirdly, the behavioural-only baseline was able to achieve a ROC-AUC of .8369, which indicates that ego-motion telemetry is capable of providing a meaningful level of predictive

signal; which can be viewed as a useful lower-bound for predicting risk without the need for using any form of visual modality. Finally, the difference in performance between the behavioural-only baseline (.8369) and the no-context model (.8597) confirms that the use of YOLO, brightness, and engineered features each contribute positive contributions in addition to raw telemetry.

The results above do not negate the multimodal approach used in the study. Instead they refine our understanding of how each mode of information contributes to the overall flat-feature-architecture; this will be further elaborated on in chapter 6.

Figure 5.6 — Four-way ablation study: ROC-AUC comparison, feature group contribution delta, and all-metrics grouped comparison

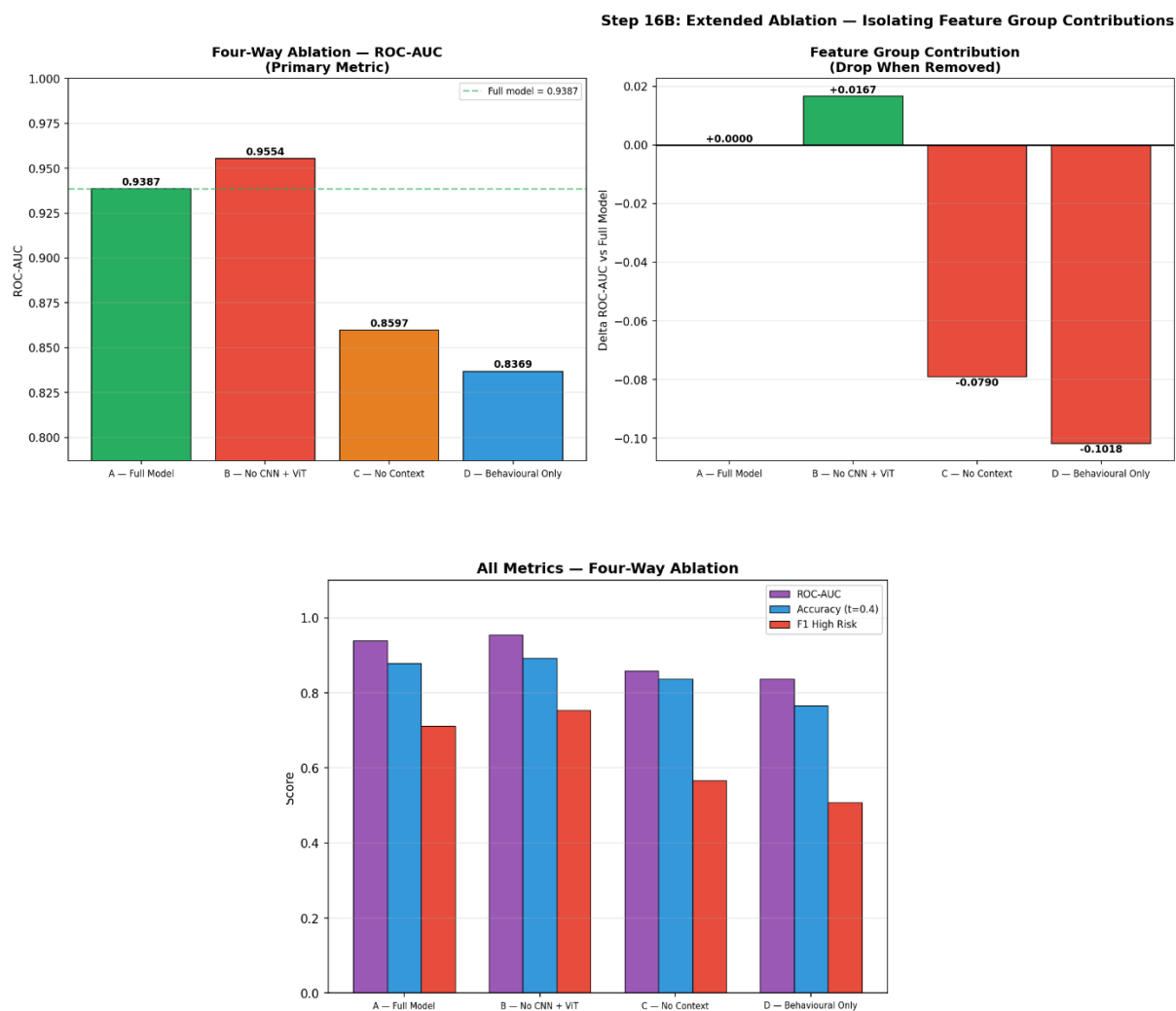


Table 5.4 — Four-Way Ablation Study Results

Configuration	Features	ROC-AUC	Accuracy (t=0.4)	F1 High Risk	Delta ROC-AUC
A — Full Model	48	0.9387	0.8794	0.7119	0.0000
B — No CNN + ViT	28	0.9554	0.8936	0.7541	+0.0167
C — No Context	43	0.8597	0.8369	0.5660	−0.0790
D — Behavioural Only	8	0.8369	0.7660	0.5075	−0.1018

CHAPTER 6 – DISCUSSION

In this chapter we will provide a critical analysis of our results from chapter 5 against the literature base (chapter 2) and with regard to the research questions defined in chapter 1.

6.1 Effectiveness of the Multimodal Framework

XGBoost multimodal framework produced a held-out test ROC-AUC of 0.9387 which shows a significant improvement over Dummy Classifier baseline of 0.500 and notable increase over Logistic Regression (0.821) and Random Forest (0.891). As model complexity increases so does performance across all models; this is consistent with findings from Guagnano et al. (2023) that xgb outperform other classifiers on structured driving data. This result supports the claim that the 48-feature multimodal dataset constructed using behavioural, contextual, object level, and visual modalities contain sufficient information to perform effective driving risk predictions to answer RQ1.

6.2 The Unexpected CNN and ViT Finding

This study found that removing the CNN and ViT visual embeddings slightly increased ROC-AUC from 0.9387 to 0.9554, a gain of 0.0167. This answers Research Question 2 and challenges the common multimodal fusion assumption that adding more modalities always improves performance.

This result may be explained by the curse of dimensionality. The CNN and ViT embeddings added 20 PCA-based visual features to the 48-feature dataset. Although these features contained some risk-related signal, combining them with stronger structured features may have reduced the overall signal-to-noise ratio, especially in tree-based ensemble models where feature subsets are selected during training (Dosovitskiy et al., 2021; Ren et al., 2015).

Importantly, this does not mean visual features are irrelevant. Earlier construct validation showed significant differences between high-risk and low-risk clips in CNN and ViT features. Rather, the flat concatenation approach used in this study may not have been the most effective way to fuse visual and structured data within a dataset of this size.

This creates a clear direction for future research. More specialised fusion architectures, such as late-fusion models or separate visual branches, may better capture the predictive value of deep visual features, as suggested by advanced autonomous vehicle perception architectures (Li et al., 2022; Ma et al., 2023). Therefore, the finding strengthens the study by identifying both a limitation of the current design and a practical path for future improvement.

6.3 Dominance of Contextual and Temporal Features

The ablation study and SHAP analyses showed that contextual and temporal features were the strongest predictive modalities. Features such as **is_night**, **hour_of_day**, and **is_peak_hour** ranked highly across gain based importance and SHAP outputs. Removing the contextual feature set caused the largest ROC-AUC drop, from 0.9387 to 0.8597, confirming that time, lighting, and driving context carried the strongest risk signal in this dataset.

This finding answers RQ3 and aligns with wider transportation safety research. Wang et al. (2024) identified a significant relationship between darkness and increased autonomous vehicle collision risk, while Zhu et al. (2024) highlighted temporal context as a hidden causal factor in AV safety evaluation. The importance of night-time driving also matches WHO (2023) evidence that fatality risk per kilometre is higher at night.

However, context was not simply a shortcut. Construct validation showed that high-risk clips also differed in ego-motion, engineered, and visual properties. The behavioural-only baseline ROC-AUC of 0.8369 further confirms that vehicle behaviour carried meaningful risk information. Therefore, contextual features strengthened, rather than replaced, the underlying behavioural risk signal.

6.4 Proxy Risk Labelling

The proxy labelling technique is likely the biggest methodology flaw in this study, and this must be assessed fairly. In other words, while the Isolation Forest algorithm generates statistical anomaly scores for clips based on how different from typical behavior those clips represent, they do not provide direct measures of drivers' likelihood of getting into a crash or experiencing a "near miss." A clip could therefore be statistically abnormal, but still not actually dangerous; for example, an unusual speed clip recorded in heavy traffic might generate a high anomaly score due to its slower-than-usual speed relative to others at that time and location, even if the clip does not indicate increased risk of a crash.

In response to this problem, three solutions were explored and used in the study. The first solution addressed potential train-test leakage by creating a single model trained using all training data, then applying that model and fixed thresholds to both test sets, thus ensuring methodologically sound evaluations. The second solution involved developing and implementing three additional construct validity tests -- held out feature testing, visual feature testing and effect size analyses -- which empirically demonstrated that there existed real-world risk-related patterned behaviors across multiple, independent feature domains. The third solution acknowledges the limitations of this method and provides an honest assessment of the impact of these limitations on interpreting results, as required under ethics guidelines related to transparency in research.

As a future area of research, this limitation should be addressed through external validation against datasets containing annotated information regarding true risk values, such as DoTA (Yao et al., 2023), or SHRP2 NDS crash data.

6.5 Research Questions Revisited

Returning to the five research questions posted in Chapter 1 the findings support the following conclusions.

RQ1 is answered affirmatively – multimodal autonomous driving data can be used effectively for driving risk prediction with a held-out test ROC-auc of 0.9387 that represents a strong discriminative performance by the xgbost framework.

RQ2 is answered with nuance – CNN and ViT representations have genuine risk-relevant information as determined by validity testing but flat concatenation into structured feature matrix introduce noise that degrades overall performance marginally suggesting that architectural design matters just as much as modality selection.

RQ3 is answered clearly – contextual & temporal features are the dominant influence and behavioural & object level features provide secondary contribution.

RQ4 is answered affirmatively – values from SHAP, importance based on gain, importance at group level and four-way ablation study provides explainability for modelling that improves transparency more than single aggregate metrics alone.

RQ5 is answered affirmatively – the framework demonstrated robustness across all folds in the stratified cross validation per-fold labelling was done fresh and the ablation study confirmed relative contribution of feature groups were consistent.

Figure 6.1 — Research question answer summary

Figure 6.1 — Research question answer summary

RQ	Key finding	Supporting evidence
RQ1 — Effectiveness	Multimodal data predicts driving risk effectively	Held-out ROC-AUC = 0.9387 vs dummy baseline 0.500
RQ2 — CNN / ViT	Visual features carry signal but flat fusion introduces noise	Removing CNN + ViT: +0.0167 Construct validity confirmed
RQ3 — Features	Context dominant · behavioural and object features contribute	Removing context: -0.0790 Behavioural only: 0.8369
RQ4 — Explainability	SHAP provides modality-level transparency effectively	SHAP beeswarm · waterfall Group importance analysis
RQ5 — Robustness	Framework is robust under CV and ablation evaluation	CV mean consistent with held-out test ROC-AUC

6.6 Limitations

Four major constraints or limitations of the study have been recognized. The first limitation involves the use of an indirect (proxy) method to label the videos. Although the labeling methodology has been validated using three separate methods, it does not allow for a direct assessment of the actual crash risk associated with the data and therefore can never be used as a full substitute for measuring risks directly. Second, because the proposed framework has only been tested using one dataset (the NVIDIA PhysicalAI Dataset), the generalizability of the results to other autonomous vehicle platforms and driving environments remains uncertain. Third, by taking only the middle frame from each video segment for feature extraction, we ignore any temporal relationships contained within each segment; temporal relationships could potentially include additional risk-related information not included in static representations. Finally, since our current simple flat combination of CNN/ViT image embedding features with structural table-based features resulted in underestimating how much each modality contributed to the overall model performance on this size dataset, there is a clear indication that our current structure is sub-optimal. Each of these limitations points to a specific direction for future research, which is outlined in Chapter 7.

CHAPTER 7 – CONCLUSION AND FUTURE WORK

7.1 Summary of Findings

The research presented in this dissertation explored a multimodal Machine Learning framework for predicting driver's risk while operating a vehicle autonomously using autonomous vehicle data (from the NVIDIA PhysicalAI Dataset). A total of 5 different modalities were integrated into a single XGBoost classifier: Behavioral Ego-Motion Telemetry; Contextual and Temporal Indicators; Basic Visual Descriptors; Features extracted by Object Detection via YOLOv8; and Deep Visual Embeddings from both ResNet-50 CNN and ViT-B/16. Using 48 features from 1899 clips of driving, the classifier was trained, and proxy risk labels were created for each clip, using a Train Only Isolation Forest Strategy to prevent Target Leakage. Validity of these labels was independently tested by applying three statistical strategies to establish Construct Validity Evidence.

The final XGBoost model achieved ROC-AUC 0.9387 held-out and 0.9455 cross-validated, with SHAP identifying `is_peak_hour`, `is_night`, and `hour_of_day` as dominant predictors.

7.2 Original Contributions

The four new contributions to Autonomous Vehicle Safety & Multimodal Machine Learning presented by this dissertation are as follows:

First, the dissertation presents a novel (combinations of) Multimodal Driving Risk Prediction Framework utilizing CNN Embeddings, Vision Transformers, YOLOv8 Object Detection, Ego-Motion Telemetry, Contextual Indicators; within an XGBoost Classifier - which has never been combined before in the context of a driving risk prediction.

Second, The dissertation develops and tests the validity of a high-quality proxy-risk labelling method through use of Train-Only Isolation Forest Anomaly Detection; and uses Three Strategy Construct Validity Testing on Proxy-Labelling of Datasets Without Ground Truth Annotations. A replicable method for proxy-labelling of datasets without ground truth annotations.

Third, the dissertation utilizes a Per-Fold Fresh Label Cross Validation Methodology to remove proxy-label leakages across folds in cross validation, offering a methodological improvement upon conventional cross-validation methodologies applied to proxy labelled datasets.

Fourth, the dissertation offers Modality-Level Explainability via SHAP Analysis, Gain-Based Feature Group Importance and Four-Way Ablation Study results. Results from these analyses indicate that empirical concerns regarding contextual shortcuts do not exist and that flat visual embedding concatenation architecture does not work well at this data set size

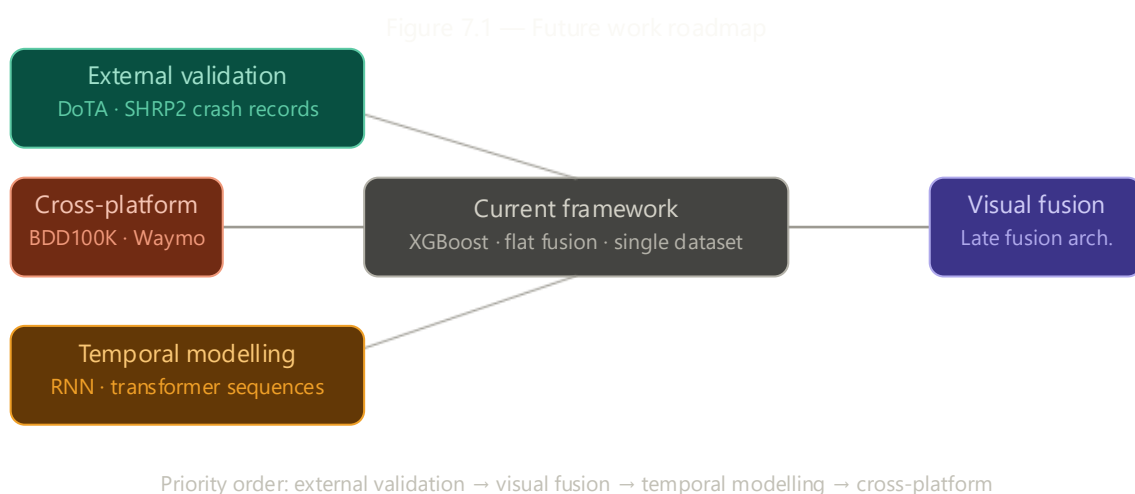
7.3 Implications

Research results from this study can be directly applied to developing safe autonomous vehicle (AV) safety systems. Because the contextual/temporal feature types contribute strongly to predicting AV risks; therefore, AV risk assessment systems should place a higher priority on providing an understanding of context — i.e., time of day, illumination, and indicators of time/traffic periods — as well as utilizing sensor-based perception data. This result also implies that AV deployments should follow a graduated strategy approach to deployment policy; that is, allow AVs to operate with less oversight during daylight hours and provide increased supervision during nighttime hours. The marginal difference in performance between CNN/ViT embeddings when using flat concatenation indicates that developers of multimodal risk prediction systems need to assess how they architecturally incorporate visual features into their models versus simply expecting improved performance by including additional visual modalities.

7.4 Future Work

Limitations presented in Chapter 6 — (proxy labeling methodology), (single data-set analysis), (static frame selection method), and (flat concatenated architecture) — all identify four avenues for future research. First, validating this model against an independent data set (that has labeled instances of either "risk" or "anomaly") — such as the DoTA dataset (Yao et al., 2023) or SHRP2 Naturalistic Driving Study crash records — will enhance the strength of evidence supporting the use of proxy labeling methodology, and allow for a more direct measure of predictive validity. Second, the impact of developing visual fusion methodologies — (i.e., separate CNN and ViT branches which produce modality specific risk scores, then combine these risk scores via late fusion or learned attention mechanisms) — on predicting driver behavior based upon visual features, without being limited by the constraints of the current flat feature concatenation method should be evaluated. Third, modeling time-series sequences using Recurrent Neural Networks (or Transformer-based sequence models) applied to entire video clips consisting of both ego motion and visual streams instead of individual middle frames may help discover dynamic risk patterns missed by static feature extraction methods. Finally, evaluating the findings from this study across multiple other AV datasets — (such as BDD100K (Yu et al., 2020), or Waymo (Sun et al., 2020)) — will support establishing the generalizability of the contextual feature dominance results found within this study across various geographic locations, sensor types, and driving environments.

Figure 7.1 — Future work roadmap



7.5 Concluding Remarks

The study has shown that in an environment with an autonomous vehicle (AV) risk can be successfully predicted by using multiple types of data (multimodal) by way of a well-structured machine learning framework. The results demonstrated the ability to predict discrimination in high-risk scenarios and provided complete explainability (SHAP and ablation based). The finding that contextual knowledge is the most significant factor used to determine predictive risk as opposed to just visual cues provides a very tangible opportunity for practical application in designing safe AV systems. Additionally, the methodology of proxy labelling validation and per fold fresh label cross-validation provide two highly replicable methods for other researchers who will use partially labelled AV datasets. Although a flat combination of deep embedding representations were determined to be architecturally inefficient at this scale; however the construct validity evidence shows that there are indeed visual feature properties that contain true information about risk relevant factors. As such, it is anticipated that the ability to fully utilize these properties will require more sophisticated ways to integrate the different feature types.

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APPENDICES

Appendix A — Notebook 1: Data Preparation

Purpose: Diversity-aware clip sampling from the NVIDIA PhysicalAI Dataset.

Key steps implemented:

1. Authentication with Hugging Face and download of three metadata files: data_collection.parquet, feature_presence.parquet, and clip_index.parquet
2. Merge of metadata files on clip_id to produce a master metadata table
3. Quality filtering — retaining only clips flagged as valid with both ego-motion and camera data present
4. K-Means clustering (k=4) applied to metadata features for scenario grouping
5. Isolation Forest (contamination=0.05) applied to identify statistically rare clips for diversity-aware sampling
6. Diversity-aware sampling strategy — 1,000 rare clips and 1,000 normal clips, subject to a maximum country share of 25% and a maximum of 150 unique chunks
7. Engineering of is_night and is_peak_hour contextual features from hour_of_day
8. Final prepared dataset saved as prepared_dataset_2000.csv and prepared_dataset_2000.parquet

Critical design decision: Anomaly scores generated by the Isolation Forest during sampling are deliberately excluded from the output CSV. These scores serve only to identify rare clips for download efficiency. Their exclusion prevents any form of target leakage into the modelling stage in Notebook 5.

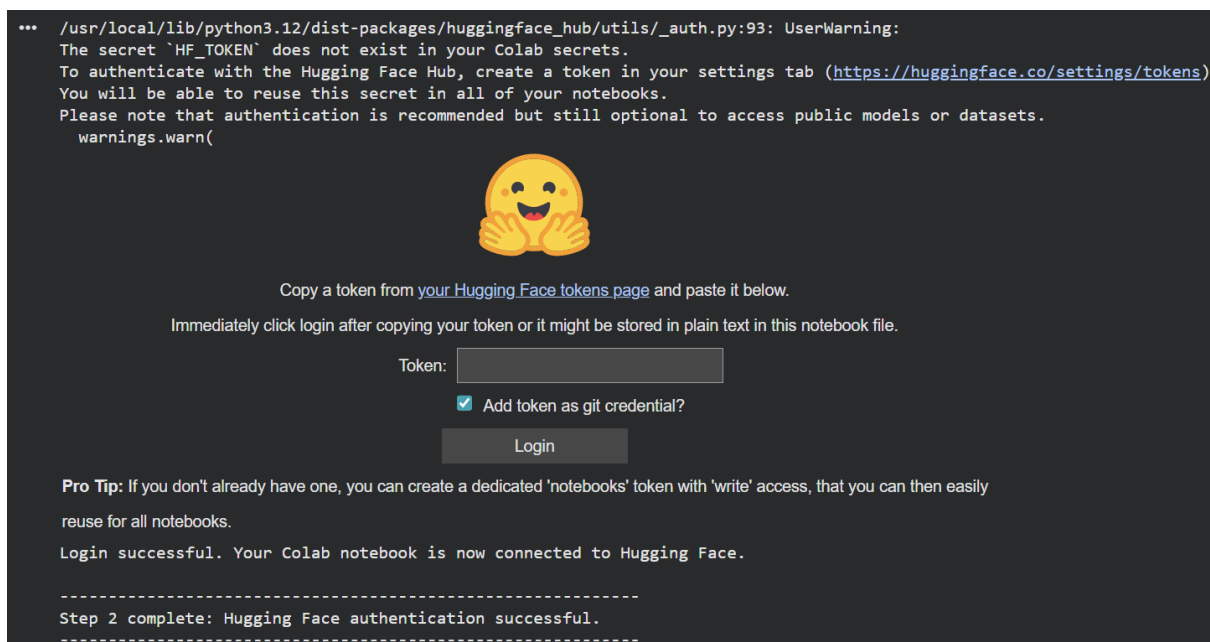
Output: prepared_dataset_2000.csv — 2,000 rows, clip-level metadata with contextual features

Google Colab path: /content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/01_data_preparation/

Key Output Screenshots:

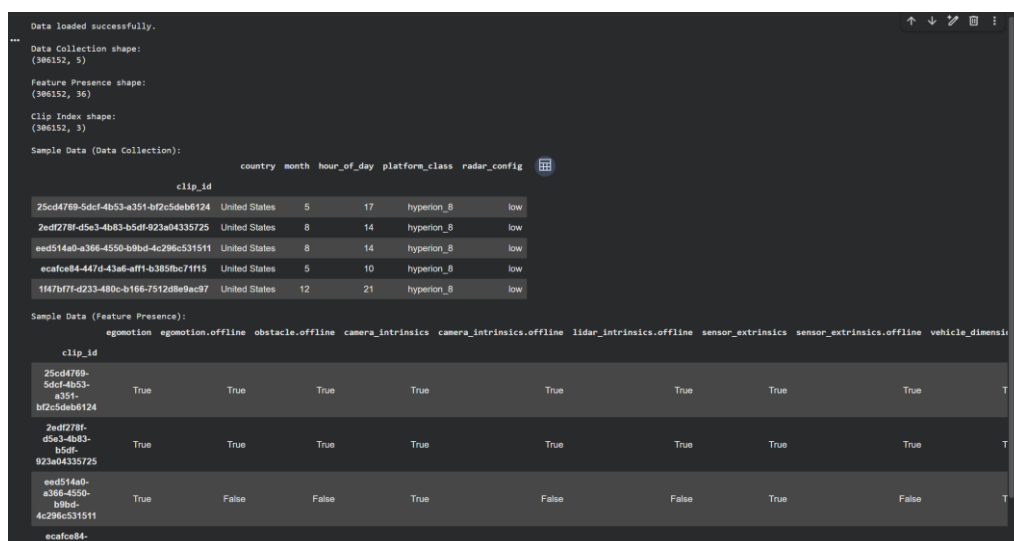
A.1 — Dataset Authentication and Library Setup

Figure A.1 — Notebook screenshot 1



A.2 — Metadata Structure

Figure A.2 — Notebook screenshot 2



A.3 — Clip Filtering Summary

Figure A.3 — Notebook screenshot 3

```

*** Valid clips filtered successfully.

Original master dataset shape:
(306152, 45)

Filtered dataset shape:
(298326, 45)

Rows removed:
7826

First 5 rows of filtered dataset:

```

	clip_id	country	month	hour_of_day	platform_class	radar_config	egomotion	egomotion.offline	obstacle.offline	camera_intrinsic	...	radar_rear_left_srr_0	radar_rear_right_mrr_2	rad
0	25cd4769-5dcf-4b53-a351-bf2c5deb6124	United States	5	17	hyperion_8	low	True	True	True	True	...	True	False	
1	2edf278f-d5e3-4b83-b5df-923a04335725	United States	8	14	hyperion_8	low	True	True	True	True	...	True	False	
3	ecafce84-447d-43a6-aff1-b385fbc71f15	United States	5	10	hyperion_8	low	True	True	True	True	...	True	False	
4	1f47bf7f-d233-480c-b166-7512d8e9ac97	United States	12	21	hyperion_8	low	True	True	True	True	...	True	False	
5	c6915a10-9f22-464d-a422-15b8960a57d7	United States	5	10	hyperion_8	low	True	True	True	True	...	True	False	

```

5 rows x 45 columns

.....
Step 8 complete: Valid clips with ego-motion and camera data selected.
.....

```

A.4 — K-Means Clustering Output

Figure A.4 — Notebook screenshot 4

```

*** K-Means clustering completed.

Cluster distribution:
cluster
0    49451
1    85433
2    92569
3    70873
Name: count, dtype: int64

First 5 rows with cluster labels:

```

	clip_id	country	hour_of_day	month	platform_class	radar_config	cluster
0	25cd4769-5dcf-4b53-a351-bf2c5deb6124	United States	17	5	hyperion_8	low	1
1	2edf278f-d5e3-4b83-b5df-923a04335725	United States	14	8	hyperion_8	low	1
3	ecafce84-447d-43a6-aff1-b385fbc71f15	United States	10	5	hyperion_8	low	1
4	1f47bf7f-d233-480c-b166-7512d8e9ac97	United States	21	12	hyperion_8	low	1
5	c6915a10-9f22-464d-a422-15b8960a57d7	United States	10	5	hyperion_8	low	1

```

.....
Step 11 complete: Scenario clusters assigned to valid clips.
.....

```

A.5 — Isolation Forest Sampling Output

Figure A.5 — Notebook screenshot 5

```
Isolation Forest rare scenario detection completed.
...
Scenario type distribution:
scenario_type
normal    283413
rare      14913
Name: count, dtype: int64

Anomaly label distribution:
anomaly_label
1         283413
-1        14913
Name: count, dtype: int64

Anomaly score summary:
count    298326.000000
mean      0.078338
std       0.043347
min       -0.104467
25%       0.049440
50%       0.080365
75%       0.114746
max       0.155856
Name: anomaly_score, dtype: float64

First 5 rows with anomaly outputs:

```

	clip_id	country	hour_of_day	month	platform_class	cluster	anomaly_label	anomaly_score	scenario_type
0	25cd4769-5dcf-4b53-a351-bf2c5deb6124	United States	17	5	hyperion_8	1	1	0.118596	normal
1	2edf278f-d5e3-4b83-b5df-923a04335725	United States	14	8	hyperion_8	1	1	0.128326	normal
3	eca6ce84-447d-43a6-aff1-b385fbc71f15	United States	10	5	hyperion_8	1	1	0.133481	normal
4	1f47bf7f-d233-480c-b166-7512d8e9ac97	United States	21	12	hyperion_8	1	1	0.021057	normal
5	c6915a10-9f22-464d-a422-15b8960a57d7	United States	10	5	hyperion_8	1	1	0.133481	normal

```
-----
Step 12 complete: Rare and normal scenario labels created.
-----
```

A.6 — Final Sampling Summary

Figure A.6 — Notebook screenshot 6

```

Diversity features created successfully.
...
Prepared dataset shape:
(2000, 51)

Scenario type distribution:
scenario_type
normal    1000
rare      1000
Name: count, dtype: int64

Unique chunks:
150

Day/night distribution:
is_night
0      1059
1       941
Name: count, dtype: int64

Peak-hour distribution:
is_peak_hour
0      1342
1       658
Name: count, dtype: int64

Country distribution top 10:
country
United States    624
Austria          495
Croatia          202
Netherlands      166
Denmark          93
Belgium          88
Sweden           63
Italy            48
Germany          45
France           39
Name: count, dtype: int64

First 5 rows with diversity features:

```

	clip_id	country	hour_of_day	is_night	is_peak_hour	scenario_type	chunk
0	bea40162-4df3-4bab-a1d2-4435767566bc	France	13	0	0	normal	2575
1	5cbf6d63-93e6-47d8-a3d0-a3d1b3ce4916	United States	0	1	0	rare	2186

Figure A.7 — Notebook screenshot 7

```
Final columns selected successfully.
***
Prepared clean dataset shape:
(2800, 16)

Unique chunks:
150

Scenario type distribution:
scenario_type
normal    1800
rare      1000
Name: count, dtype: int64

Columns included:
['clip_id', 'chunk', 'split', 'clip_is_valid', 'country', 'month', 'hour_of_day', 'platform_class', 'radar_config', 'camera_front_wide_120fov', 'egomotion.offline', 'cluster', 'anomaly_label', 'scenario_type', 'is_night']

First 5 rows:
```

	clip_id	chunk	split	clip_is_valid	country	month	hour_of_day	platform_class	radar_config	camera_front_wide_120fov	egomotion.offline	cluster	anomaly_label	scenario_type	is_night
0	bea40162-4df3-4bab-a1d2-4435767566bc	2575	val	True	France	1	13	hyperion_8.1	med	True	True	0	1	normal	
1	5cbf6d63-93e6-47d8-a3d0-a3d1b3ce4916	2186	test	True	United States	1	0	hyperion_8	low	True	True	1	-1	rare	
2	8ff39926-33c1-4956-8738-26c4ed674c1e	326	test	True	United States	10	20	hyperion_8	low	True	True	1	1	normal	
3	a1cdaa7b-4633-42a7-b735-3e90ad3fc135	3045	test	True	Austria	11	18	hyperion_8.1	med	True	True	0	-1	rare	
4	1fa7d88f-f330-4817-bd02-4f4d703a95e6	1430	test	True	Netherlands	8	19	hyperion_8.1	None	True	True	3	1	normal	

```

-----
Step 15 complete: Clean prepared dataset created for export.
NOTE: anomaly_score excluded - risk labels created in Notebook 5.
-----

```

A.7 — Prepared Dataset Export Confirmation *Screenshot the Step 15 output showing the successful save of prepared_dataset_2000.csv and prepared_dataset_2000.parquet to Google Drive.*

Figure A.8 — Notebook screenshot 8

```
*** Mounted at /content/drive
Prepared dataset saved successfully.

CSV file saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/01_data_preparation/prepared_dataset_2000.csv

Parquet file saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/01_data_preparation/prepared_dataset_2000.parquet

File existence check:
CSV exists: True
Parquet exists: True

-----
Step 16 complete: Prepared dataset saved to Google Drive.
-----
```

Figure A.9 — Notebook screenshot 9

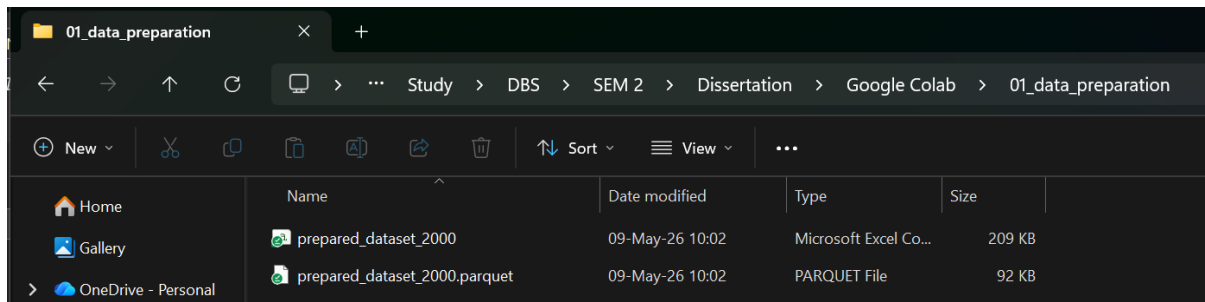


Figure A.10 — Notebook screenshot 10

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	clip_id	chunk_split	clip_is_valid	country	month	hour_of_day	platform_class	radar_config	camera_front_wide_120fov	egomotion_offline	cluster	anomaly_label	scenario_type	is_night	is_peak_hour	
2	bea40162-dfd3-4bab-a1d2-4435767566bc	2575 val	TRUE	France	1	13	hyperion_8.1	med	TRUE	TRUE	0	1	normal	0	0	
3	5cbfd6d3-93e6-47d8-a3d0-a3d1b3ce4916	2186 test	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
4	8ff39926-33c1-4956-8738-26c4ed674c1e	326 test	TRUE	United States	10	20	hyperion_8	low	TRUE	TRUE	1	1	normal	1	0	
5	a1cdaa7b-4633-42a7-b735-3e90ad3f1c135	3045 test	TRUE	Austria	11	18	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	1	
6	1fe7d88f-f330-4817-bd02-4f4d703a95e6	1430 test	TRUE	Netherlands	8	19	hyperion_8.1		TRUE	TRUE	3	1	normal	0	1	
7	39c7ad5c-clcf-43ab-a2fd-ebf6d11d991d	2836 train	TRUE	Croatia	8	3	hyperion_8.1		TRUE	TRUE	3	1	normal	1	0	
8	0949b67-64e5-4dce-92b6-9e9b83593b6	3045 test	TRUE	Austria	11	18	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	1	
9	64712e09-b370-4d0d-babb-d3b2585e99e1	2960 val	TRUE	Croatia	8	23	hyperion_8.1		TRUE	TRUE	3	1	normal	1	0	
10	2c08f463-6d13-4c44-8f5d-d1fab72ea8d7	2960 val	TRUE	Austria	11	14	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	0	
11	f405fbc-8eb9-485b-ba04-10035da076f3	1430 test	TRUE	Netherlands	9	17	hyperion_8.1		TRUE	TRUE	3	1	normal	0	1	
12	1f02b15b-a6ad-4c07-b476-f2584ad30298	221 val	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
13	d1c42eed-d5ee-4fa5-b27b-3fb319ca1c5e	296 val	TRUE	Croatia	8	22	hyperion_8.1		TRUE	TRUE	3	1	normal	1	0	
14	ff53ba6d-e497-4df4-b380-clbdf1042259	1044 val	TRUE	United States	1	17	hyperion_8.1	high	TRUE	TRUE	2	1	normal	0	1	
15	d4e92ae37-5958-4bd8-9926-ca108b3f4cdd	1992 val	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
16	7b9dca4d-b673-418c-b0aa-c05efe882fc3	1717 train	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
17	23e4189a-4e82-4ad1-98f5-0585b38d6ed7	3044 test	TRUE	Austria	10	10	hyperion_8.1		TRUE	TRUE	3	-1	rare	0	1	
18	26c1ad5d-7412-4314-b56e-7415b877330e	1045 val	TRUE	United States	5	0	hyperion_8.1	high	TRUE	TRUE	2	-1	rare	1	0	
19	7256aa47-b0b5-472a-a712-fbe1bcf06d0f	3089 val	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
20	1716ea3b-6edd-4e40-bb83-de79a10e2fce	1049 val	TRUE	United States	1	7	hyperion_8.1	high	TRUE	TRUE	2	1	normal	0	1	
21	d96145ef-f0ca-4c00-9042-5cee0a31631e	2925 val	TRUE	Germany	9	12	hyperion_8.1		TRUE	TRUE	3	1	normal	0	0	
22	ac7d5141-e2a9-47fc-bca5-22f9d567f3e8	44 train	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
23	2a536ffe-98d3-48f3-89a4-1bb36783e89b	3056 test	TRUE	Czechia	12	13	hyperion_8.1		TRUE	TRUE	3	1	normal	0	0	
24	786c3012-5956-4c20-98e4-ba53ca19fb9b	1703 train	TRUE	United States	9	12	hyperion_8	low	TRUE	TRUE	1	1	normal	0	0	
25	b9e63b98-ce4e-43eb-a9e8-c074d44bf09	2493 train	TRUE	Denmark	5	23	hyperion_8.1		TRUE	TRUE	3	1	normal	1	0	
26	32b8f8f1-6d5b-4ba5-9927-5269c6c9a677	805 train	TRUE	Netherlands	11	21	hyperion_8.1	med	TRUE	TRUE	0	1	normal	1	0	
27	5de7cc06-8608-4d48-9694-ec4de0be3d69	163 train	TRUE	Belgium	3	13	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	0	
28	11ff9334-d889-4332-a433-c4b90851786f	2493 train	TRUE	Slovakia	7	6	hyperion_8.1		TRUE	TRUE	2	1	normal	1	0	
29	90265336-1d4e-423c-8c78-61d35f73ee9	3045 test	TRUE	Austria	11	18	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	1	
30	bcf28a8d-43d0-44ba-ac27-71d6df114745	3089 val	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
31	52960728-4106-40e7-bc4c-67a053488555	2594 val	TRUE	Austria	11	20	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	1	0	
32	de150cce-b14a-4341-838b-673cd5d5d526	3044 test	TRUE	Austria	1	23	hyperion_8.1		TRUE	TRUE	3	-1	rare	1	0	
33	735e9733-adda-40d9-bc74-8e6157e0909c	1422 test	TRUE	Austria	4	4	hyperion_8.1	high	TRUE	TRUE	3	-1	rare	1	0	
34	5ee9ac33-6e26-42a9-9025-b5a17bfe0a57	172 train	TRUE	Austria	8	19	hyperion_8.1	med	TRUE	TRUE	0	-1	rare	0	1	
35	9a2aca1-f431-4ce2-a1c4-40ae075a814c	312 test	TRUE	United States	1	0	hyperion_8	low	TRUE	TRUE	1	-1	rare	1	0	
36	5b943573-973c-44c7-ad25-9690f59ddbf9	1021 val	TRUE	United States	1	17	hyperion_8.1	med	TRUE	TRUE	0	1	normal	0	1	
37	7dc4b007-bb69-43aa-845c-d03766654bf7	1422 test	TRUE	Austria	5	7	hyperion_8.1	high	TRUE	TRUE	3	-1	rare	0	1	
38	eeefef57c-5d85-489c-86d1-4c16df43c4a	2575 val	TRUE	France	1	21	hyperion_8.1	med	TRUE	TRUE	0	1	normal	1	0	

prepared_dataset_2000

Appendix B — Notebook 2: Ego-Motion Feature Engineering

Purpose: Extraction of eight behavioural telemetry features from frame-level vehicle ego-motion data.

Key steps implemented:

1. Download of ego-motion parquet files for all required chunks from Hugging Face
2. Frame-level ego-motion extraction per clip from parquet files
3. Aggregation of frame-level records into eight clip-level behavioural features — avg_speed, max_speed, avg_acceleration, max_acceleration, avg_jerk, max_jerk, braking_events, and hard_braking_events
4. Hard braking events defined as frames with acceleration below -3 m/s^2 , consistent with established driving behaviour analysis conventions
5. Ego-motion features saved as ego_features_2000.csv

Output: ego_features_2000.csv — 1,899 rows, 8 behavioural features per clip

Google Colab path: /content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/02_ego_features/

Table B.1 — Ego-Motion Features: Definitions and Computation Method

Feature	Description	Threshold
avg_speed	Mean vehicle speed across all clip frames (m/s)	—
max_speed	Maximum vehicle speed recorded in the clip (m/s)	—
avg_acceleration	Mean vehicle acceleration across clip frames (m/s^2)	—
max_acceleration	Maximum vehicle acceleration recorded (m/s^2)	—
avg_jerk	Mean rate of change of acceleration (m/s^3)	—

Feature	Description	Threshold
max_jerk	Maximum jerk recorded in the clip, clipped at ± 50 (m/s^3)	± 50
braking_events	Count of frames with acceleration below threshold	-1 m/s^2
hard_braking_events	Count of frames with acceleration below threshold	-3 m/s^2

All eight features are aggregated from frame-level ego-motion telemetry parquet files. Hard braking events are defined consistent with established driving behaviour analysis conventions (Wang et al., 2019; Guagnano et al., 2023).

Key Output Screenshots:

B.1 — Required Chunks Identified

Figure B.1 — Notebook screenshot 11

```
Mounted at /content/drive
Prepared dataset loaded successfully.

'''
Input file path:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/01_data_preparation/prepared_dataset_3000.csv

Prepared dataset shape:
(2000, 16)

Scenario type distribution:
scenario_type
normal    1000
rare      1000
Name: count, dtype: int64

Unique chunks:
150

First 5 rows:
   clip_id  chunk  split  clip_is_valid  country  month  hour_of_day  platform_class  radar_config  camera_front_wide_120fov  egomotion_offline  cluster  anomaly_label  scenario_type  is_night
0  bea40162-4d93-4bab-a1d2-4435767566bc  2575  val      True      France      1      13      hyperion_8.1  med      True      True      0      1      normal      False
1  5cbf6d63-93e6-47db-a3d0-a3d1b3ce4916  2186  test      True      United States      1      0      hyperion_8  low      True      True      1     -1      rare      False
2  8f39926-33c1-4956-8738-26c4ed674c1e  326  test      True      United States      10     20      hyperion_8  low      True      True      1      1      normal      False
3  a1cd4a7b-4633-42a7-b735-3e90ad3fc135  3045  test      True      Austria      11     18      hyperion_8.1  med      True      True      0     -1      rare      False
4  1fe7d88f-f330-4817-b002-4f4d703a95e6  1430  test      True      Netherlands      8      19      hyperion_8.1  NaN      True      True      3      1      normal      False

Step 3A complete: Updated prepared dataset loaded from Google Drive.
```

Figure B.2 — Notebook screenshot 12

```

*** Required chunks identified successfully.

Total unique chunks needed:
150

First 10 chunk IDs:
[np.int64(3), np.int64(8), np.int64(12), np.int64(14), np.int64(30), np.int64(42), np.int64(44), np.int64(45), np.int64(53), np.int64(60)]

-----
Step 4 complete: Required ego-motion chunks identified.
-----

```

B.2 — Ego-Motion Feature Extraction Output

Figure B.3 — Notebook screenshot 13

```

labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.18MB/s]
...
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.20MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.16MB/s]
Downloaded 140 / 150 chunks
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.65M/1.65M [00:01<00:00, 8.29MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.65M/1.65M [00:01<00:00, 8.25MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.66M/1.66M [00:01<00:00, 8.30MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.23MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.22MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.21MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.21MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.17MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.66M/1.66M [00:01<00:00, 8.30MB/s]
labels/egomotion.offline/egomotion.offli(...): 100% ██████████ 1.64M/1.64M [00:01<00:00, 8.22MB/s]
Downloaded 150 / 150 chunks

Download completed successfully.

Total chunks downloaded:
150

First 5 entries:

```

	chunk	file_path
0	3	/root/.cache/huggingface/hub/datasets--nvidia-...
1	8	/root/.cache/huggingface/hub/datasets--nvidia-...
2	12	/root/.cache/huggingface/hub/datasets--nvidia-...
3	14	/root/.cache/huggingface/hub/datasets--nvidia-...
4	30	/root/.cache/huggingface/hub/datasets--nvidia-...

```

-----
Step 5 complete: Ego-motion chunks downloaded efficiently.
-----

```

B.3 — Ego-Motion Feature Summary Statistics

Figure B.4 — Notebook screenshot 14

```
Processed 140 / 150 chunks
Processed 150 / 150 chunks

Improved ego-motion extraction completed.

Ego features shape:
(2000, 10)

Unique clips processed:
2000

Missing clips:
0

First 5 rows:
```

	clip_id	avg_speed	max_speed	avg_acceleration	max_acceleration	avg_jerk	max_jerk	braking_events	hard_braking_events	chunk
0	31856298-2960-4d45-a906-1b489066c452	15.474187	17.625619	0.006037	2.189002	-0.007385	30.761935	7	0	3
1	18bab3f7-41e4-4ebc-b701-67569de3709b	8.976852	14.930298	-0.013062	4.165472	0.033925	45.713046	50	4	3
2	4ed4b27e-6b5a-4468-a8a6-b54feec34c6f	11.680353	13.049555	0.019714	2.616180	0.126845	47.692112	21	2	3
3	784f877-ec64-4896-8560-19b3c686d2cd	8.510242	14.600693	-0.339619	8.531664	0.266604	50.000000	55	7	3
4	34860342-d420-4ec4-8353-6d79ae77f83c	19.600180	20.694840	0.047169	1.894011	-0.155498	43.172932	7	0	3

```
Summary statistics:
```

	avg_speed	max_speed	avg_acceleration	max_acceleration	avg_jerk	max_jerk	braking_events	hard_braking_events	chunk
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	11.675544	15.079039	0.024579	3.326604	0.003434	38.052197	16.678000	2.983000	1682.439500
std	8.075036	7.593572	0.325497	2.156485	0.204846	11.660184	16.789737	7.113439	1109.338681
min	0.028834	0.064190	-1.162090	0.588448	-1.751889	12.025594	0.000000	0.000000	3.000000
25%	5.608803	9.757625	-0.127900	2.031255	-0.081035	27.536673	3.000000	0.000000	319.500000
50%	9.397453	13.513143	0.011214	2.759259	0.006415	39.319172	11.500000	0.000000	1704.000000
75%	15.571871	19.379730	0.182123	3.747810	0.099928	50.000000	26.000000	2.000000	2683.000000
max	36.411681	38.154803	1.338420	14.924023	1.527859	50.000000	80.000000	59.000000	3107.000000

```
-----
Step 6 complete: Improved ego-motion features extracted.
-----
```

B.4 — Ego-Motion Features Export Confirmation

Figure B.5 — Notebook screenshot 15

```
*** Ego-motion features saved successfully.

CSV file saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/02_ego_features/ego_features_2000.csv

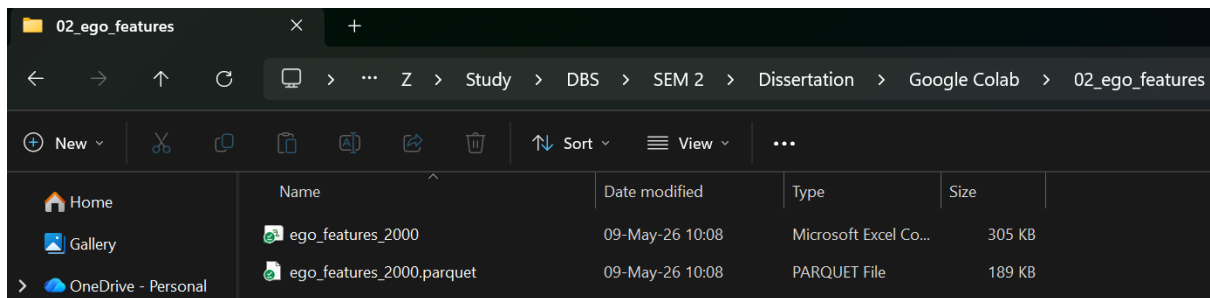
Parquet file saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/02_ego_features/ego_features_2000.parquet

File existence check:
CSV exists: True
Parquet exists: True

Ego feature dataset shape:
(2000, 10)

-----
Step 8 complete: Ego-motion features saved to Google Drive.
-----
```

Figure B.6 — Notebook screenshot 16



The screenshot shows a file explorer window with the following details:

- Path:** > ... > Z > Study > DBS > SEM 2 > Dissertation > Google Colab > 02_ego_features
- Files:**
 - ego_features_2000**: Microsoft Excel Co..., 305 KB, modified 09-May-26 10:08
 - ego_features_2000.parquet**: PARQUET File, 189 KB, modified 09-May-26 10:08

Name	Date modified	Type	Size
ego_features_2000	09-May-26 10:08	Microsoft Excel Co...	305 KB
ego_features_2000.parquet	09-May-26 10:08	PARQUET File	189 KB

Figure B.7 — Notebook screenshot 17

	A	B	C	D	E	F	G	H	I	J
1	clip_id	avg_speed	max_speed	avg_acceleration	max_acceleration	avg_jerk	max_jerk	braking_events	hard_braking_events	chunk
2	31856298-2960-4d45-a906-1b489066c452	15.4741869	17.62561903	0.006036733	2.189001524	-0.007384594	30.76193504	7	0	3
3	18bab3f7-41e4-4ebc-b701-67569de3709b	8.976852224	14.93029803	-0.013062057	4.165471515	0.033925029	45.71304566	50	4	3
4	4ed4b27e-6b5a-4468-a8a6-b54fec34c6f	11.68035332	13.04955467	0.019713545	2.616179884	0.126844706	47.69211247	21	2	3
5	784f8f77-ec64-4896-8560-19b3c686d2cd	8.51024225	14.60069344	-0.339618867	8.531664162	0.26660449	50	55	7	3
6	34860342-d420-4ec4-8353-6d79ae77f83c	19.60017991	20.69484045	0.047169033	1.894011088	-0.155497541	43.17293195	7	0	3
7	81174ca6-3a75-4b21-b156-e983eedafc45	3.348503538	13.38546815	-0.67704843	1.503594774	-0.031898962	26.8018644	40	0	8
8	42388dff-6173-428e-9d41-8161a18a827	17.8251271	19.03763532	-0.244030936	1.762045637	-0.103656237	34.13218629	7	0	8
9	91679561-a777-44ea-b910-25a1735bea49	18.44484249	26.54388985	0.65306416	6.355669729	0.347005922	50	3	0	8
10	17d52f2d-6ba0-4b7a-a3bc-6d3c5a13f468	5.054314181	15.23867905	0.767373017	3.394260432	0.132630361	28.85385936	1	0	8
11	0a701173-f30e-4a84-a938-6d15d4cf9c2f	12.44631588	14.16161143	0.080673703	1.550544765	-0.015934431	29.37846027	0	0	8
12	5e3e0875-51a3-49e7-81ad-3f1d2329df27	18.8851334	21.66718211	0.123568637	4.113242452	0.099646248	50	18	0	12
13	460491e4-0ea6-47a0-ac93-b4be13e88d4e	10.79854881	15.24296555	-0.441933413	1.696295476	-0.097503706	23.74336622	7	0	12
14	3f709ec5-34f3-436d-8b48-74838bab242a	7.078664904	12.84844004	0.639836193	3.470962095	-0.037236308	42.47049887	1	0	12
15	3d45c5a3-29e4-4c66-bda3-41da3231fab2	19.86035891	20.40231609	0.033462363	2.721740184	-0.137786253	50	6	0	12
16	a84a5e42-89e9-4a2c-b0f7-8173b0176970	16.43656869	17.78147521	-0.099200732	5.635794754	-0.064443017	50	33	11	14
17	4297877f-9c47-4208-b157-9dd6d3ba2025	12.80947279	13.90567021	0.02461629	4.147333756	-0.144804658	50	19	4	14
18	e38aff51-b752-4195-9a08-eeb48fce433d	4.54992554	12.174983	0.609611061	3.279829975	-0.028659174	44.61020351	0	0	14
19	3e59c628-efec-4fea-ab6d-867a503c0fe9	9.312641938	10.08199977	-0.027079302	1.369515178	-0.206925045	24.58232882	2	0	14
20	a74fd7e5-2b68-48d5-8c32-077d3601fa43	3.458498122	7.36270917	0.219836608	3.484885818	0.0714376	50	8	0	14
21	2696bb1c-bfef-4318-9d02-cffa41d3120a	12.66354479	14.21274692	0.158633752	6.145147678	0.001001815	50	13	3	30
22	9dad2d41-e027-4ac7-a5b9-2cebfa55539	14.96928553	15.86502121	0.084539187	4.06341039	0.200633079	50	10	3	30
23	178bd5b3-ef04-4eb0-be13-486e5d660c05	3.344786294	7.091265362	-0.106652028	2.076202777	-0.025080332	34.60707068	7	1	30
24	96425b1d-bfdd-4b4f-ae95-2be804b86152	5.781908483	10.0694788	0.123528136	2.660846865	-0.06889095	32.27591787	18	2	30
25	0da00f1b-64e1-45e3-8cdd-a55c1067de93	7.553049959	10.46626343	-0.211496575	2.740471858	0.271045079	50	36	0	30
26	4165b24b-2772-41c9-b374-c87a9476ce35	12.57803447	14.01949199	0.090002364	3.030928411	-0.034456222	45.71137749	9	0	30
27	1be837c3-f738-4868-9d0c-e20383c96e04	17.04401563	19.60145684	0.173375368	2.664623363	0.012319289	33.10382365	0	0	42
28	69dcbd81-877b-4bac-8420-a7c12370b52a	15.45804877	16.34483158	0.052416388	4.946035002	0.461361872	50	25	4	42
29	5ab8a10a-25be-4b6d-bbf2-597010cefd1d7	5.870431246	8.437373185	-0.024568249	2.135883178	0.057557523	40.38010281	27	0	42
30	51264d9d-c9d8-43a9-9834-173a2fc396e2	14.17742756	14.6757463	-0.195422918	1.982798328	-0.448231976	36.02203142	13	3	42
31	44866f90-fe22-424d-b420-3533ff2b852	12.53582877	15.80377782	-0.577804717	5.622308894	0.296966688	50	62	16	42
32	b6ad6df5-9d14-4dcf-9df2-b14f418616ea	5.120780342	8.361313781	0.162627879	2.546927366	0.030955024	33.79962709	19	0	42
33	5067533c-e396-44cc-b5e7-3f642a060469	8.06895511	13.35718837	0.176478768	2.691065122	0.091331511	31.26026065	38	3	42
34	326c225c-8e59-4956-ad12-de1d580e2ae7	4.150937429	13.33812267	-0.674106579	2.157150427	0.002415661	47.45576231	40	2	42
35	28c0d702-cf3e-493a-802b-0e6c4dd77479	14.58063915	15.43423149	0.074599266	2.101086813	0.000807682	32.96889483	2	0	42
36	55c033d0-c197-411b-a3f7-7ede68ebc2a6	11.74613168	13.63228217	0.073471787	1.854623513	0.134275302	39.17581464	3	0	44
37	ad3cf7d5-2c4a-4557-a778-8c7c95af525a	15.3270971	15.73326823	0.023507518	2.591644916	0.192984209	50	12	0	44
38	262444e9-f04a-42c2-b932-1ef92867fbf1	7.515980731	13.58814386	0.649221011	4.367087751	0.304742296	50	0	0	44

Appendix C — Notebook 3: Visual Feature Extraction

Purpose: Extraction of visual features from the middle frame of each clip across four modalities — basic visual descriptors, YOLOv8 object detection, ResNet-50 CNN embeddings, and ViT-B/16 embeddings.

Key steps implemented:

1. Resumable middle frame extraction from front-wide-120fov camera footage saved individually to Google Drive
2. Basic visual feature extraction — mean brightness and RMS contrast via OpenCV grayscale conversion
3. YOLOv8x object detection — nine object count features per clip across eight categories plus total objects

4. ResNet-50 CNN embedding extraction — 2,048-dimensional penultimate feature vector reduced to ten PCA components
5. ViT-B/16 Vision Transformer embedding extraction — 768-dimensional class token output reduced to ten independent PCA components
6. PCA explained variance diagnostics reported for both CNN and ViT reductions
7. Coverage check before merging confirming all four modalities cover the same clips
8. All visual feature tables merged on clip_id and saved as visual_features_final.csv

Output: visual_features_final.csv — 1,899 rows, 31 visual feature columns

Google Colab path: /content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/03_visual_features/

Key Output Screenshots:

C.1 — Frame Extraction Progress

Figure C.1 — Notebook screenshot 18

```
Number of image files found:
1899

Rebuilt frame log shape:
(1899, 3)

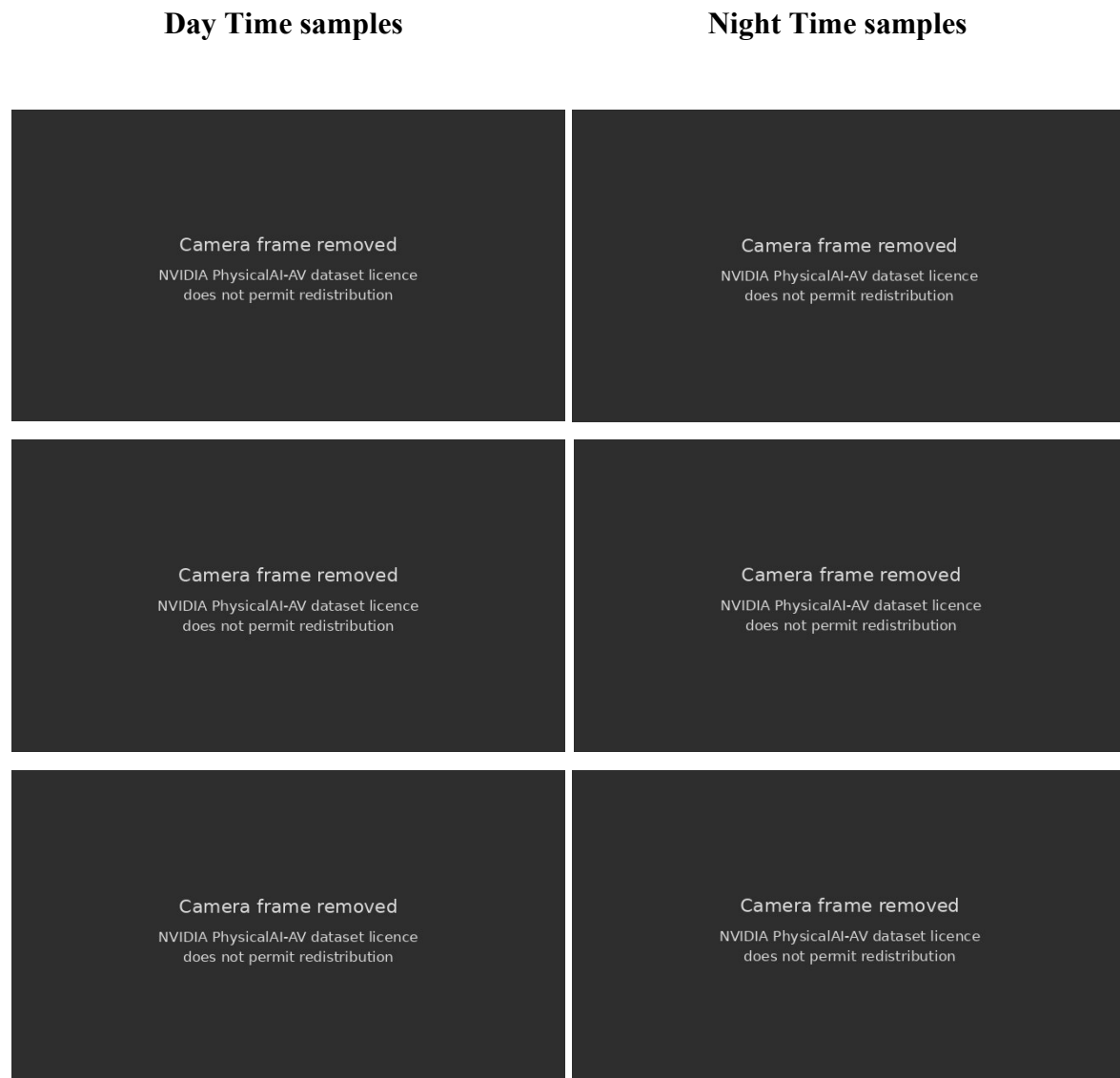
First 5 rows:
```

	clip_id	frame_path	frame_extracted
0	ace2f414-4145-4808-b103-5df9337addfa	/content/drive/My Drive/Z/Study/DBS/SEM 2/Diss...	True
1	6a530d88-fb41-4db4-9c5b-73cd57106489	/content/drive/My Drive/Z/Study/DBS/SEM 2/Diss...	True
2	5c0c6f37-3996-486e-8367-5426d992f254	/content/drive/My Drive/Z/Study/DBS/SEM 2/Diss...	True
3	126bd5cc-dadd-4277-b7af-18bb99d7b30e	/content/drive/My Drive/Z/Study/DBS/SEM 2/Diss...	True
4	77be1f14-010b-470d-8289-d61d1605f885	/content/drive/My Drive/Z/Study/DBS/SEM 2/Diss...	True

```
Updated frame log saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/03_visual_features/frame_extraction_log_rebuilt.csv
```

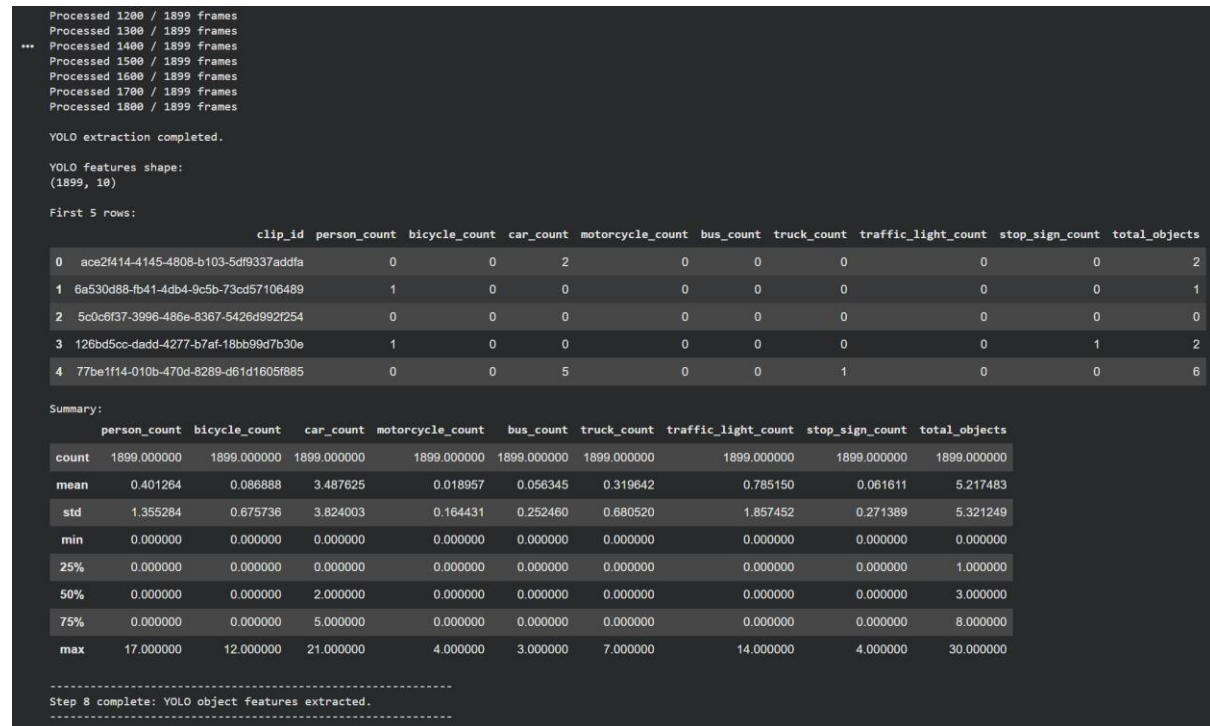
C.2 — Sample Extracted Frames

Figure C.2 — Notebook screenshot 19



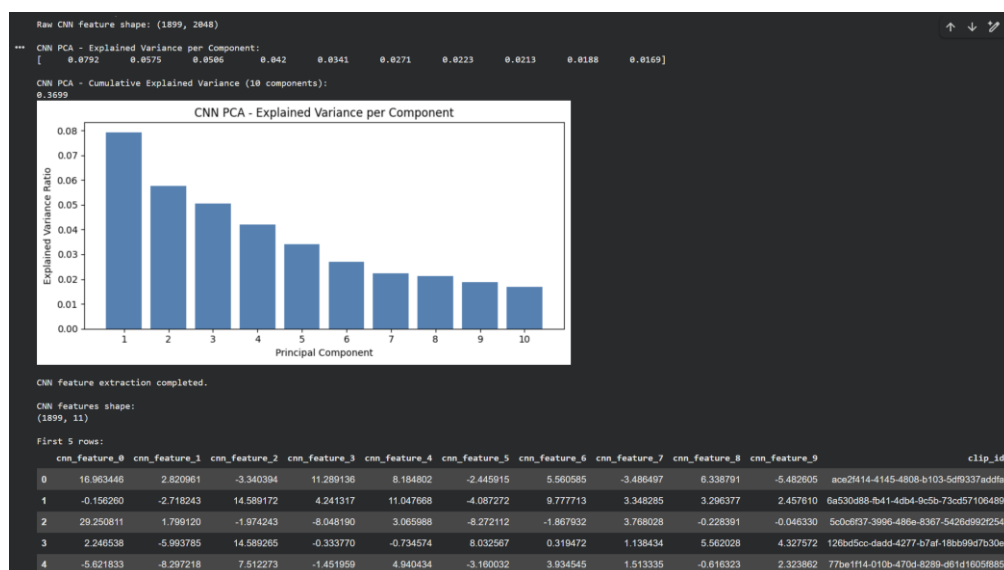
C.3 — YOLOv8 Detection Output

Figure C.3 — Notebook screenshot 20



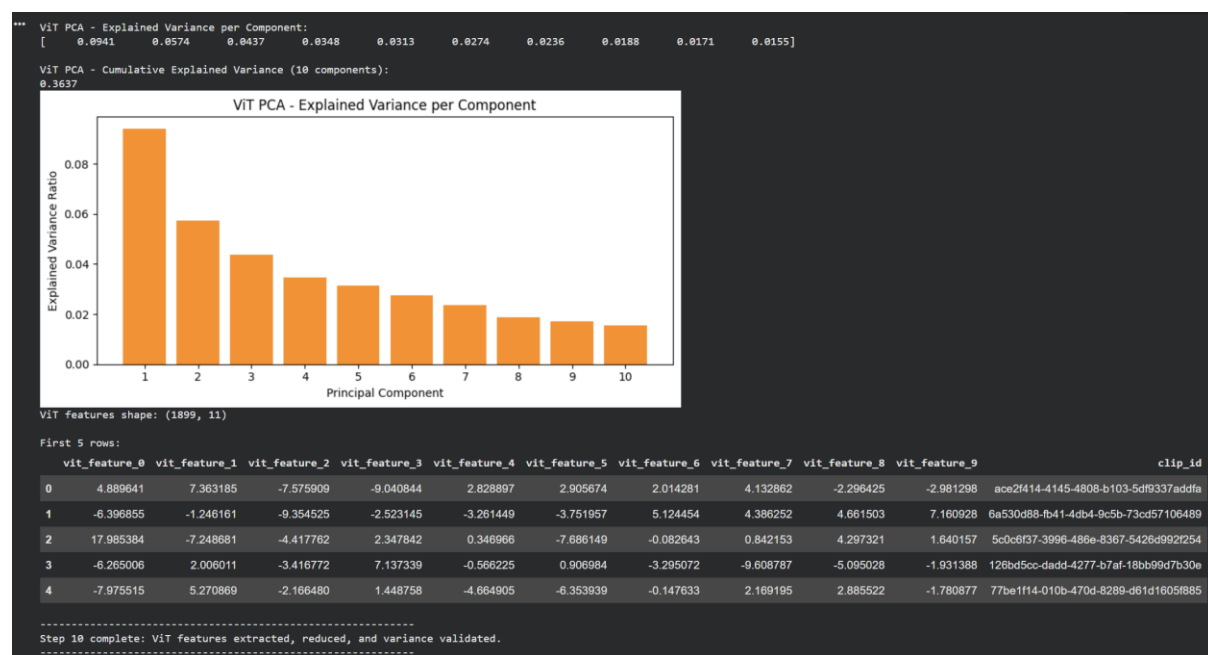
C.4 — CNN PCA Explained Variance

Figure C.4 — Notebook screenshot 21



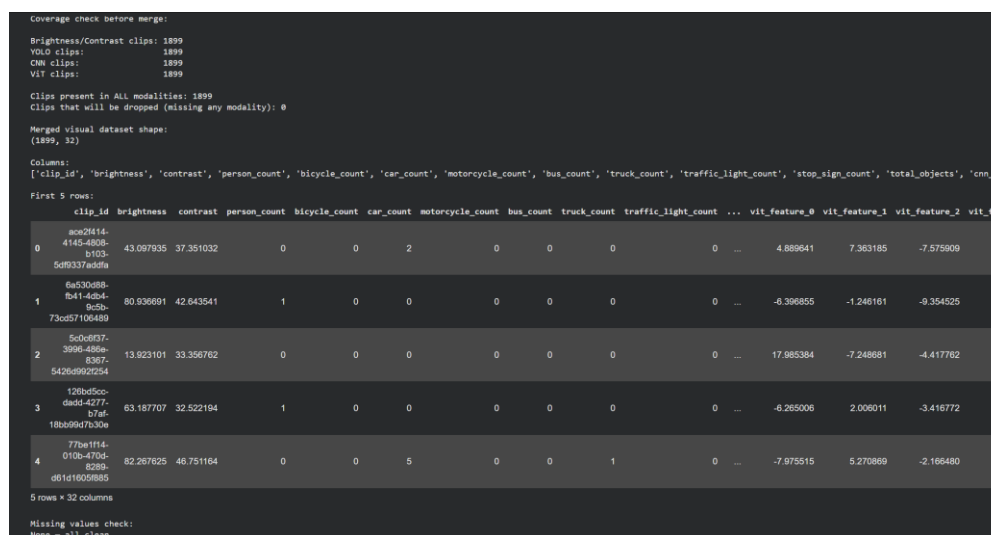
C.5 — ViT PCA Explained Variance

Figure C.5 — Notebook screenshot 22



C.6 — Visual Feature Coverage Check

Figure C.6 — Notebook screenshot 23



C.7 — Visual Features Export Confirmation *Screenshot the Step 11 output showing the successful save of visual_features_final.csv to Google Drive including the final merged DataFrame shape.*

Figure C.7 — Notebook screenshot 24

```
Saved files:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/03_visual_features/visual_features_final.csv
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/03_visual_features/visual_features_final.parquet

-----
Step 11 complete: Visual features merged with coverage verified.
-----
```

Figure C.8 — Notebook screenshot 25

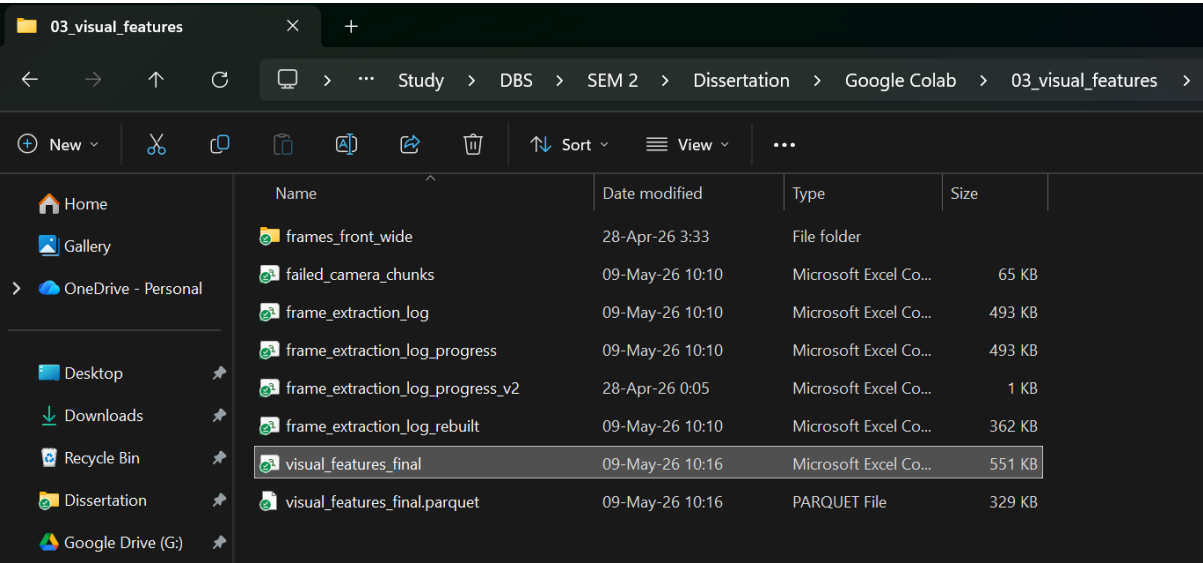


Figure C.9 — Notebook screenshot 26

		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
	clip_id	brightness	contrast	person_count	bicycle_count	car_count	motorcycle_count	bus_count	truck_count	traffic_light_count	stop_sign_count	total_objects	cnr_feature_0	cnr_feature_1	cnr_feature_2	cnr_feature_3	
1	ae03f414-4145-4808-b103-5df9337adffa	43.09793547	37.3510318	0	0	2	0	0	0	0	0	2	16.963446	2.8209612	3.3403935	11.289136	
2	6a530d88-fb41-4db4-9c5b-73cd57106489	80.93669078	42.64354102	1	0	0	0	0	0	0	0	1	-0.15625995	-2.7182431	14.589172	4.2413173	
3	5c0cfd37-3996-486e-8367-5426d992f254	13.92310137	33.35676214	0	0	0	0	0	0	0	0	0	29.25081	1.7991196	-1.9742434	-8.04819	
4	126bd5cc-dadd-4277-b7af-18bb99d7b30e	63.18770689	32.52219357	1	0	0	0	0	0	0	1	2	2.246538	-5.9937854	14.589265	-0.33377022	
5	77be1f14-010b-470d-8289-d61d1605f885	82.26762539	46.75116407	0	0	5	0	0	0	0	0	6	-5.6218333	-8.297218	7.5122733	-1.4519594	
6	864c7e5b-e9f9-4dad-aae8-0c789088043e	60.31747926	30.86232854	0	0	1	0	0	0	0	0	1	-5.570134	6.5553765	11.403972	2.7289226	
7	6da278fb-d5df-4903-8508-bc3281d0092d	68.4165513	28.58961494	0	0	3	0	0	0	0	0	4	3.4213333	2.5980408	-17.410185	-1.5248783	
8	f7f6dccc-82a1-47ab-8577-87d2ba44d028	63.69274306	47.45129444	0	0	3	0	0	0	0	0	3	-0.7851778	-8.737805	6.177149	-14.481373	
9	6d8a2da0-e716-4ce8-b133-0f7034f9224b	97.86615934	58.43173321	3	0	0	0	0	0	0	0	3	-8.923638	-20.250402	12.685905	-4.354589	
10	8b20fd9f-6748-4c8d-ac0d-7fdad605eab5	80.50833864	45.00951951	0	0	0	0	0	0	0	0	0	-7.0696926	-17.025743	6.911981	-10.205281	
11	f833dfe-aa28-4ae7-b3af-52c26cc489e7	62.84041088	28.20333561	0	0	1	0	0	0	0	1	1	-7.3813834	-11.736141	10.069765	-7.213719	
12	6db69fb3-c5f0-441f-b1e1-6cc1b77bd842	106.2929012	56.17420842	0	0	0	0	0	1	0	0	1	-8.538169	-6.3696623	-3.1614902	4.409802	
13	2c217036-dc54-4a4f-9634-1ffdefd8d088	33.77086806	37.34500908	0	0	0	0	0	0	0	0	0	18.886497	4.0923166	-12.114427	13.813104	
14	4b01b5cd-528b-4e48-8d4e-29a4d65870a1	70.335856	28.83625606	0	0	0	0	0	0	0	0	0	0.8907784	-0.21862172	21.746628	5.20177	
15	ff9e4113-4709-4da6-e214-1f94b6c0611d	65.30965856	31.70153489	0	0	3	0	0	0	0	0	3	6.3368025	8.612731	-12.056222	2.7934697	
16	407b0284-8a82-4909-90d9-0dd84007084e	44.7414048	40.63953464	0	0	0	0	0	0	0	0	0	11.751301	-1.7079707	-0.46192235	11.447679	
17	facb8d79-6a09-44a0-9d8a-02c799c94f7e	72.78173081	46.51331195	0	0	7	0	0	0	0	0	10	-16.838337	-14.015165	-12.140203	-18.849083	
18	f757cd86-befa-4b02-aa12-63df7ab74337	66.47613764	38.81847052	1	0	6	0	0	2	0	0	9	-6.3042526	2.0390885	2.957615	-11.97867	
19	0dc09c6b-ae40-417a-8a74-c7299e8e891c	36.44253954	32.91350212	0	0	4	0	0	0	0	1	1	4.749477	-9.301349	4.0307555	2.1292834	
20	1a86854ed-5983-49b7-82a6-df88a5d2a945	44.25871046	34.19644256	0	0	0	0	0	0	0	0	0	9.526746	-6.3549724	10.041794	-6.550375	
21	527168d1-1497-4562-b34c-29df7dcdd49	74.15722174	47.75525238	0	14	0	0	0	0	0	18	18	-10.257269	-9.032023	1.1483245	-14.140232	
22	0ed7cd93-8375-4912-90a0-3a83293b3377	63.72918981	31.88181986	1	0	1	0	0	1	0	0	3	-5.8670554	-4.2626963	6.0002856	-11.040227	
23	cd23c382-d926-4f2d-8e94-6f6ecce6e15a	65.51503762	38.826333	0	0	5	0	0	0	3	0	8	-7.9181447	-3.8472836	2.8571038	-9.418116	
24	355ce0aa-6f97-4125-bd55-2a5de07fc03b	54.32676649	40.3747839	0	0	1	1	0	0	0	3	3	10.64912	-11.79495	-6.457637	4.336752	
25	338d4dd6-d569-4154-9a7e-d053cfe9f616	70.29582224	28.24906555	0	0	6	0	0	0	5	0	11	5.4759903	-4.0855637	-15.325796	-18.091845	
26	6b1bd1c1-e077-4c8d-ab84-519464c9ab3c	44.92148052	33.14807227	0	0	2	0	0	0	0	0	3	4.810803	-2.4675736	5.20101	4.3011155	
27	3c0cd84f-1dc8-4bae-a45f-e253ce226e67	63.64061535	24.17427963	0	0	3	0	0	0	2	0	4	18.165604	-5.485276	-9.144451	0.7021639	
28	6d2a2af7-e939-4e47-93ec-30b1127bd6fb	83.80574026	41.26758372	0	0	1	0	0	2	0	0	3	-1.5151603	0.90826097	9.179214	6.8117775	
29	3915d2d0-06be-4b56-8e51-a75c8f16ada	72.60810812	42.10212269	0	0	4	0	0	1	0	0	5	-10.342437	-12.610694	5.0780873	-4.3618407	
30	14d5cfc8-8eb9-485b-ba04-10035da076f3	103.2039873	80.82971616	0	0	2	0	0	0	0	0	2	-3.5230439	-1.6241088	19.378725	15.017331	
31	4cae3df1-d4d4-4a35-9ff2-20a4847dc5c	100.4725029	69.01419792	0	0	1	0	0	0	0	1	1	-0.5729423	0.30208907	13.575023	3.9706492	
32	d2a3b7ba-b1fc-48d8-b65b-dc460f6dd0bf	81.84680218	38.99788869	0	0	2	0	0	0	0	0	3	0.15099438	-0.03437154	11.493493	12.451038	
33	42d0162c-b769-474a-a55c-a6eaa9f9c5ed	82.00478974	44.1786372	0	0	2	0	0	0	0	0	3	-5.5327506	2.087857	-4.5158987	15.484574	
34	7b996af1-21c1-49ee-9054-841ced9a9e0	83.48355324	41.32312434	0	0	3	0	0	0	3	0	6	-12.209767	-7.8533387	7.906246	4.2505045	
35	9b2180c8-09fe-4a76-af6d-e11d27e34fc2	81.60677035	57.36295879	0	0	8	0	0	0	0	8	8	-10.502587	-9.225577	3.9645512	6.7924576	
36	0274b515-5c01-4483-b02e-74f6c778707e	82.61357784	56.96513589	5	1	9	0	0	0	0	2	0	17	-14.6952915	-13.684507	-3.6973665	-14.871934
37	920a549f-c1c1-4152-b7c1-e9b637136338	66.19880353	42.62317517	0	0	7	0	0	0	0	0	7	-7.6191783	-1.6244287	5.8724613	10.3705515	
38	visual_features_final																

Appendix D — Notebook 4: Multimodal Dataset Assembly

Purpose: Integration of all feature groups from Notebooks 1, 2, and 3 into the final unified multimodal dataset of 1,899 clips and 48 features.

Key steps implemented:

1. Load of prepared metadata, ego-motion features, and visual features from Google Drive
2. Drop of chunk column from ego-motion dataset before merge to prevent column conflicts
3. Sequential inner joins on clip_id — metadata with ego-motion, then with visual features
4. Engineering of aggressive_score using StandardScaler-normalised components to prevent scale dominance
5. Engineering of env_complexity as a weighted sum of detected object counts
6. Engineering of object_density as total objects normalised by environmental complexity
7. Creation of lighting_condition categorical variable from frame brightness and is_poor_lighting binary flag

8. Programmatic check confirming anomaly_score is absent from the assembled dataset before saving
9. Final dataset saved as final_dataset.csv — 1,899 rows $\times$ 48 features

Critical design decision: The programmatic absence check for anomaly_score in Step 5 ensures that no leakage from the Notebook 1 sampling process has propagated into the modelling dataset. This check is a mandatory quality gate before any data is saved.

Output: final_dataset.csv — 1,899 rows $\times$ 48 features

Google Colab path: /content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/04_dataset/

Key Output Screenshots:

D.1 — Dataset Assembly Merge Output

Figure D.1 — Notebook screenshot 27

```

After ego + visual merge: (989, 40)
After metadata merge: (989, 50)

Columns: ['clip_id', 'avg_speed', 'max_speed', 'avg_acceleration', 'max_acceleration', 'avg_jerk', 'max_jerk', 'braking_events', 'hard_braking_events', 'brightness', 'contrast', 'person_count', 'b:

Missing values per column:
radar_config    284
dtype: int64

First 5 rows:

```

	clip_id	avg_speed	max_speed	avg_acceleration	max_acceleration	avg_jerk	max_jerk	braking_events	hard_braking_events	brightness	...	chunk	country	month	hour_of_day	platform_classification
0	31856298-2960-4d45-a906-1b489066c452	15.474187	17.625619	0.006037	2.189002	-0.007385	30.761935	7	0	90.334996	...	3	United States	1	0	hyperion
1	4ed4b27e-6b5a-4468-a8a6-b54feec34c6f	11.680353	13.049555	0.019714	2.616180	0.126845	47.692112	21	2	87.408020	...	3	United States	1	0	hyperion
2	784f8f77-ec64-4896-8560-19b3c686d2cd	8.510242	14.600693	-0.339619	8.531664	0.266604	50.000000	55	7	51.222828	...	3	United States	1	0	hyperion
3	34860342-d420-4ec4-8353-6d79ae77f63c	19.600180	20.694840	0.047169	1.894011	-0.155498	43.172932	7	0	92.022001	...	3	United States	1	0	hyperion
4	42388dff-6173-428e-9d41-81617a18a827	17.825127	19.037635	-0.244031	1.762046	-0.103656	34.132186	7	0	90.761515	...	8	United States	1	0	hyperion

```

5 rows x 50 columns

-----
Step 2 complete: Datasets merged, anomaly_score excluded.
-----

```

D.2 — Final Dataset Shape, Column List and Anomaly Score Absent Check

Figure D.2 — Notebook screenshot 28

```
*** Missing values before cleaning:
radar_config      284
dtype: int64

Rows dropped: 284
Final dataset shape: (705, 55)

Confirmed: anomaly_score NOT present. No leakage.

Final columns:
['clip_id', 'avg_speed', 'max_speed', 'avg_acceleration', 'max_acceleration', 'avg_jerk', 'max_jerk', 'braking_events', 'hard_braking_events', 'brightness', 'contrast', 'person_count', 'bicycle_co

Step 5 complete: Dataset cleaned. No target leakage present.
```

D.3 — Engineered Features Summary

Figure D.3 — Notebook screenshot 29

```
Feature engineering complete.

aggressive_score stats:
count      9.890000e+02
mean      -4.598052e-16
std        2.822428e+00
min       -5.102318e+00
25%       -1.886469e+00
50%       -4.970707e-01
75%        1.175929e+00
max        1.460927e+01
Name: aggressive_score, dtype: float64

env_complexity stats:
count      989.000000
mean         5.969666
std         6.958458
min         0.000000
25%         1.000000
50%         4.000000
75%         9.000000
max        54.500000
Name: env_complexity, dtype: float64

-----
Step 3 complete: Engineered features created with proper scaling.
-----
```

D.4 — Final Dataset Export Confirmation

Figure D.4 — Notebook screenshot 30

```
*** Saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/04_dataset/final_dataset.csv
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/04_dataset/final_dataset.parquet
```

Figure D.5 — Notebook screenshot 31

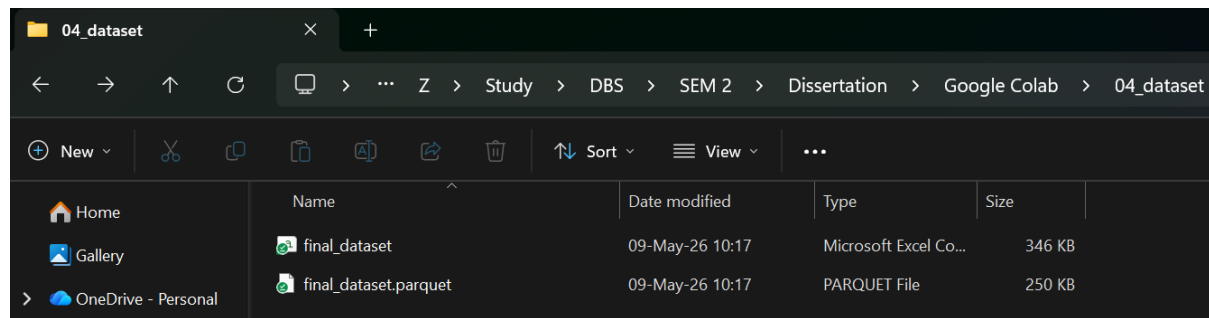


Figure D.6 — Notebook screenshot 32

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	clip_id	avg_speed	max_speed	avg_acceleration	max_acceleration	avg_jerk	max_jerk	braking_events	hard_braking_events	brightness	contrast	person_count	bicycle_count	car_count	motorcycle_count	bus_count
2	31856298-2960-4d45-a906-1b489066c452	15.4741869	17.62561903	0.006036733	2.189001524	-0.007384594	30.76193504	7	0	90.33499614	41.24066878	0	0	0	0	0
3	4ed4b27e-6b5a-4468-a8a6-b54fec34c6f	11.68035332	13.04954607	0.019713545	2.616179884	0.126844706	47.69211247	21	2	87.40801987	45.17522293	0	0	0	0	0
4	784f8f77-e0d4-489e-8560-19b3c868d2cd	8.51024225	14.60069344	-0.339618867	8.531664162	0.26660449	50	55	7	51.22282841	26.85418259	0	0	1	0	0
5	34860342-4d20-4ec4-8353-6d79ae77883c	19.60017991	20.59484045	0.047169033	1.894011088	-0.155497541	43.17293195	7	0	92.02200135	31.03705756	0	0	3	0	0
6	42388df6-6173-428e-9d41-81617a18a827	17.8251271	19.03763532	-0.244030936	1.762045637	-0.103656237	34.13218629	7	0	90.76151524	42.72353452	0	0	1	0	0
7	91679561-a777-44ea-b910-25a1735bea49	18.44484249	26.54388985	0.65306416	6.355669729	0.347005922	30	3	0	89.73710021	39.2658904	0	0	0	0	0
8	7f6572d4-6b40-4b7a-a3bc-6d3c5a13f468	5.054314181	15.23867905	0.767373017	3.394260432	0.132630361	28.85385936	1	0	79.90117814	34.24538814	0	0	5	0	0
9	0a701173-f30e-4a84-a938-6d15d4cf9c2f	12.44631588	14.16161143	0.080673703	1.550544765	-0.015934431	29.37846027	0	0	87.22033227	41.04135536	0	0	1	0	0
10	5e3a0875-51a3-49e7-81ad-3f1d2329d27	18.8851334	21.66718211	0.123568637	4.113242452	0.099646248	50	18	0	90.636386	47.14360669	0	0	6	0	0
11	4049f1e4-dea6-47a0-ac93-b4be13e886de	10.79854881	15.24269555	-0.441913413	1.696295476	-0.097503706	23.74316622	7	0	94.19588059	36.59400411	0	0	3	0	0
12	3f709ac5-34f3-436d-8c4b-74838bab242a	7.07864904	12.84844004	0.639836193	3.470962095	-0.037236308	42.47049887	1	0	95.40478884	45.50870825	1	2	5	0	0
13	3d45c5a3-29e4-4c66-bda3-41da3231fab2	19.86035891	20.40231609	0.033462363	2.721740184	-0.137786253	50	6	0	98.9781086	55.66098196	0	0	3	0	0
14	2696bb1c-bfef-4318-9d02-cffa41d3120a	12.66354479	14.21274692	0.158633752	6.145147678	0.001001815	50	13	3	73.13565538	24.2740099	0	0	10	0	0
15	9dad2d41-e027-4ac7-a5b9-2cebfa55539	14.96928553	15.86502121	0.084539187	4.06341039	0.200633079	50	10	3	82.35802276	25.78326848	0	0	3	0	0
16	178bd5b3-ef04-4eb0-be13-486e5d660c05	3.344786294	7.091265362	-0.106652028	2.076202777	-0.025080332	34.60707068	7	1	88.95560716	29.72932564	0	0	8	0	0
17	96425bd1-bfd4-4b4f-ae95-2be804b86152	5.781908483	10.0604788	0.123528136	2.660846865	-0.06889095	32.7591787	18	2	80.383356	31.90686019	0	0	4	0	0
18	0a400f1b-5d4e-45e3-8c0d-a55c1067de93	7.55204959	10.46626343	-0.211866575	2.740471858	0.271045079	50	36	0	91.1610706	43.73067676	0	0	4	0	0
19	1be8937c-f738-4868-9d0e-c20383c9b6d4	17.04401563	19.60145684	0.117375368	2.664623363	0.012319289	33.10382365	0	0	90.79519387	39.7704819	0	0	1	0	0
20	69dcdb81-877b-4ba8-8420-a7c12370b52a	15.45804877	16.34483158	0.052416388	4.946035002	0.461361872	50	25	4	93.69662037	45.39848663	0	0	2	0	0
21	5ab8a10a-25be-4b6d-bbf2-59701c3cf1d7	8.570431246	8.437373185	-0.024568249	2.135883178	0.057557523	40.38010281	27	0	98.97284144	42.94501694	0	0	6	0	0
22	51264d9d-c9d8-43a9-9834-173a2fc396e2	14.17742756	14.6757463	-0.195422918	1.982798328	-0.448231976	36.02203142	13	3	82.8750815	42.36056774	0	0	4	0	0
23	44866f90-fe22-424d-b420-3535ff2b852	12.53582877	15.80377782	-0.577804717	5.622308894	0.296966688	50	62	16	53.50788194	33.73358374	0	0	0	0	0
24	5067533c-e396-44cc-b5e7-3f642a060469	8.06895911	13.35718837	0.176478768	2.691065122	0.091331511	31.26026065	38	3	57.51583767	34.72810514	0	0	1	0	0
25	55c033d0-1197-411b-a3ff-7ede68ebc2a6	11.74613168	13.63228217	0.073471787	1.854623513	0.134275302	39.17581464	3	0	94.03687838	42.3643869	0	0	6	0	0
26	af3d47f5-3c4a-4557-a77b-8c7c95af525a	15.3270971	15.73326823	0.0235057518	2.591644916	0.192984209	50	12	0	92.42129811	50.37980112	0	0	7	0	0
27	7c2a872c-a7f7-481b-9e63-04f717639151	29.35804861	29.77443952	0.003600422	5.859121353	-0.29625584	50	20	2	84.61364632	34.60837155	0	0	2	0	0
28	0a0762e3-ca63-4a55-ac76-00b313c3cf58	15.2707208	16.72189311	0.142153718	3.368228339	0.090253697	50	6	2	94.96919608	42.91100266	0	0	3	0	0
29	ac7d5141-e2a7-47fc-bca5-22f9d567f13e	5.335917785	10.67401426	0.407484874	3.073532262	0.09079578	32.2853921	7	0	65.01307581	32.83639974	0	0	4	0	0
30	5b6d1f36-03d6-47ad-9d2b-9da452acce77	13.14153384	13.64910522	0.050469287	1.433200739	-0.039902867	24.57057597	1	0	63.74345293	26.20553328	0	0	4	0	0
31	06c082b2-13f5-45e8-9410-45be4c63ce08	4.68137675	8.018708296	0.348487424	2.380091011	-0.065052932	27.55759453	2	0	93.5883941	40.09008977	3	0	11	0	0
32	1bd0a6d5-09c2-4e26-bd19-08cb11d48ff2	25.2868547	27.08797542	-0.1533867194	10.07734431	-0.472930816	50	27	4	76.60475694	28.26362575	0	0	3	0	0
33	c51343b3-a00a-4c0a-acd3-78e5afafac9	13.73154542	14.64565258	0.110548691	2.561457336	-0.100527524	48.69952532	2	0	87.07907552	38.25883391	0	0	0	0	0
34	a2e7c1fa-d57c-4748-899c-bf48d6144db3	15.37715541	16.76634216	-0.090581213	3.829695727	-0.036671526	50	14	0	88.80521653	40.03332248	0	0	12	0	0
35	87a4a6aa-f987-4015-b1d9-7cf3371cb888	11.96273774	15.83925149	-0.815350688	1.530926493	-0.165575279	35.01188937	55	9	92.33548611	35.78214696	0	0	0	0	0
36	7eb4c075-d3ba-4f87-903a-54767d6d423	17.72390074	18.29559772	0.006165469	4.26450585	0.32350728	50	10	2	89.00560667	37.6846612	0	0	4	0	0
37	d69add23-cb5c-406a-9ca0-269600247dbd	17.93436201	19.12190573	0.016146431	2.035721289	0.059114293	43.84628562	12	0	98.52368875	40.80925514	0	0	3	0	0
38	ac97d7dd-4189-a87a-bb6e0503daad	3.925800281	9.110308317	-0.41919647	3.006301819	-0.208059576	50	31	8	86.37541184	38.11344266	0	0	7	0	0

Appendix E — Notebook 5: Modelling and Evaluation

Purpose: Complete modelling, evaluation, and explainability pipeline — proxy risk labelling, hyperparameter tuning, XGBoost training, cross-validation, SHAP explainability, and four-way ablation study.

Key steps implemented:

1. Load of final_dataset.csv and 80/20 stratified train-test split before any label creation — train size 564, test size 141
2. Isolation Forest proxy risk labelling fitted exclusively on training set — 141 high-risk clips (25.0%) and 423 low-risk clips (75.0%) assigned
3. Three-strategy proxy label validation — Mann-Whitney U tests, visual feature significance, Cohen's d effect sizes
4. Feature preparation — LabelEncoder for lighting_condition fitted on train only, StandardScaler fitted on train applied to both sets
5. Baseline model evaluation — Dummy Classifier (0.500), Logistic Regression (0.879), Random Forest (0.971)
6. Two-phase gap-aware hyperparameter optimisation — Phase 1 (CV 0.9450, gap 0.0550), Phase 2 (CV 0.9455, gap 0.0545)
7. Final XGBoost model training with GPU acceleration — ROC-AUC 0.9387 at threshold 0.4
8. Stratified five-fold cross-validation with per-fold fresh Isolation Forest labels — mean CV ROC-AUC 0.9577, std 0.0099
9. Threshold analysis across 0.3, 0.4, 0.5, 0.6 — threshold 0.4 selected
10. SHAP explainability — TreeExplainer SHAP values, beeswarm, bar, waterfall, group importance
11. Feature group importance analysis across seven groups
12. Four-way ablation study — Full (0.9387), No CNN+ViT (0.9554), No Context (0.8597), Behavioural Only (0.8369)

Critical design decision: Cross-validation uses per-fold fresh Isolation Forest label creation — each fold independently fits its own Isolation Forest on fold training data, derives its own threshold, and applies it to fold validation data. This eliminates proxy label leakage between folds and represents a more rigorous evaluation than standard cross-validation with pre-assigned labels.

Output directory: /content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/05_modelling_outputs/

Saved artefacts: final_xgboost_model.pkl, model_comparison.csv, feature_importance.csv, feature_group_importance.csv, shap plots, ablation plots, tuning plots, proxy validation plots

Key Output Screenshots and Figures:

E.1 — Train-Test Split Confirmation and Proxy Label Distribution

Figure E.1 — Notebook screenshot 33

```

... Train size: 564
    Test size: 141

Risk label distribution - TRAIN:
risk_label
0    423
1    141
Name: count, dtype: int64
risk_label
0    75.0
1    25.0
Name: proportion, dtype: float64

Risk label distribution - TEST:
risk_label
0    113
1     28
Name: count, dtype: int64
risk_label
0    80.14
1    19.86
Name: proportion, dtype: float64

Anomaly score threshold used: 0.0

-----
Step 2 complete: Risk labels created on training data only.
Same model and threshold applied to test data - no leakage.
-----

```

E.2 — Mann-Whitney U Validation Results

Figure E.2 — Notebook screenshot 34

```

... High risk clips: 169
    Low risk clips: 536

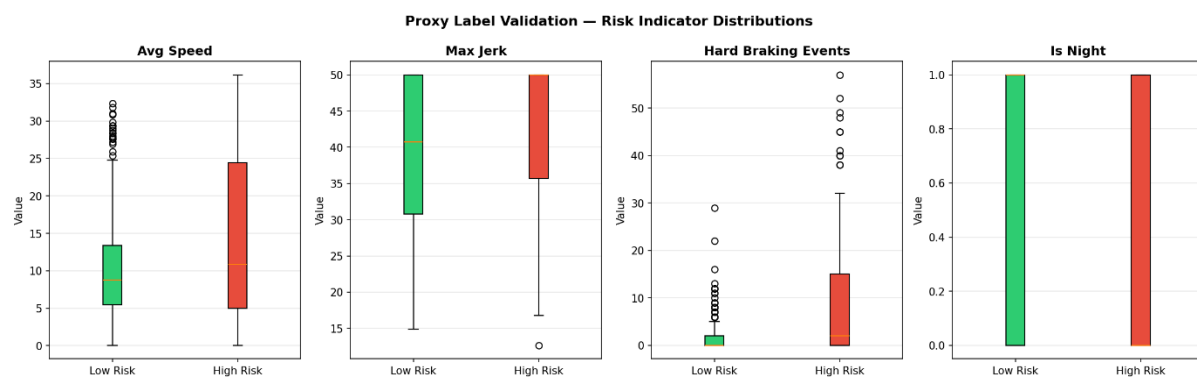
Proxy Label Validation - Statistical Comparison:

```

	Feature	High Risk Mean	Low Risk Mean	Difference	Direction	Mann-Whitney U	p-value	Significant
0	avg_speed	14.2730	10.3103	3.9628	Higher in High Risk	52553.0	0.001661	Yes
1	max_speed	18.4354	13.4271	5.0083	Higher in High Risk	58857.0	0.000000	Yes
2	avg_jerk	-0.0017	0.0027	-0.0044	Lower in High Risk	48215.0	0.205530	No
3	max_jerk	42.6895	39.3517	3.3378	Higher in High Risk	54493.0	0.000035	Yes
4	hard_braking_events	9.0000	1.3787	7.6213	Higher in High Risk	64403.5	0.000000	Yes
5	is_night	0.4024	0.7295	-0.3271	Lower in High Risk	30476.5	0.000000	Yes
6	is_peak_hour	0.5030	0.1399	0.3630	Higher in High Risk	61734.5	0.000000	Yes

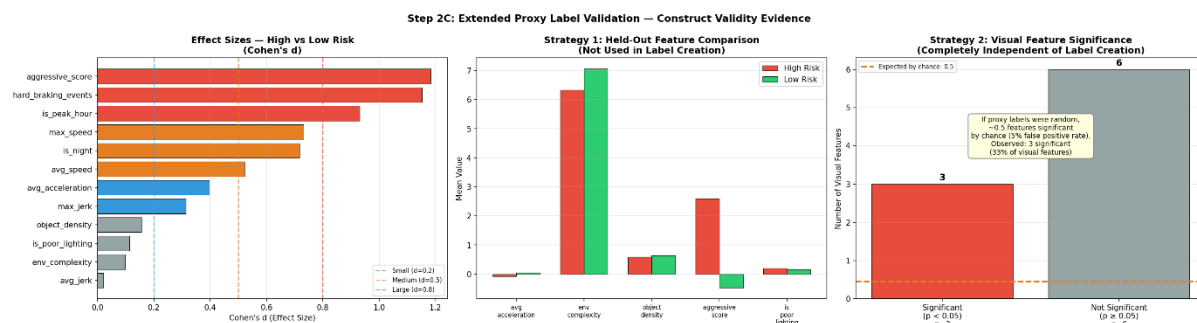
E.3 — Proxy Label Validation Box Plots

Figure E.3 — Proxy label validation: distribution comparison across average speed, maximum jerk, hard braking events, and night driving between high-risk and low-risk clips (Step 2B)



E.4 — Extended Proxy Label Validation

Figure E.4 — Extended proxy label validation: Cohen's d effect sizes, held-out feature comparison, and visual feature significance across three independent strategies (Step 2C)



E.5 — Feature Preparation Confirmation

Figure E.5 — Notebook screenshot 35

```
... lighting_condition encoded successfully.  
Classes: ['good', 'moderate', 'poor']  
  
X_train shape: (564, 48)  
X_test shape: (141, 48)  
  
Train class distribution:  
risk_label  
0    423  
1    141  
Name: count, dtype: int64  
  
Test class distribution:  
risk_label  
0    113  
1     28  
Name: count, dtype: int64  
  
Feature group sizes:  
Ego/Behavioural: 8  
Visual (brightness/contrast): 2  
YOLO Object: 9  
Context/Time: 6  
CNN: 10  
ViT: 10  
  
-----  
Step 3 complete: Feature matrices prepared.  
lighting_condition encoded and included.  
-----
```

E.6 — Baseline Model Classification Reports *Screenshot the Step 5A, 5B, and 6 outputs showing classification reports for Dummy Classifier (ROC-AUC 0.500), Logistic Regression (ROC-AUC 0.879), and Random Forest (ROC-AUC 0.971).*

Figure E.6 — Notebook screenshot 36

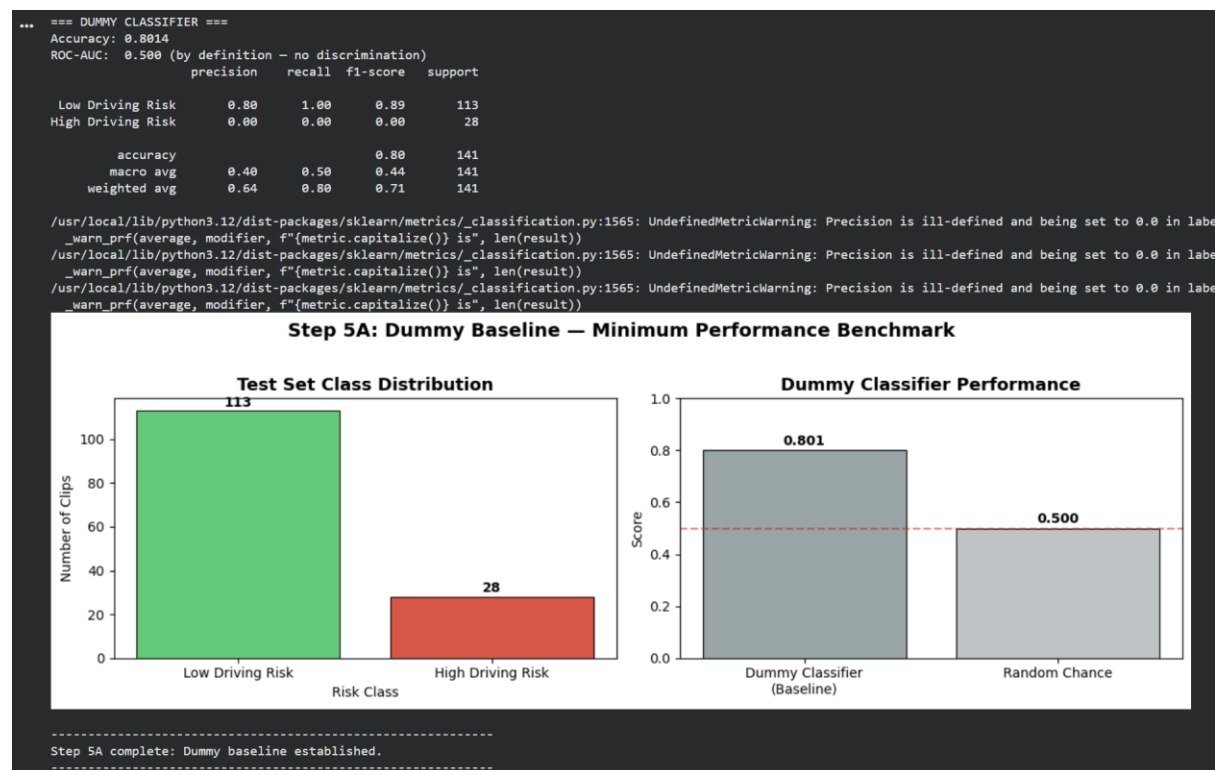


Figure E.7 — Notebook screenshot 37

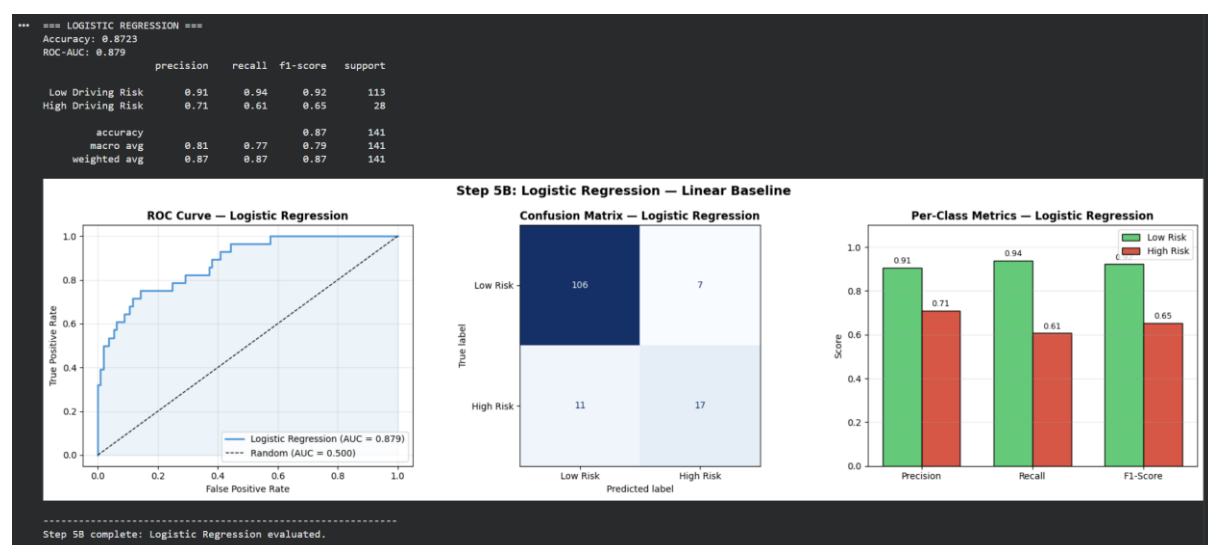
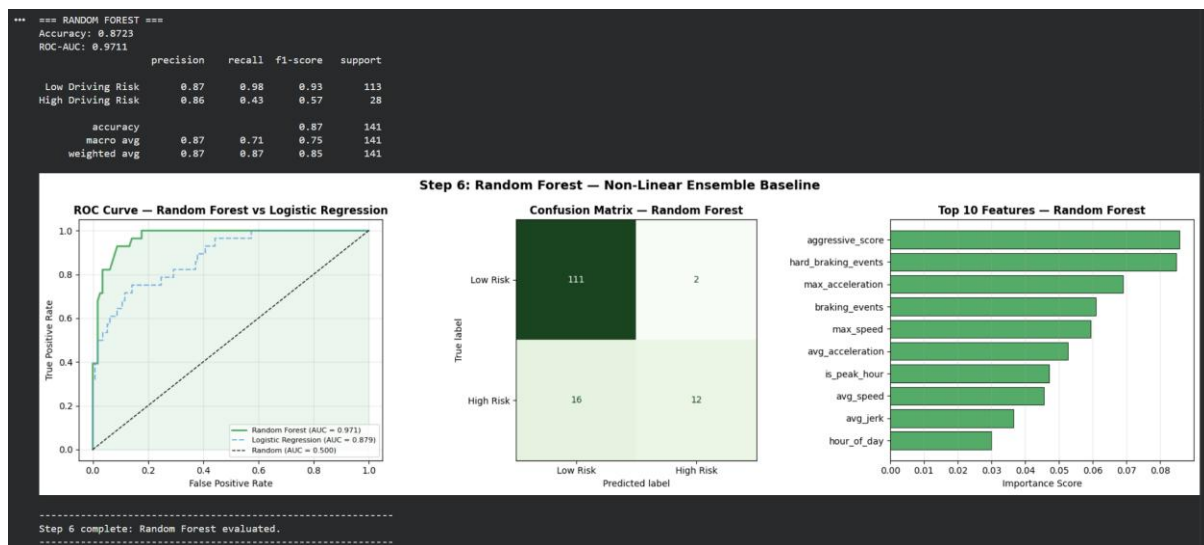
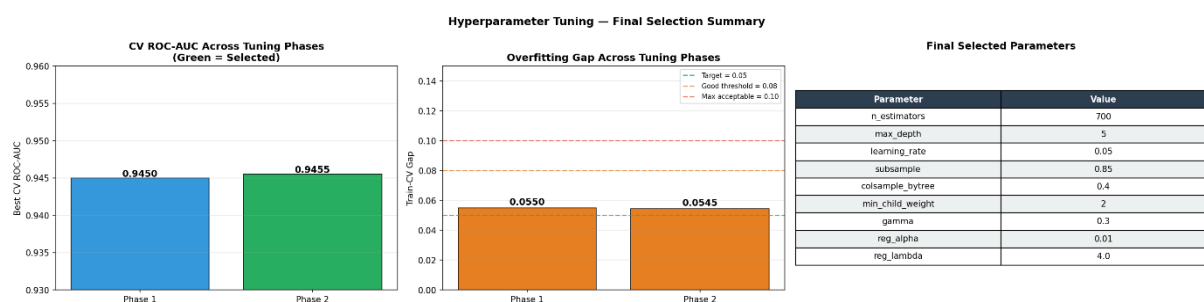


Figure E.8 — Notebook screenshot 38



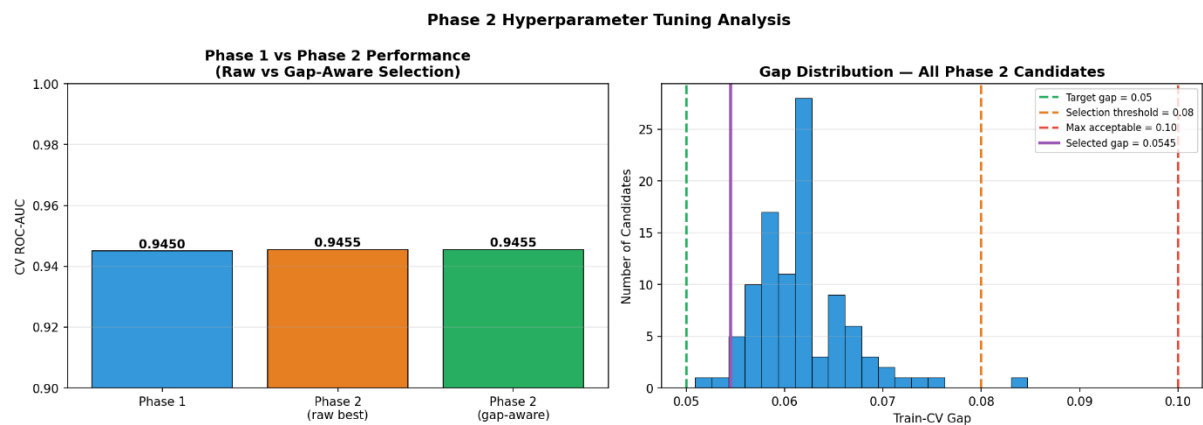
E.7 — Hyperparameter Tuning Final Selection

Figure E.9 — Hyperparameter tuning final selection: CV ROC-AUC across Phase 1 and Phase 2, overfitting gap comparison, and final selected parameter values (Step 6C)



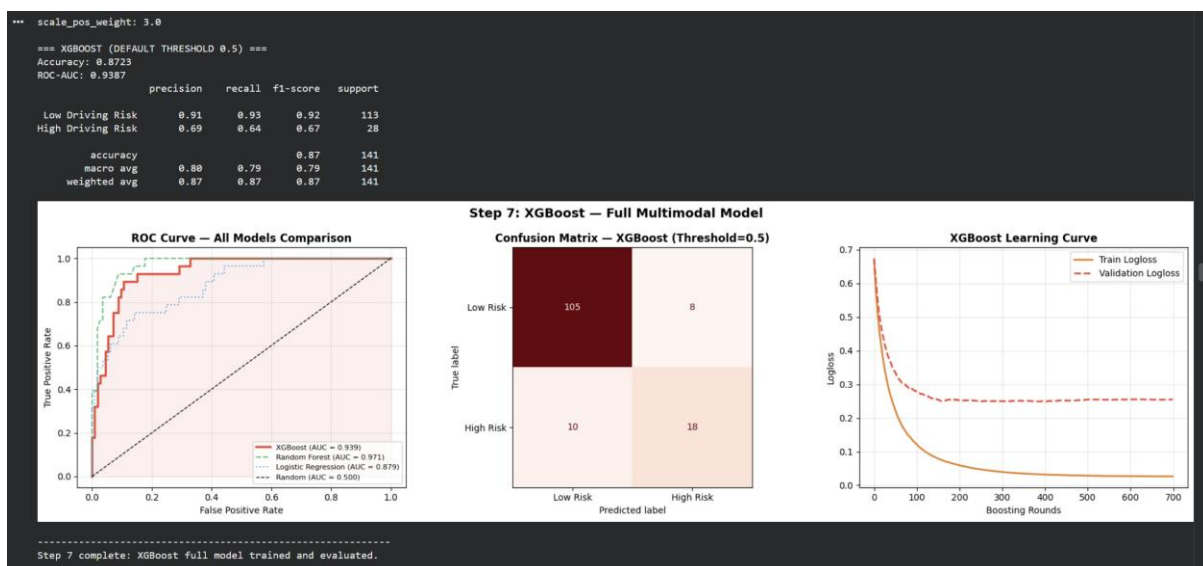
E.8 — Phase 2 Tuning Gap Analysis

Figure E.10 — Phase 2 tuning analysis: phase comparison and gap distribution across all 100 candidates confirming gap-aware selection at 0.0545 (Step 6B)



E.9 — XGBoost Final Model Classification Report

Figure E.11 — Notebook screenshot 39



E.10 — Cross-Validation Per-Fold Results

Figure E.12 — Notebook screenshot 40

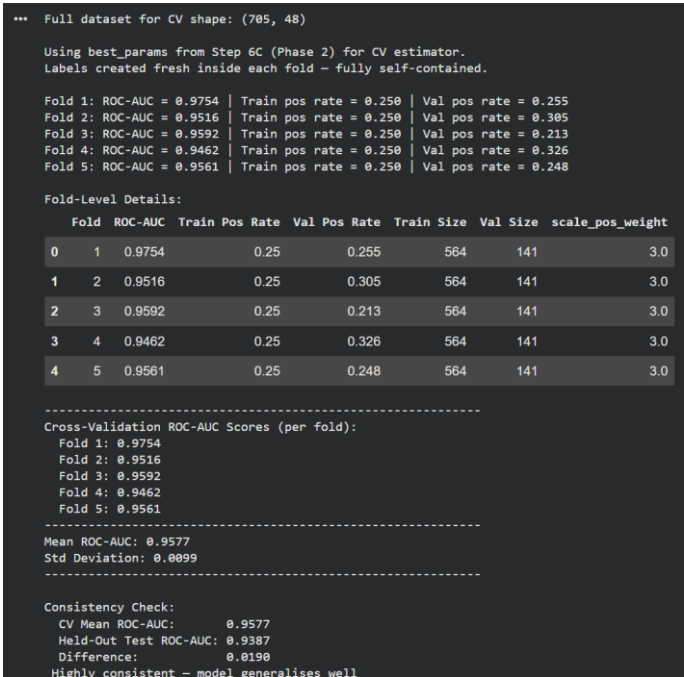
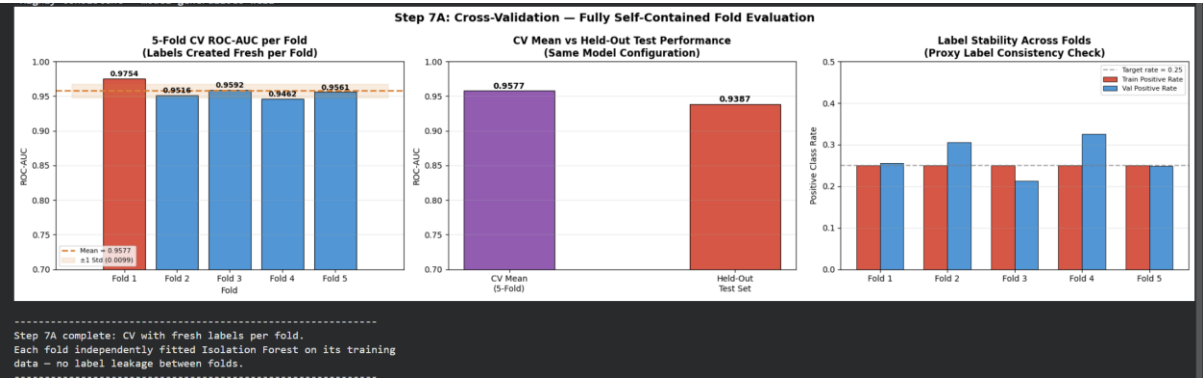


Figure E.13 — Notebook screenshot 41



E.11 — Threshold Analysis Table

Figure E.14 — Notebook screenshot 42

```
=== Threshold: 0.3 ===
```

	precision	recall	f1-score	support
Low Driving Risk	0.97	0.88	0.93	113
High Driving Risk	0.66	0.89	0.76	28
accuracy			0.89	141
macro avg	0.81	0.89	0.84	141
weighted avg	0.91	0.89	0.89	141

```
=== Threshold: 0.4 ===
```

	precision	recall	f1-score	support
Low Driving Risk	0.94	0.91	0.92	113
High Driving Risk	0.68	0.75	0.71	28
accuracy			0.88	141
macro avg	0.81	0.83	0.82	141
weighted avg	0.88	0.88	0.88	141

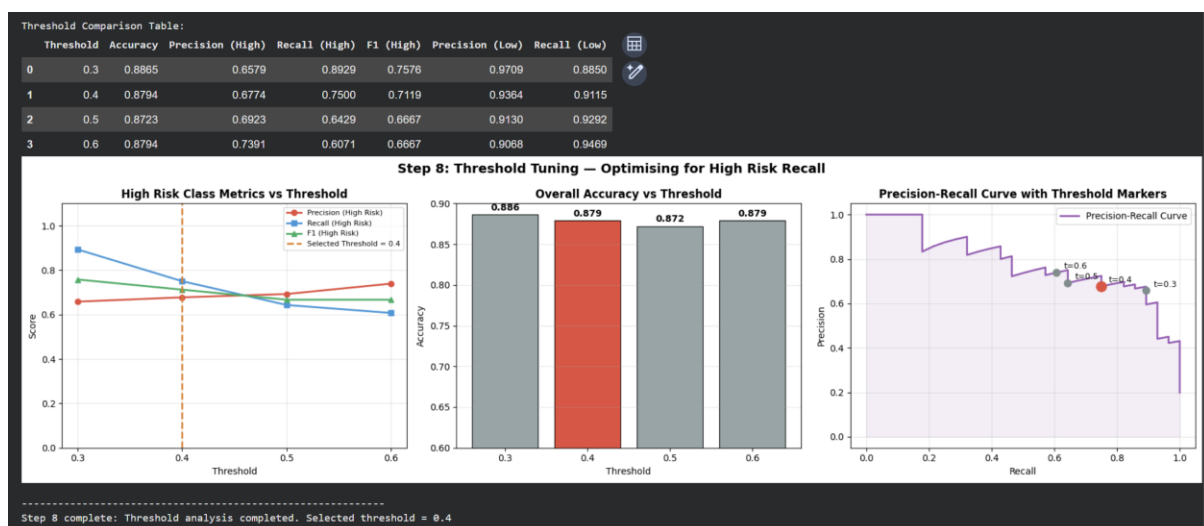
```
=== Threshold: 0.5 ===
```

	precision	recall	f1-score	support
Low Driving Risk	0.91	0.93	0.92	113
High Driving Risk	0.69	0.64	0.67	28
accuracy			0.87	141
macro avg	0.80	0.79	0.79	141
weighted avg	0.87	0.87	0.87	141

```
=== Threshold: 0.6 ===
```

	precision	recall	f1-score	support
Low Driving Risk	0.91	0.95	0.93	113
High Driving Risk	0.74	0.61	0.67	28
accuracy			0.88	141
macro avg	0.82	0.78	0.80	141
weighted avg	0.87	0.88	0.87	141

Figure E.15 — Notebook screenshot 43



E.12 — Final Model at Threshold 0.4

Figure E.16 — Notebook screenshot 44

```

... === FINAL MODEL (THRESHOLD = 0.4) ===
Accuracy: 0.8794326241134752
          precision    recall  f1-score   support

   Low Driving Risk      0.94      0.91      0.92       113
   High Driving Risk      0.68      0.75      0.71        28

         accuracy
    macro avg      0.81      0.83      0.82       141
    weighted avg      0.88      0.88      0.88       141

-----
Step 9 complete: Final model selected.
-----

```

E.13 — Gain-Based Feature Importance Top 15

Figure E.17 — Notebook screenshot 45

```

... Top 15 Important Features:

```

	feature	importance
0	is_peak_hour	0.093200
1	is_night	0.092568
2	aggressive_score	0.072130
3	hard_braking_events	0.061131
4	max_acceleration	0.047233
5	braking_events	0.034380
6	max_speed	0.032260
7	hour_of_day	0.030587
8	avg_acceleration	0.028946
9	avg_speed	0.027354
10	avg_jerk	0.025548
11	vit_feature_8	0.025030
12	max_jerk	0.023779
13	truck_count	0.019392
14	cnn_feature_4	0.019282

```

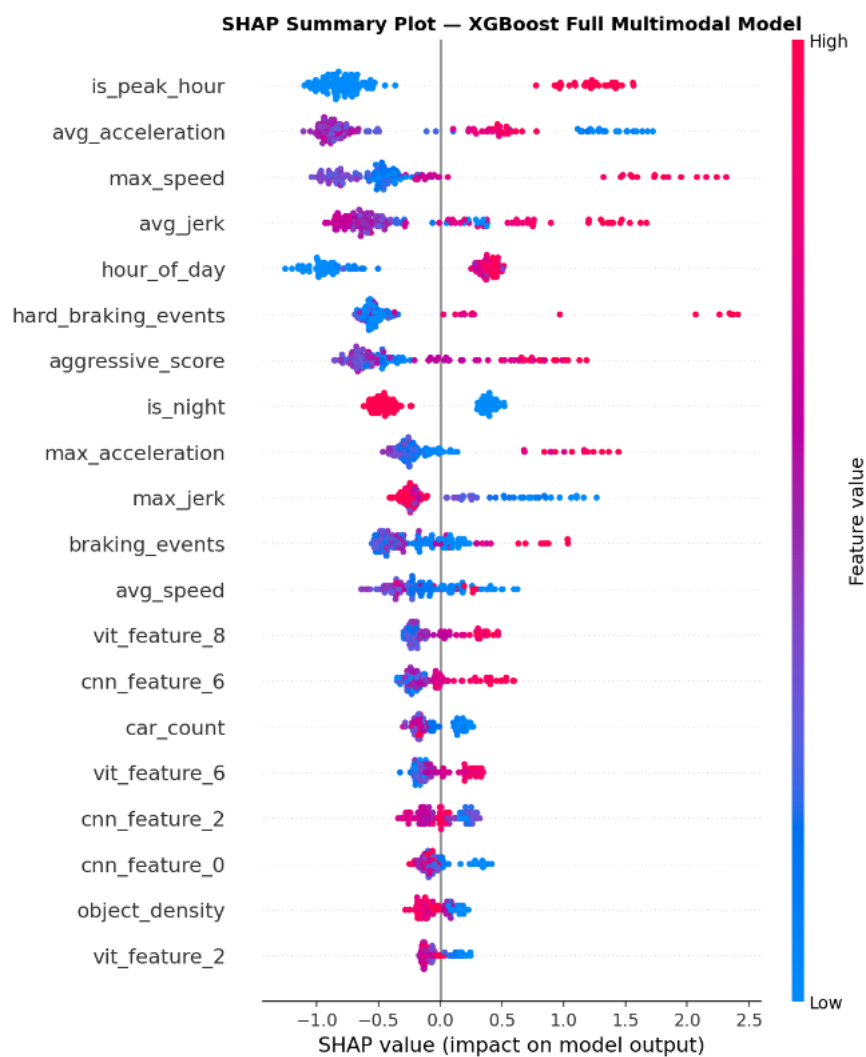
Total features ranked: 48

-----
Step 12 complete: Feature importance extracted.
-----

```

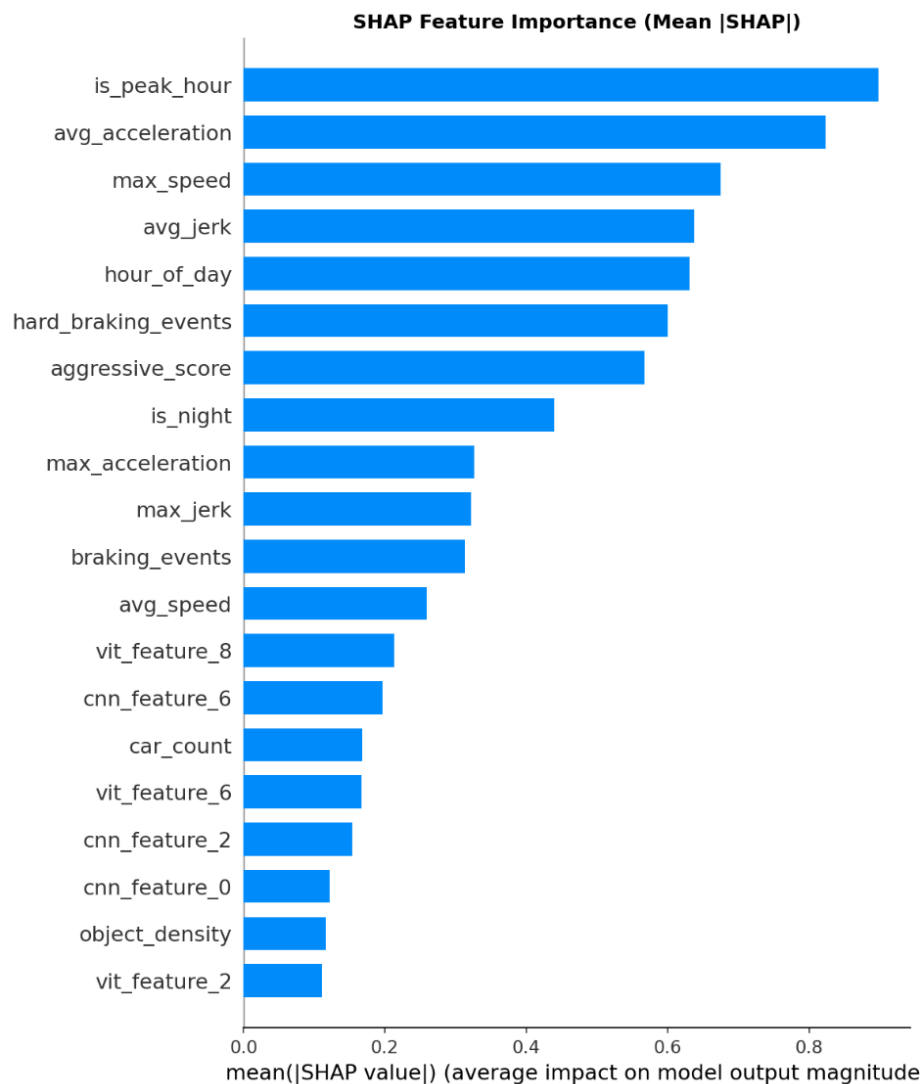
E.14 — SHAP Summary Beeswarm Plot

Figure E.18 — SHAP summary beeswarm: individual feature contributions across all 141 test clips ranked by mean absolute SHAP value (Step 13B)



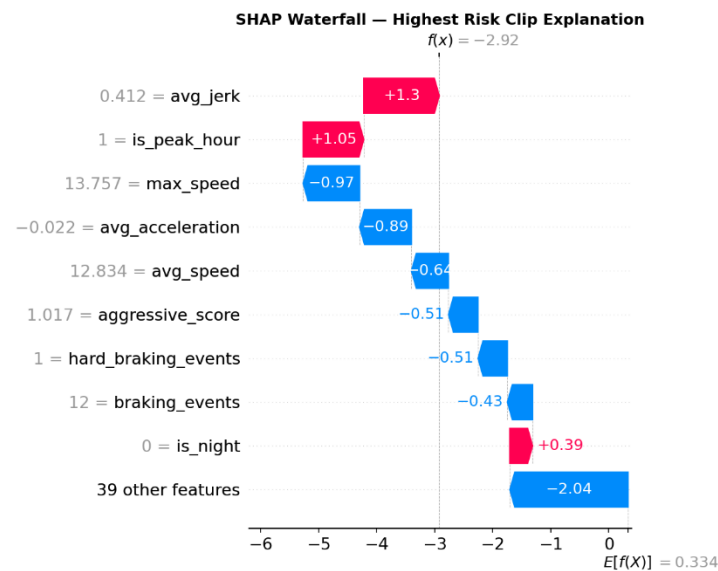
E.15 — SHAP Feature Importance Bar Chart

Figure E.19 — SHAP feature importance: top 20 features by mean absolute SHAP value, `is_peak_hour` confirmed as most influential (Step 13B)



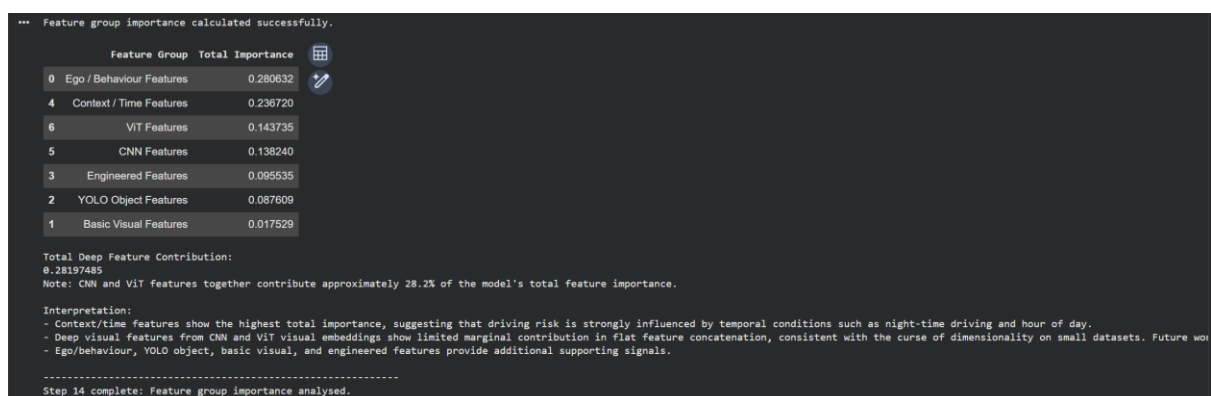
E.16 — SHAP Waterfall Plot

Figure E.20 — SHAP waterfall: highest-risk clip explanation — avg_jerk (+1.3) and is_peak_hour (+1.05) as strongest contributors toward high risk (Step 13B)



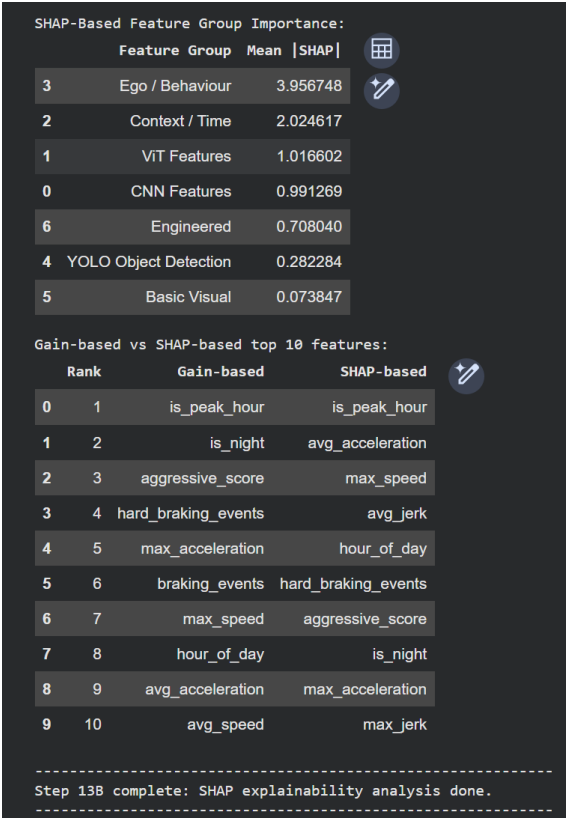
E.17 — Feature Group Importance Table

Figure E.21 — Notebook screenshot 46



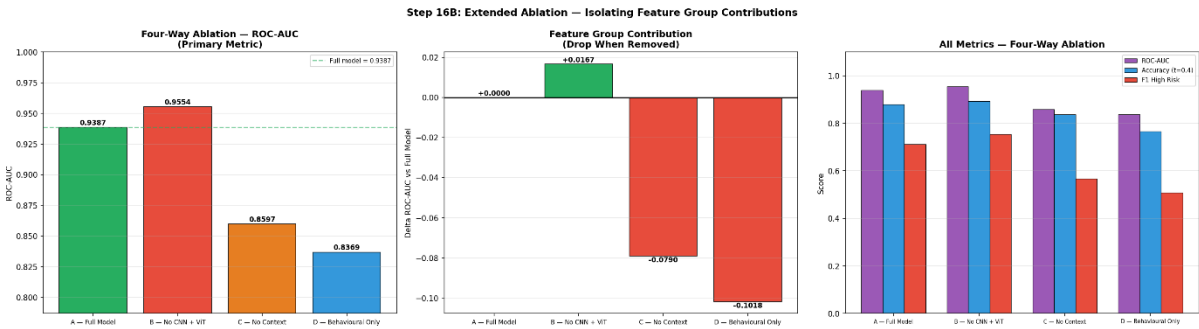
E.18 — SHAP Group Importance and Gain vs SHAP Comparison

Figure E.22 — Notebook screenshot 47



E.19 — Four-Way Ablation Results

Figure E.23 — Four-way ablation: ROC-AUC comparison, delta from full model, and all-metrics grouped comparison (Step 16B)



E.20 — Final Model Comparison Table

Figure E.24 — Notebook screenshot 48

```
... === FINAL MODEL COMPARISON ===
```

	Model	ROC-AUC	Accuracy (t=0.4)	F1 High Risk (t=0.4)
0	Dummy Classifier	0.500	0.801	0.000
1	Logistic Regression	0.879	0.851	0.618
2	Random Forest	0.971	0.936	0.836
3	XGBoost — Full Multimodal	0.939	0.879	0.712
4	XGBoost — Ablation (No CNN+ViT)	0.955	0.894	0.754

E.21 — Model Export Confirmation

Figure E.25 — Notebook screenshot 49

```
... Modelling outputs saved successfully.

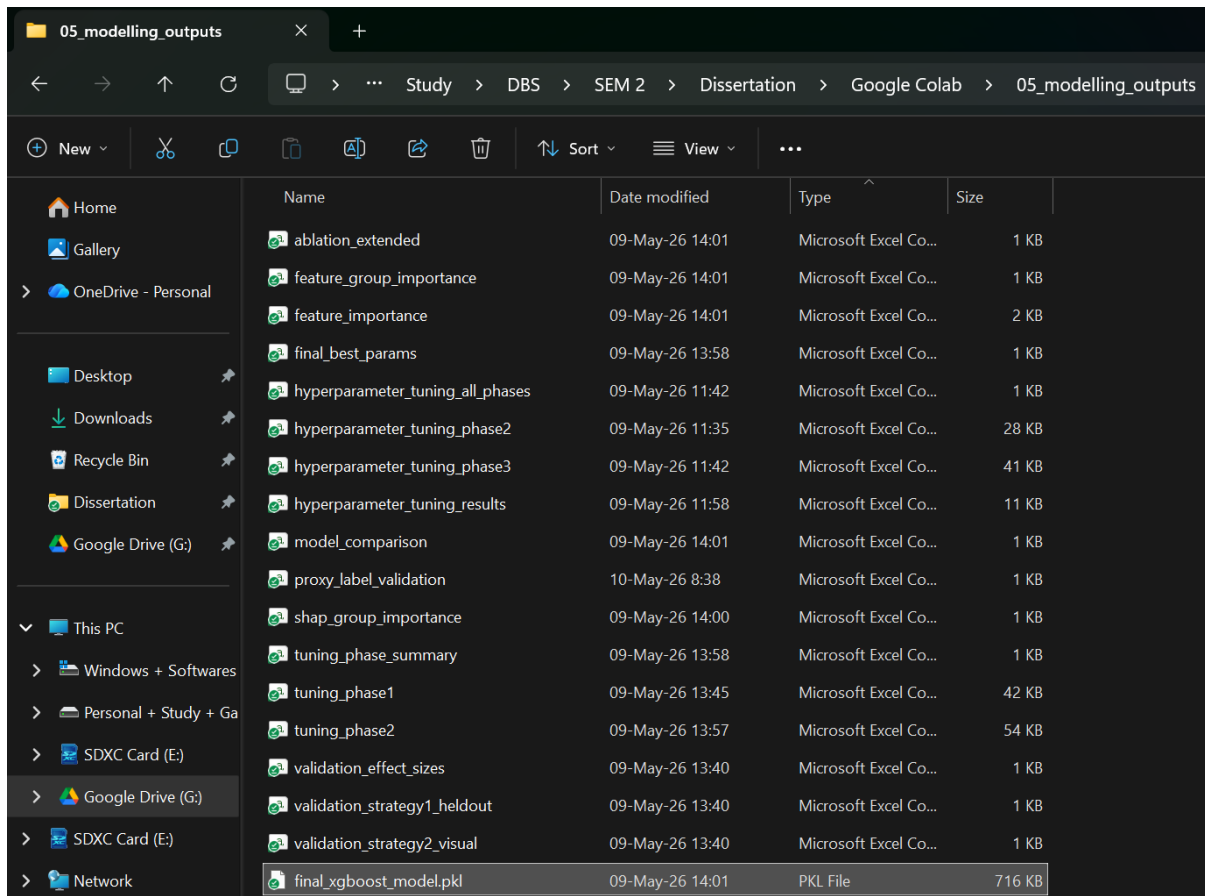
Model comparison saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/05_modelling_outputs/model_comparison.csv

Feature importance saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/05_modelling_outputs/feature_importance.csv

Feature group importance saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/05_modelling_outputs/feature_group_importance.csv

Final XGBoost model saved at:
/content/drive/My Drive/Z/Study/DBS/SEM 2/Dissertation/Google Colab/05_modelling_outputs/final_xgboost_model.pkl

-----
Step 18 complete: Modelling outputs saved.
-----
```

Figure E.26 — Notebook screenshot 50


Name	Date modified	Type	Size
ablation_extended	09-May-26 14:01	Microsoft Excel Co...	1 KB
feature_group_importance	09-May-26 14:01	Microsoft Excel Co...	1 KB
feature_importance	09-May-26 14:01	Microsoft Excel Co...	2 KB
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hyperparameter_tuning_phase3	09-May-26 11:42	Microsoft Excel Co...	41 KB
hyperparameter_tuning_results	09-May-26 11:58	Microsoft Excel Co...	11 KB
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tuning_phase_summary	09-May-26 13:58	Microsoft Excel Co...	1 KB
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tuning_phase2	09-May-26 13:57	Microsoft Excel Co...	54 KB
validation_effect_sizes	09-May-26 13:40	Microsoft Excel Co...	1 KB
validation_strategy1_heldout	09-May-26 13:40	Microsoft Excel Co...	1 KB
validation_strategy2_visual	09-May-26 13:40	Microsoft Excel Co...	1 KB
final_xgboost_model.pkl	09-May-26 14:01	PKL File	716 KB

Appendix F — Extended Model Metrics

This appendix presents the complete classification metrics, threshold sensitivity analysis, and cross-validation results for the final XGBoost model.

Table F.1 — Full Classification Report: XGBoost Final Model (Threshold = 0.4)

Class	Precision	Recall	F1-Score	Support
Low Risk	0.94	0.91	0.92	113
High Risk	0.68	0.75	0.71	28
Macro Average	0.81	0.83	0.82	141
Weighted Average	0.88	0.88	0.88	141

Table F.2 — Threshold Sensitivity Analysis: XGBoost Full Model

Threshold	Accuracy	Precision (High Risk)	Recall (High Risk)	F1 (High Risk)
0.3	0.8865	0.6579	0.8929	0.7576
0.4	0.8794	0.6774	0.7500	0.7119
0.5	0.8723	0.6923	0.6429	0.6667
0.6	0.8794	0.7391	0.6071	0.6667

Threshold 0.4 selected — highest recall for the high-risk class while maintaining acceptable precision and overall accuracy. Safety-oriented priority: missed high-risk scenarios carry greater consequence than false alarms.

Table F.3 — Cross-Validation Per-Fold Results (Per-Fold Fresh Labels)

Fold	ROC-AUC	Train Positive Rate	Validation Positive Rate
Fold 1	0.9754	0.250	0.255
Fold 2	0.9516	0.250	0.305
Fold 3	0.9592	0.250	0.213
Fold 4	0.9462	0.250	0.326
Fold 5	0.9561	0.250	0.248
Mean	0.9577	0.250	0.269
Std Dev	0.0099		

Consistent positive class rate of 0.250 across all training folds confirms the per-fold fresh labelling strategy produces stable proxy labels regardless of which clips fall in each fold. Validation rates vary naturally around 0.25 due to the smaller validation set size.

Table F.4 — Confusion Matrix Summary: All Models (Threshold = 0.4)

Model	True Negatives	False Positives	False Negatives	True Positives
Dummy Classifier	113	0	28	0
Logistic Regression	106	7	11	17
Random Forest	111	2	16	12
XGBoost Full Model	103	10	7	21

XGBoost achieves the highest true positive count (21 out of 28 high-risk clips correctly identified) at the cost of 10 false positives — an acceptable trade-off in a safety-oriented context where missed high-risk scenarios are more consequential than false alarms.

Figure F.1 — Hyperparameter tuning progression across all phases: phase comparison and top 20 candidates with CV score versus overfitting gap (supplementary).

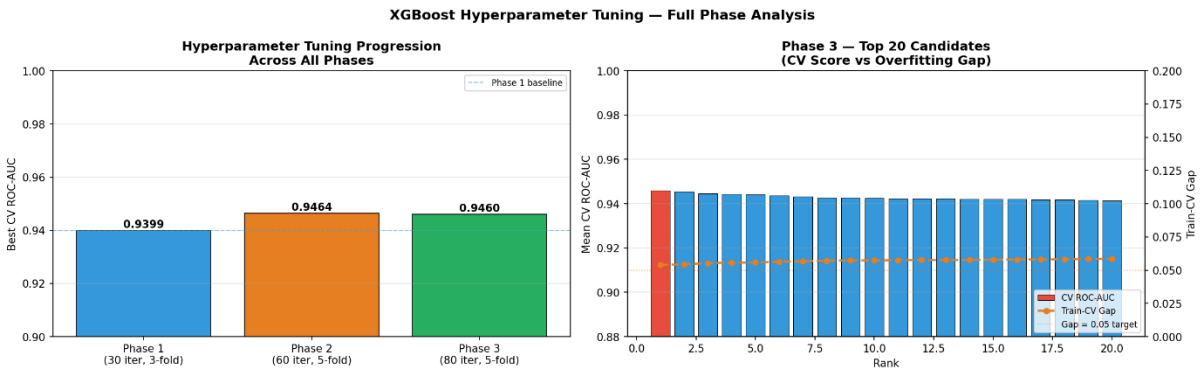
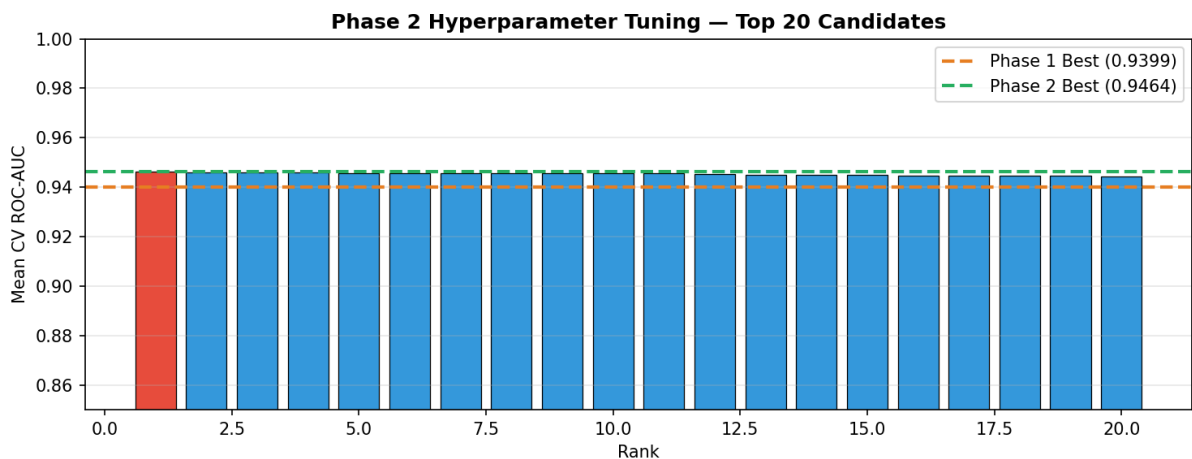


Figure F.2 — Phase 2 top 20 candidates: mean CV ROC-AUC by rank showing all candidates exceed Phase 1 baseline of 0.9399.



Appendix G — Parameter Configurations

This appendix documents the complete parameter configurations for all models and processes used in this research.

Table G.1 — Final XGBoost Hyperparameters

Parameter	Value	Rationale
n_estimators	700	Sufficient trees without structural memorisation risk at this dataset size
learning_rate	0.05	Balances convergence speed and generalisation
max_depth	5	Capped across all phases — deeper trees cause train score = 1.0 on ~1,500 samples
subsample	0.85	Row-level stochastic sampling reduces variance
colsample_bytree	0.4	Reduces feature correlation between trees
min_child_weight	2	Primary mechanism for reducing overfitting on small datasets
gamma	0.3	Minimum loss reduction required to split — acts as pruning

Parameter	Value	Rationale
reg_alpha	0.01	Light L1 regularisation
reg_lambda	4.0	Strong L2 regularisation — most effective paired with low <code>colsample_bytree</code>
scale_pos_weight	3.0	Ratio of negative to positive training instances — addresses class imbalance
tree_method	hist	GPU-optimised histogram-based tree construction
device	cuda	T4 GPU acceleration — Google Colab
random_state	42	Fixed seed for full reproducibility
eval_metric	logloss	Binary cross-entropy loss monitoring

Table G.2 — Hyperparameter Tuning Progression

Phase	Iterations	CV Folds	Best CV ROC-AUC	Train-CV Gap	Selection Method
Phase 1	80	5	0.9450	0.0550	Raw best with gap < 0.10
Phase 2	100	5	0.9455	0.0545	Gap-aware (gap < 0.08)

Delta between phases = +0.0005 CV ROC-AUC, −0.0005 gap — convergence confirmed. No further tuning phases required.

Table G.3 — Baseline Model Configurations

Model	Key Parameters	Purpose	ROC-AUC	Accuracy (t=0.4)	F1 High Risk
DummyClassifier	strategy = most_frequent	Minimum performance threshold	0.500	0.801	0.000
Logistic Regression	max_iter = 1000, random_state = 42	Linear interpretable baseline	0.879	0.851	0.618
Random Forest	n_estimators = 200, random_state = 42	Non-linear ensemble baseline	0.971	0.936	0.836
XGBoost	See Table G.1	Final fusion model	0.939	0.879	0.712
XGBoost Ablation (No CNN+ViT)	Same as G.1, 28 features	Final fusion model ((No CNN+ViT))	0.955	0.894	0.754

Table G.4 — Isolation Forest Configuration (Proxy Labelling)

Parameter	Value
contamination	0.25
random_state	42
Features used	avg_speed, max_speed, avg_jerk, max_jerk, hard_braking_events, is_night, is_peak_hour
Fitted on	Training set only (564 clips) — test set never seen during fitting
Threshold	25th percentile of training anomaly scores
Training labels assigned	141 high risk (25.0%), 423 low risk (75.0%)
Test labels assigned	28 high risk (19.9%), 113 low risk (80.1%)

Table G.5 — Feature Group Importance: Gain-Based vs SHAP

Feature Group	Features	Gain Importance	SHAP Mean ABS	Gain Rank	SHAP Rank
Ego / Behaviour	8	0.2806	3.9567	1	1
Context / Time	6	0.2367	2.0246	2	2
ViT Features	10	0.1437	1.0166	3	3
CNN Features	10	0.1382	0.9913	4	4
Engineered	3	0.0955	0.7080	5	5
YOLO Object Detection	9	0.0876	0.2823	6	6
Basic Visual	2	0.0175	0.0738	7	7

Gain-based and SHAP-based rankings are in complete agreement across all seven feature groups, validating the robustness of the feature importance findings. Ego/Behaviour and Context/Time are consistently the two dominant modalities by both methods.

Table G.6 — Top 10 Feature Rankings: Gain-Based vs SHAP

Rank	Gain-Based	SHAP-Based
1	is_peak_hour	is_peak_hour
2	is_night	avg_acceleration
3	aggressive_score	max_speed
4	hard_braking_events	avg_jerk
5	max_acceleration	hour_of_day
6	braking_events	hard_braking_events

Rank	Gain-Based	SHAP-Based
7	max_speed	aggressive_score
8	hour_of_day	is_night
9	avg_acceleration	max_acceleration
10	avg_speed	max_jerk

Both methods place `is_peak_hour` as the single most important feature. All top 10 features by both methods are behavioural or contextual — no CNN, ViT, YOLO, or visual features appear in the top 10 by either measure, consistent with the ablation findings.

Appendix H — Ethical Considerations and Data Access

This research was conducted exclusively using publicly available secondary data from the NVIDIA PhysicalAI Autonomous Vehicle Dataset, accessible via authenticated Hugging Face research access. No primary data collection was conducted, no human participants were involved, and no personally identifiable information was processed. The dataset does not contain facial imagery or individual driver identifiers. All dataset usage is consistent with the terms under which NVIDIA makes the data available for academic research.

The research was conducted in full accordance with Dublin Business School's academic integrity policy. All external sources are cited throughout in DBS Harvard referencing style. The five Google Colab notebooks constituting the research artefact are fully reproducible, with `random_state = 42` applied consistently across all stochastic components including dataset sampling, train-test splitting, model training, cross-validation, and hyperparameter search.

Appendix I — Gantt Chart

Figure I.1 — Gantt Chart

